



भारत सरकार

Government Of India

विद्युत मंत्रालय

Ministry Of Power

पूर्वी क्षेत्रीय विद्युत समिति

Eastern Regional Power Committee

14 Golf Club Road, Tollygunje-70033 Website: www.erpc.gov.in

NO. ERPC/EE/OPERATION/2023/1312

DATE: 26.12.2023

To

As per list enclosed/ संलग्न सूची के अनुसार.

विषय... 15.12.2023 (शुक्रवार) को ईआरपीसी सचिवालय, कोलकाता में भौतिक रूप से आयोजित 210वीं ओसीसी बैठक का कार्यवृत्त - संबंध में।

Sub: Minutes of 210th OCC Meeting held on 15.12.2023 (Friday) physically at ERPC secretariat, Kolkata - reg.

Sir/ महोदय

कृपया अपनी जानकारी और आवश्यक कार्रवाई के लिए 15.12.2023 (शुक्रवार) को ईआरपीसी सचिवालय, कोलकाता में 10:30 बजे आयोजित 210वीं ओसीसी बैठक के संलग्न कार्यवृत्त देखें। यह ईआरपीसी वेबसाइट (www.erpc.gov.in) पर भी उपलब्ध है।

Please find enclosed minutes of 210th OCC Meeting held on 15.12.2023 (Friday) physically at ERPC secretariat, Kolkata at 10:30 hrs for your kind information and necessary action. The same is also available at ERPC website (www.erpc.gov.in).

टिप्पणियाँ, यदि कोई हों, कृपया यथाशीघ्र इस कार्यालय को अग्रेषित करें।

Observations, if any, may please be forwarded to this office at the earliest.

इसे सदस्य सचिव के अनुमोदन से जारी किया जाता है।

This issues with the approval of Member Secretary.

Regards/ सम्मान,

Yours faithfully/ आपका विश्वासी,

 26/12/2023

(S.Kejriwal)

SE (Operation)

एसई (ऑपरेशन)

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CC:

Chief Engineer, OPM, CEA	Chief Engineer, NPC, CEA	ASSISTANT SECRETARY,ERPC
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ERPC:: Kolkata



MINUTES
OF
210TH OCC MEETING

Date: 15.12.2023

Eastern Regional Power Committee

14, Golf Club Road, Tollygunge

Kolkata: 700033

Contents

1. PART-A	3
1.1. Confirmation of Minutes of 209th OCC Meeting held on 23rd November 2023 physically at ERPC Secretariat, Kolkata	3
2. PART-B: ITEMS FOR DISCUSSION	3
2.1 Resource adequacy plan of states should be finalized at the earliest in consultation with CEA & MOP may be informed of the status of finalization: CEA	3
2.2 Capacity addition by State GENCOs & IPPs to meet growing energy Demand-CEA4	4
2.3 Implementation of Draft phasing plan of R&M/LE for Identified coal based thermal units-CEA	5
2.4 Seeking Project Appraisal from ERPC for PSDF Grant for Renovation and Modernization of intra-state transmission systems for relieving congestion in the state of West Bengal-WBSETCL	7
2.5 Procurement, Erection and Commissioning of Two Nos. 400 KV 125 MVAR (Each) Reactors at KTPS and Two Nos. 220 KV 125 MVAR (Each) Reactors (With New 220 KV Bays) at DSTPS-DVC	8
2.6 Shutdown proposal of generating units for the month of January'2024-ERPC	8
2.7 Re-conductoring and earth wire replacement in 220kV Lines (more than 35 years in service) commissioned under CTS.- POWERGRID ER-II	11
2.8 Discrepancy regarding resolution of availability declaration data on WBES portal- JITPL	12
2.9 Implementation of SPS for LILO arrangement/Power evacuation of 350 MW -IBEUL	13
2.10 Proposed shutdown of both circuits of OPGC-Sundergarh line for LILO connection of Ind Barath 350 MW unit -IBEUL	14
2.11 Reactive capability assessment- ERLDC	15
2.12 Furnishing of data for various studies mandated under IEGC 2023-ERLDC	16
2.13 Review of UFR setting- ERLDC	18
2.14 1200 MW Teesta III HE projects, Sikkim-Revision of schedule for due to force majeure on 04.10.2023	21
2.15 Installation of Three phase transformer bank at 400 kV New Chanditala sub-station- WBSETCL	22
2.16 Long outage of reactor at Jeerat S/S imposing high VAR charges on all DISCOMs of West Bengal_ WBSETCL	22
3. PART-C: ITEMS FOR UPDATE/FOLLOW-UP	24
3.1. ER Grid performance during November 2023.	24
3.2. Ensuring the healthiness of ADMS	24
3.3. Commissioning status of ADMS	25
3.4. Primary frequency response of generating units in ER	26

4. PART-D: OPERATIONAL PLANNING	28
4.1. Anticipated power supply position during January 2024	28
4.2. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on 12-12-2023)	28
4.3. Commissioning of new units and transmission elements in Eastern Grid in the month of November-2023.....	32
4.4. UFR operation during the month of November 2023.	34

EASTERN REGIONAL POWER COMMITTEE

MINUTES OF 210TH OCC MEETING TO BE HELD ON 15.12.2023 (FRIDAY) AT 10:30 HRS

Member Secretary, ERPC chaired the 210th OCC meeting. On welcoming all the participants, he outlined the performance of ER grid during November 2023 and highlighted the following points:

- *In November -2023, energy consumption of ER was **12,972.5 MU** which is **3.4 % more** than November -2022.*
- *In November -2023, Peak demand met of ER was **23,216 MW** which is **6.8 % more** than November -2022.*
- *During November -2023, **74.4 %** of the time, grid frequency was in IEGC band(49.90 Hz-50.05Hz)*
- *Thermal PLF of ER during November 2023 was **69.3 %**.*
- *Generating stations whose PLF is more than 90% during November -2023:*

Utility	Generating station	PLF(%)
WBPDC	Bakreswar TPS	97
NTPC	Darlipalli STPS	99

Transmission line (132 kV & above) commissioned during November-2023:

220 kV Muzaffarpur(PG)-Anmourn-1(ckt 1) D/C (ACSR Zebra) 65.4 ckt.km by BSPTCL.

There is no marked improvement in coal stock as stock position is in sub-optimal range in most generating units especially Jharkhand (TVNL) (only 4% stock), WBPDC (Kolaghat TPS having only 6% stock) along with Bokaro 'A' TPS (DVC)(with 28% stock) need to diligently focus on building their actual coal stock as per their normative requirement.

Coal stock position (As on 10.12.2023) is detailed as follows:

SL.	Name of States/Power Stns.	% of Actual Stock vis-à-vis Normative Stock
1.	Jharkhand (TVNL)	4%
2.	Odisha/IBTPS	66 %
3.	WBPDC	22 % (Min. Kolaghat TPS-6%, Max. Santaldih TPS-34 %)
4.	DVC	41 % (Min.Bokaro 'A' TPS-28 %; Max Chandrapura TPS - 53%)
5.	NTPC	66 % (Min.Barh STPS – 41 %; Max. North karanpura TPP -166%)

In this regard he highlighted that this is the most suited time to build coal stock keeping in view impending high demand season(Summer 2024) and thus advised all concerned GENCOs to strive for building optimum coal stock on sorting out various operational, commercial and administrative issues pertinent to coal linkage of respective generating units.

GM, ERLDC at the outset congratulated SLDC Bihar for being bestowed with best medium SLDC award in 3rd LDC excellence award ceremony organised by Grid-India in association with FITT, IIT Delhi and thereafter underscored operational issues like opening of transmission lines at higher bus voltage, conducting proper mock black start exercises and availing planned shutdown of generating units beforehand instead of encountering frequent forced outages. He also apprised of winter preparedness meeting convened on 30.11.23 by ERLDC to enhance awareness among all GENCOs as well as SLDCs about maintaining optimum VAR level in the grid so as to mitigate frequent overvoltage problem and also providing adequate primary frequency response by free governor control(FGMO) . He also urged all constituents to open transmission lines as last resort of mitigating overvoltage keeping system redundancy and reliability under consideration rather than on commercial grounds.

In this regard, West Bengal SLDC representative aptly pointed out the lack of adequate VAR support from the generators as a major contributing factor to overvoltage frequently encountered in many WBSETCL S/S and thus lightly loaded lines(below SIL) have to be opened to lessen overvoltage problem. He also apprised the forum that the initiative of installing 3 nos of 125 MVAR reactors by WBSETCL for VAR compensation and installation of bus reactors at 220 kV level besides 400 kV is also under consideration. It was also highlighted that installation of reactors has to be carried out in phases with proper planning to avoid overcompensation and in this regard variable excitation control of the generators can provide support to reduce overvoltage.

1. PART-A

1.1. Confirmation of Minutes of 209th OCC Meeting held on 23rd November 2023 physically at ERPC Secretariat, Kolkata

The minutes of 209th Operation Coordination sub-Committee meeting held on 23.11.2023 was circulated vide letter dated 01.12.2023.

Members may confirm the minutes of 209th OCC meeting.

Deliberation in meeting

Members confirmed the minutes of 209th OCC meeting with following modifications:

- In item no:2.10 of Minutes of 209th OCC meeting:*

OCC, on request from NTPC, suggested timelines of submitting digital proof against DC revision as follows:

Digital evidence against each instance of DC revision for partial outage occurring in a particular week(Monday-Sunday) shall be submitted by GENCOs to ERLDC by Wednesday of the immediate following week.

- Observations on item no:2.7 of Minutes of 209th OCC meeting have been deliberated at in item no:2.9*

2. PART-B: ITEMS FOR DISCUSSION

2.1 Resource adequacy plan of states should be finalized at the earliest in consultation with CEA & MOP may be informed of the status of finalization: CEA

In exercise of powers conferred under the rule 16 of Electricity(Amendment)Rules,2022, Ministry of power in consultation with CEA has issued the guidelines for **Resource adequacy**.

The guidelines were issued in June 2023 after stakeholder consultations. The guidelines aim to institutionalize the Resource adequacy framework to be followed by distribution licensees for power procurement & capacity contracting.

The Resource adequacy framework lays down the optimal capacity mix required to minimise the total cost in meeting the projected demand for the future.

This includes determining new generation capacities to meet future demand.

All individual states are requested to finalize Resource adequacy Plan at the earliest in consultation with CEA and Ministry of Power and may also inform the status of finalization.

Members may discuss.

Deliberation in meeting

ERPC representative delivered a comprehensive presentation (Annexure B.1) on guidelines related to resource adequacy planning framework in India , highlighting the following key facets:

Need and backdrop of resource adequacy planning in a fast growing major economy like India. Important terminologies of resource adequacy like planning reserve margin, expected energy not served, marginal cost of load reduction, loss of load probability, capacity credit and their significance in steering resource planning by the respective utilities and timelines for implementation of Resource adequacy as per CEA guidelines.

It was informed that CEA shall prepare Long term National resource adequacy plan (LT-NRAP) while NLDC is entrusted with responsibility of resource adequacy in short term (ST-NRAP)

In this regard, concerns were raised from OPTCL as well as WB SLDC regarding practical constraints to forecast 10 years plan amid highly fluctuating demand scenario in successive years, thus adequacy planning on year ahead basis sounds to be pragmatic.

On acknowledging this concern, OCC underscored the paramount significance of visionary planning amid rapid economic progress as well as India's commitment in COP 27 of 50% variable RE integration by 2030 and thereby advised all utilities(STU/SLDCs) to furnish forecasted demand vis-a vis availability details for next 10 years to CEA on rolling basis (by May of every year) for preparation of Long term National resource adequacy plan (LT-NRAP).

2.2 Capacity addition by State GENCOs & IPPs to meet growing energy Demand-CEA

Currently, India is the fifth largest economy in the world & is poised to become the third largest economy by 2030. This would entail availability of sufficient electricity to meet power demand of a growing economy. India's electrical energy demand has grown at CAGR of 4.5% while peak demand has grown at CAGR OF 5.6% during 2018-23. In the recent past, the energy demand growth has been much higher & it was 9.5% during 2022-23 & is showing a sustained rising trend. A peak demand of 243GW was recorded on 1st September 2023.

The future demand growth rates are likely to be higher primarily due to increased electrification of the economy, penetration of EVs etc. As per 20th Electric power survey(EPS), peak electricity demand & energy requirement is poised to grow at CAGR of 6.42% & 6.67% during 2022-27 & at a CAGR of 5.74% & 5.33% during 2027-32. respectively. Further as per the same survey, the projected All India peak electricity demand & electrical energy requirement is likely to be 277.2GW & 1907.8 BU respectively.

Therefore, adequate generation capacity addition is required matching with the growth in demand. As per the National Electricity plan 2022-32 notified by CEA, the likely installed capacity in the country for the year 2026-27 & 2031-32 is estimated to be 609.5GW(including 235.1GW from coal & 52.4GW from Large Hydro) & 900.4 GW(including 283GW from coal & 62.1GW from Large Hydro) respectively.

To meet the increased power demand, capacity addition in thermal & hydro sectors has been undertaken by the GOI. A total of 68.8 GW central sector capacity additions are likely to be achieved by 2029-30 through construction of Thermal & Hydro power plants by the GOI. A list of under construction/planned thermal & Hydro power plants in the central sector is enclosed for your information at **Annexure B.2.**

However the growing power needs of country can't be met without active participation from all stake holders viz state govts & IPPs. It is therefore essential that state Govts & its PSUs should also immediately plan & make concerted efforts to achieve capacity addition programme.

For those states which have already finalized & initiated the capacity addition activity, it is requested to inform the status of their capacity addition programme at the earliest to Ministry of Power.

GENCOs may update. Members may discuss.

Deliberation in meeting

*OCC, on underlining the significance of thermal generating units in enhancing inertia of interconnected grid, apprised all concerned utilities that to meet the increased power demand, capacity addition in thermal & hydro sectors has been undertaken by the GOI. A total of 68.8 GW central sector capacity additions are likely to be achieved by 2029-30 through construction of Thermal & Hydro power plants by the GOI. A list of under construction/planned thermal & Hydro power plants in the central sector is enclosed for your information at **Annexure B.2.***

OCC, thereby opined that all states as well as IPPs of ER may accordingly plan for capacity addition programme in long term to keep pace with nationwide demand escalation and share the same with CEA for monitoring of under-construction generating units.

2.3 Implementation of Draft phasing plan of R&M/LE for Identified coal based thermal units-CEA

The Central Electricity Authority (CEA) has prepared a guideline for Renovation and Modernization (R&M) in 2023, which was circulated to power utilities in August 2023. A draft phasing plan of identified thermal units for R&M/LE until 2030 was prepared and discussed in meetings held on 12th, 13th, 14th, 15th, and 16th of June 2023 at the CEA office in New Delhi.

After receiving suggestions from stakeholders, the phasing plan for R&M/LE of identified coal-based thermal units has been revised. It is therefore requested to provide inputs/comments on revised draft phasing plan for R&M/LE for identified coal based thermal units (upto 2030) at earliest convenience.

Communication from CEA in this regard and draft phase-wise plan for identified coal based thermal units are attached at **Annex B.3.1** and **Annex B.3.2** respectively for perusal and ready reference.

GENCOs may update. Members may discuss.

Deliberation in meeting

A brief overview of phase-wise R&M plan for ER generating units as drafted by CEA was presented by ERPC representative. (Annexure B.3.1) Generating units planned for R&M were illustrated in 4 phases for period spanning from Dec'2025 to Feb'2034.

OCC opined that phase-wise R&M plan(for 3 or 5 months) shall enable generating units to obtain mandatory clearances in an organised manner.

OCC also urged all generating units to explore R&M plan in their units to make them capable of operating even at 40% of MCR without oil support, thus accordingly the GENCOs may prepone their unit R&M plan from phase-3 or phase-4 to phase-1 with consent of OCC and intimation to CEA.

WBPDCCL representative submitted that some R&M activities including ESP renovation have already being undertaken at Kolaghat U#3(210 MW) during annual maintenance. It was also updated that Bandel TPS U#5 , having recently underwent R& M , doesn't require R&M activities in near future while Kolaghat U#5(210 MW) may be considered for R&M in place of Bandel TPS U#5 in phase-1 of CEA draft plan. OCC advised WBPDCCL to reassess healthiness of all its generating units prior to ascertaining the need of R&M and accordingly apprise CEA on R&M plan.

CESC representative affirmed that Southern generating unit(135 MW) does not require R&M as of now while OCC advised CESC to figure out possibility of carrying out R&M at Titagarh generating unit(240 MW).

NTPC representative asserted issuance of upcoming tariff regulations(2024-2029) as pre-requisite to better planning of R&M activities in regards to cost implications. He further underlined ascertaining the need of renovation of auxiliaries / main equipment prior to finalizing R&M time frame.

OCC opined that for ensuring optimum availability or PLF irrespective of tariff regulations, NTPC should explore carrying out R&M under O&M cost and accordingly submit R&M plan to CEA after conducting residual life assessment of all generating units.

DVC representative intimated about ongoing residual life assessment at Mejia TPS(U#1,2,3)

OCC advised WBPDCCL, DVC and NTPC to share comprehensive R&M plan with information to CEA.

WBPDCCL vide mail dated 20.12.2023 shared R&M plan for their respective units(Annexure B.3.2)

2.4 Seeking Project Appraisal from ERPC for PSDF Grant for Renovation and Modernization of intra-state transmission systems for relieving congestion in the state of West Bengal-WBSETCL

Intra-state transmission networks of West Bengal have constraints in meeting the demand for power in several pockets/ regions within the State of West Bengal. Whereas new lines/ connectivity could not be established due to severe ROW problem, replacement of existing conductor by high performance conductor has been envisaged to relieve the congestion.

The use of High Temperature and Low Sag (HTLS) Conductors is an attractive method of increasing transmission line thermal rating. The conventional ACSR Conductors are able to withstand a continuous temperature of 75 deg C to 85 deg C. In case of emergency, for a short duration the conductors can work up to 105 deg C without any sign of deformation. In order to increase the thermal rating of existing lines, one method involves replacing ACSR conductors with special “High-Temperature Low-Sag” (HTLS) conductors having approximately the same diameter as the original ACSR but being capable of operation at temperatures as high as 250 deg C, with less thermal elongation than ACSR. Ideally, these special HTLS conductors can be installed and operated without the need for extensive modification of the existing structures and foundation. With the use of this new generation HTLS conductor, power carrying capacity will be raised almost double from its existing capacity.

Following projects have been considered in present proposal:

a) Replacement of existing ACSR Zebra Conductor by high capacity HTLS Conductor along with replacement of 220 KV Feeder and TBS bay CT of the following transmission lines:

1. KTHP-Food Park 220 KV D/C Transmission Line (R.L. $\approx$ 51.0 KM).
2. Food Park-Jangalpur 220 KV D/C Transmission Line (R.L. $\approx$ 6.0 KM).
3. Barasat-Kasba 220 KV D/C Transmission Line (R.L. $\approx$ 39.1 KM).
4. Subhasgram-Kasba 220 KV D/C Transmission Line (R.L. $\approx$ 23.0KM).
5. Jeerat-Barasat 220 KV D/C Transmission Line (R.L. $\approx$ 23.3 KM).

In this regard a DPR has been prepared for grant of funding from PSDF as per Clause No 6.3(i) as per guideline of GoI, MoP issued vide no 10/1/2014-OM Dtd. 18.09.2014. The in-principle approval of the competent authority of WBSETCL has already been obtained.

The DPR has been placed before ERPC on 03-11-2023 for appraisal on the project as per the requirement under PSDF Application Format-A5.

WBSETCL may update. Members may discuss.

Deliberation in meeting

WBSETCL representative submitted that the total estimated cost of the proposed R&M of intra-state transmission system is ₹252.34 Cr out of which ₹189.25 Cr is being sought as grant from PSDF. The rest shall be met from its internal fund. He requested for technical concurrence of ERPC under PSDF Application Format-A5.

Considering merit of the proposal and keeping in view the congestion in the intra state network of West Bengal as deliberated earlier in various previous ERPC fora, OCC gave technical go-ahead to the proposal and advised WBSETCL to approach PSDF nodal agency (NLDC) for seeking further approval.

2.5 Procurement, Erection and Commissioning of Two Nos. 400 KV 125 MVAR (Each) Reactors at KTPS and Two Nos. 220 KV 125 MVAR (Each) Reactors (With New 220 KV Bays) at DSTPS-DVC

The 400 KV bus voltage of DVC system occasionally crosses the higher operational voltage limit of 103% i.e. 412 KV as stipulated in IEGC-2010. Moreover, with the introduction of charges for exchange of VAR during violation of either the lower limit (97% of 400 KV) or the higher limit (103% of 400 KV) as a disciplinary measure it has become important to operate the system buses within these two limits. Hence to maintain 400 KV Bus voltage within 412 KV it has become necessary to install and commission 2 nos. of 400 KV, 125MVAR Reactor at KTPS and 2 nos. of 220 KV 125 MVAR Reactors at DVC, DSTPS.

In line with above CTU has already extended their consent vide MOM No. CTU/E/00/DVC dated 26.06.23 towards installation of suitable number of bus Reactors of 125MVAR at various generation switchyards to control the voltage excursions.

The detailed study report (**Annex B.5.1**) and the CTU MOM(**Annex B.5.2**) are attached herewith for scrutiny and subsequent approval towards implementation of the proposal.

DVC may update. Members may discuss.

Deliberation in meeting

DVC representative submitted that bus reactor capacity has been increased from 50 MVAR to 125 MVAR as the same to be installed at 400 kV level instead of earlier proposed 220 kV level.

*He clarified that 2x125MVAR reactor has already been considered in Project of Koderma TPS and the existing 2x50MVAR reactor from KTPS will be used at MTPS 400KV system. He further intimated that present case is related with DSTPS only and accordingly submitted the VAR data and reactive capability curve of DSTPS. (**Annexure B.5.1**)*

*ERLDC informed that despite MVAR absorption by DSTPS unit, bus voltage still touching 417 kV and thus to bring the same below 412 kV, more 250 MVAR absorption is needed thereby justifying the need of 2X125 MVAR bus reactor at DSTPS. Study results on reactive performance for DSTPS units attached at **Annexure B.5.2**.*

OCC finally granted consent in favour of installing 2X125 MVAR bus reactor at DSTPS including associated bays at 400 kV level.

2.6 Shutdown proposal of generating units for the month of January'2024-ERPC

Maintenance Schedule of Thermal Generating Units of ER during 2023-24 in the month of January '2024							
System	Station	Unit No.	Capacity (MW)	Period (as per LGBR 2023-24)		No. of Days	Reason
				From	To		
DVC	Mejia TPS	3	210	21.01.2024	14.02.2024	25	AOH-Blr, LPT, FGD
	Mejia TPS	7	500	01.01.2024	28.01.2024	28	AOH-Blr, LPT, Gen
WBPDC	Santalidih TPS	5	250	04.01.2024	23.01.2024	20	AOH/BOH
	APPROVED DATE (as per 208 th OCC)			26.01.2024	14.02.2024	19	AOH/BOH
	Kolaghat TPS	3	210	26.01.2024	14.02.2024	20	AOH/BOH
NTPC	KhSTPS	4	210	01.12.2023	04.01.2024	35	Boiler + HP + IP + Generator
	APPROVED DATE (as per 209 th OCC)			01.12.2023	04.01.2024	35	Boiler + HP + IP + Generator
	KhSTPS	5	500	01.12.2023	14.01.2024	45	Boiler+LP+Gen.+ Combustion Modification (Approved in LGBR 2022-23)
	APPROVED DATE (as per 209 th OCC)			Shutdown not to be availed as per 209 th OCC			
	KhSTPS	7	500	10.01.2024	08.02.2024	30	Boiler Maintenance
	Barh-I	2	660	01.12.2023	14.01.2024	45	Boiler +Generator
	Barh-II	5	660	20.01.2024	18.02.2024	30	BLR Modification Balance work
	Darlipalli STPS	2	800	15.11.2023	13.01.2024	60	COH

BRBCL	Nabinagar TPS	4	250	01.12.2023	09.01.2024	40	Boiler+ LPT O/H + Generator rotor thread out and checking +NOX Work
DPL	DPL	8	250	01.01.2024	10.01.2024	10	Boiler License Renewal
HEL	HEL	1	300	06.01.2024	20.01.2024	15	AOH/BOH
	APPROVED DATE (as per 209th OCC)			02.12.2023	20.12.2023	19	AOH & De-NOx modification
MPL	MPL	2	525	17.12.2023	25.01.2024	40	AOH/COH
	Decided to be postponed as per 209th OCC						

Members may discuss.

Deliberation in meeting

DVC representative sought for postponing planned shutdown of DSTPS U#2 from 19.12.2023 for 28 days due to FGD works cum annual overhauling, RTPS U#2 from 16.01.2024 to 12.02.2024 for boiler overhauling cum FGD works and Mejia TPS U#5 from 14.02.2024 to 15.03.2024 for capital overhauling cum FGD works. Shutdown of Mejia TPS U#3 and U#7 were informed not to be availed by DVC.

WBPDCCL representative submitted shutdown requests for Bandel U#5 from 07.01.2024 to 31.01.2024 and Kolaghat TPS U#3 from 02.02.2024 to 23.02.2024 and intimated preponing already approved shutdown of Santaldih TPS U#5 from 03.01.2024 to 22.01.2024

NTPC representative apprised of adhering to already approved shutdown schedule of Kahalgaon TPS U#4 and U#5 while pleaded for availing shutdown of FSTPP III U#6 from 10.01.2024 to 25.02.2024 for foundation recasting of HP turbine subjected to bearing failure. Barh-I(U#1), subjected to forced outage since 19.11.2023 due to BTL, was intimated to be restored by 05.01.2023 while shutdown of Barh-I(U#2) as well as Barh-II(U#5) was confirmed not to be availed by NTPC.

HEL representative acceded to already approved shutdown of HEL U#1.

DPL representative apprised of not availing shutdown of DPL U#8 as per approved LGBR for FY 2023-24 while DPL U#7, which is under shutdown for ESP works cum exciter fault, is expected to be restored to service by 31.01.2024. It was further intimated that owing to HP turbine failure DPL U#8 is also currently out of service since 25.11.2023 and expected to be revived in last week of December'2023. Raising serious concern on frequent outage of DPL units, OCC advised DPL to emphasize rigorous predictive maintenance as well as maintaining optimum coal stock.

OCC granted consent to all above mentioned planned shutdown requests and urged all concerned generating units of ER to meticulously execute maintenance and overhauling related works in upcoming winter so that no planned shutdown entails to be availed from March 2024 to May 2024 in line with MOP guidelines.

The detailed shutdown schedule as approved by OCC is provided at **Annexure B.6**.

2.7 Re-conductoring and earth wire replacement in 220kV Lines (more than 35 years in service) commissioned under CTS.- POWERGRID ER-II

In 209th OCC Meeting , POWERGRID ER-II was advised to submit a detailed survey report along with health assessment report from NSIC of conductor installed in old 220kV Lines commissioned under CTS.

Accordingly, vide mail dated 12.12.2023, Powergrid ER-II has shared detail test report of old 220kV Lines , carried out by NSIC Howrah.(**Annex B.7.1**)

From the test report, it can be observed that the conductor sample has failed in mostly all Technical parameters. A summary of the TS requirements on which the conductor has failed during Testing are stated below:-

SI No.	Tests performed	Observation/Findings	Remarks
1	Freedom from defects (Visual Inspection)	Conductor found blackened & surface not smooth	Condition Poor
2	Surface Condition Test	Upon applying 50% of UTS Load, the diameter of the conductor measured at 4 places are more than the sum of minm. Specified diameter of the individual Al & Steel strands	Failed
3	Ultimate Breaking Load (Whole conductor & individual Aluminium Strands also)	Conductor strength found (114.25kN) way below the minm. UTS requirement of 130.32kN	Failed
4	DC Resistance test	Average value of resistance observed (0.12846 ohm/KM) is way above the Max. allowable DC resistance (0.06868 ohm/KM) of conductor as per TS.	Failed

Standard technical specifications for ACSR conductors is attached at **Annex B.7.2**

In addition, POWERGRID ER-II has also conducted detailed field survey at Alipurduar, Birpara, Binaguri, Dalkhola & Malda Transmission Line maintenance offices entrusted for maintenance of those lines. From the survey report, it is evident that condition of conductor & earth-wire in 220kV Lines under CTS have really deteriorated resulting in repeated conductor & earth-wire snapping incidents which is causing untoward line trippings as well as chances of any major accident cannot be ruled out in future.

Considering the increase in conductor & earth-wire snapping, it is proposed to consider the re-conductoring of 220kV Lines commissioned under CTS under the ADDCAP 2024-2029 tariff block of Chukha Transmission System.

POWERGRID ER-II may update. Members may discuss.

Deliberation in meeting

POWERGRID ER-II representative, with submission of test report by NSIC as well as field survey report, underscored the need of re-conductoring the Chukha Transmission system, comprising of transmission lines totalling 470 ckt. km from existing ACSR Zebra conductor to HTLS. He reiterated that the said transmission lines are more than 37 years old and are past their useful life.

ERLDC representative submitted that considering the strategic importance of the corridor and RoW constraint due to the small corridor in the chicken neck area, re-conductoring of the lines may be done by the HTLS conductor for some sections where high loading is expected. It will increase power evacuation capacity from future hydro projects of hydro-rich Bhutan & North-Eastern region.

OCC found merit and technically agreed to the proposal of Powergrid.

OCC advised POWERGRID ER-II to explore various funding options including PSDF funding for re-conductoring of the above proposed lines and submit the same to next CCM for further deliberation.

2.8 Discrepancy regarding resolution of availability declaration data on WBES portal-JITPL

Jindal India Thermal Power Limited (JITPL) has 2x600 MW thermal power plant located in the village of Derang, District Angul, State Odisha.

As per IEGC Regulations, 2023 Clause 49 (vi) the availability declaration by regional entity generating station shall have a resolution of two decimal (0.01) MW and three decimal (0.001) MWh but still the availability declaration by regional entity generating station is showing a resolution of six decimal in MW and MWh on WBES portal Full Schedule and at the same portal in GNA report the availability declaration by regional entity generating station is showing

a resolution of two decimal in MW and resolution of six decimal in MWh due to this mismatch in availability declaration it is very confusing that what schedule to be considered for the purposes of scheduling, accounting and billing.

In this regard, it is requested to kindly consider the same resolution of two decimal in MW and three decimal in MWh as per IEGC Regulations, 2023 in every report uploaded on WBES portal for the purposes of scheduling, accounting and billing.

JITPL and ERLDC may update. Members may discuss.

Deliberation in meeting

ERLDC representative submitted that the issue of resolution in availability declaration on WBES portal i.r.o. JITPL has been resolved and the same has also been comprehensively conveyed to JITPL.

JITPL representative acceded to the submission by ERLDC.

2.9 Implementation of SPS for LILO arrangement/Power evacuation of 350 MW - IBEUL

With reference to 209th OCC meeting held on 23.11.2023, it has been decided that line loading of each circuit of D/C 400 kV OPGC-Lapanga line will be restricted upto 1000 MW and accordingly SPS to be designed for power evacuation of 350 MW through LILO connection with one circuit of OPGC-Sundergarh 400 kV line, while mentioned as 1000 MVA in MOM.

Also, ERPC is requested vide mail dated 27.11.2023 on observation on SPS logic, that in place of alarm triggering and unit tripping on MW signal, it should be considered on basis of current signal as there may be constraint in relay for MW signal. Tripping on MW or current logic can be finalized after engineering to be carried out considering available installed relay at OPGC end.

In this regard, OPGC is requested by IBEUL vide letter no. IBEUL/23-24/0184 dtd 08.12.2023 for providing data of 400 kV Lapanga line and other data required for LILO connection.

OCC is requested to intervene in this matter to facilitate requisite support from OPGC as well as getting access to OPGC end for SPS implementation.

IBEUL may update. Members may discuss.

Deliberation in meeting

IBEUL representative requested for modification in SPS logic from MW to current as actuating signal for the numerical relay.

As confirmed by SLDC Odisha and OPTCL, line loading of any one circuit of 400kV-OPGC-Lapanga-DC should be restricted to 1000MW.

OCC, agreeing to this request suggested modified SPS logic as follows:

It was finally decided that SPS shall be implemented in three stages where the current in Ampere per conductor of any of the 400 KV OPGC lapanga double ckt lines shall be used to trip one unit of Ind- bharat. The earlier decided MW value which is 950 for alarm and 1000 for trip will be converted to amps considering 0.95 Pf

Stage 1 (Alarm)

- The current of OPGC-Lapanga S/C is greater than 1440 Amps and less than 1520 Amps*

Stage 2 (trip with 10 minutes delay)

- The current of OPGC-Lapanga S/C is greater than 1520 Amps and less than 1550 Amps.*

Stage 3 (trip with 10 Second delay)

- The current of OPGC-Lapanga S/C is greater than 1550 Amps*

OCC also advised OPGC to extend all necessary support to IBEUL for implementing SPS logic as well as for proposed LILO connection.

2.10 Proposed shutdown of both circuits of OPGC-Sundergarh line for LILO connection of Ind Barath 350 MW unit -IBEUL

Ind Barath is in advance stage of its LILO agreement with CTU and subsequent approval from CEIG and ERLDC for the first time charging and may be ready for physical connection in January (any day).

So it has been proposed for shutdown of both circuits of OPGC-Sundergarh for LILO connection of Ind Barath both circuit with OPGC-Sundergarh Circuit#2. After physical connection, OPGC-Sundergarh Circuit#1 may be charged while circuit#2 may be kept off for protection testing.

The proposed shutdown duration is as per below:

OPGC-Sundergarh Circuit#1 & 2	2 days
OPGC-Sundergarh Circuit#2	7 days

Presently, exact shutdown date could not be provided but OCC is requested to consider shutdown of OPGC-Sundergarh line in January and it is assured that Ind- Barath may submit shutdown request at least 7 days before availing the same.

IBEUL may update. Members may discuss.

Deliberation in meeting

IBEUL representative apprised of availing emergency shutdown of OPGC-Sundergarh lines on intimation to and obtaining consent from ERLDC.OCC acceded to the same.

On enquiry from OCC, it was also affirmed that 40% of DTL for IBEUL plant is already completed and extra manpower has been deployed by IBEUL for its final completion by mid-Februray'2024.

2.11 Reactive capability assessment- ERLDC

As per IEGC 2023 39.3

All generating stations connected to the grid shall generate or absorb reactive power as per instructions of the concerned RLDC or SLDC, as the case may be, within the capability limits of the respective generating units, where capability limits shall be as specified by the OEM.

Periodic testing is mandated by **IEGC 2023 clause no 40.2(c)** and The tests shall be performed once every five (5) years or whenever major retrofitting is done. **If any adverse performance is observed during any grid event, then the tests shall be carried out even earlier, if so advised by SLDC or RLDC or NLDC or RPC, as the case may be.**

Suitable extract of testing requirement for synchronous generators are mentioned below,

Power System Elements	Tests	Applicability
Synchronous Generator	(1) Real and Reactive Power Capability assessment.	Individual Unit of rating
	(2) Assessment of Reactive Power Control Capability as per CEA Technical Standards for Connectivity	100MW and above for Coal/lignite,
	(3) Model Validation and verification test for the complete Generator and Excitation System model including PSS.	50MW and above gas turbine and 25
	(4) Model Validation and verification of Turbine/Governor and Load Control or Active Power/ Frequency Control Functions.	MW and above for Hydro.
	(5) Testing of Governor performance and Automatic Generation Control.	

As discussed in winter preparedness meeting on 30 November 2023 and based on the performance the list of non-performing units for reactive support is mentioned below which should take corrective measures to provide suitable reactive support as per capability curve .Not fulfilling the same units may have to go under revised **reactive capability assessment as per IEGC Clause 40.2(C)**.

1. JITPL Unit 1& 2 ,600MW Each
2. Mejia B Unit 7&8 ,500Mw Each
3. BRBCL Units 2&3 , 250 Mw each
4. NPGC Unit 1, 660 MW
5. Kolaghat Units , 210 MW Each

ERLDC may update. Members may discuss.

Deliberation in meeting

*ERLDC representative shared a concise presentation (**Annexure B.11**) illustrating reactive performance of some generating units i.e JITPL (U#1 and U#2), BRBCL(U#2 &U#3),Mejia B(U-08), Kolaghat U#4 and NPGC U#1. It was also highlighted that periodic testing of all generating units to be carried out every 5 years or earlier if any adverse performance is observed during any grid event and accordingly revised capability curve should be submitted to ERLDC.*

WB SLDC representative stressed upon the fact that despite paying for both active and reactive power to the generators as full time beneficiaries, DISCOMs are unduly penalized with huge amount of reactive energy charges owing to overvoltage menace, especially in West Bengal with extremely high peak to off-peak demand ratio. Thus the necessity of optimization between technical requirements and commercial implications was pointed out.

WBPDC representative intimated VAR absorption threshold of Kolaghat U#4 to be 20 MVAR while the same was confirmed to be 50-55 MVAR by WB SLDC as per capability curve.

NPGC U#1 was advised by OCC to explore possibility of tap change by OCC to mitigate VAR absorption deficit by the generator.

OCC advised ERLDC to put up the agenda in next OCC meeting for comparison of reactive performance of poor performing generating units between Jan'2023 and Jan'2024 and accordingly the need of reactive capability assessment of the respective generating units shall be detailed out.

2.12 Furnishing of data for various studies mandated under IEGC 2023-ERLDC

As per IEGC 2023, various studies are mandated to be done at SLDC/RLDC and NLDC levels. They broadly comprise of **operational planning studies, interconnection studies and transfer capability assessment studies.**

As per IEGC 2023 Clause no 31, **Operational planning** shall be carried out in advance by NLDC, RLDCs and SLDCs within their respective control areas in various timelines ranging from monthly and yearly time horizons in co-ordination with CTU, RPCs or STUs, to Intra-day, Day Ahead, Weekly time horizons. Methodology of carrying out such studies has been described under section 2.4 of Operating procedures for Eastern region. The roles and timelines are again reproduced in the **Annexure B.12.1**.

Interconnection study is to be performed at SLDC, RLDC and NLDC level which involves analyzing the impact of all elements to be commissioned within six months on a rolling basis each month and evaluating the impact these elements on the transfer capability of import/export of state, at interstate, intra and inter regional levels. '**Procedure for carrying Out Inter-Connection Studies of New Power System Elements**' published by Grid-India in September,2023, lays down the guidelines for data submission and performing the interconnection studies for new power system elements to be integrated into the Indian grid. All Users ,which may include, Concerned Transmission Licensee (s)/ Settlement Nodal Agency (SNA), ISTS Licensee, Intra- state (In STS) Licensee through SLDC, Generation Entity/ Load Serving Entity / Connectivity Grantee responsible for providing the necessary data for first time integration of elements shall furnish the information each month on a rolling basis The format specified for submitting the data is given at **Annexure-I** of the **interconnection procedure** mentioned. The necessary technical and modelling data of all the elements expected to be energized in the “M” month shall be submitted as per the formats specified in Grid-India/NLDC’s procedure for **“First Time Charging /Energization (FTC) an Integration of New or Modified Power System Element”**, as amended from time to time, for carrying out the necessary studies.

The role specified for SLDCs, RLDCs and NLDC after receipt of the above data is taken from the above mentioned **Interconnection procedure** and **Procedure for transfer capability capability assessment methodology** ,both published by Grid India in September 2023. Respective SLDC, in consultation with respective RLDC and STU, shall carry out a joint study on the impact of new elements to be energized in the intra-state system within its jurisdiction in the next six (6) months and shall share the results with the concerned RLDC. Respective RLDC, in consultation with concerned SLDCs, STUs, NLDC and CTU, shall carry similar joint study in (a) the ISTS of the region and (b) the intra-state system on the inter-state system and shall share the results of the studies with NLDC while NLDC shall carry out similar study on the impact of new for (a) inter-regional system, (b) cross-border links and (c) intra-regional system on the inter-regional system and share the results with the RLDCs/SLDCs. The overall workflow with timelines at M-7th and M-6th month for month M are shown in **Annexure B.12.2**. **Item NO. B.23 and annexure B.23 of 208th OCC** briefly touched upon these aspects.

On 208th OCC, under ITEM NO. B.22, the commencement of twelve month ahead transfer capability assessment as mandated by IEGC 2023 was intimated. The requirement of base case submission by SLDC to RLDC and assessing of transfer capability (TTC/ATC) and sharing its results, on a rolling basis, were introduced in three separate sets of timelines which are

- 1) twelve month in advance with elements already commissioned,
- 2) Review of the above case one month ahead with updation of elements commissioned .
- 3) six month ahead including the upcoming elements considered for interconnection study

The aspects of TTC assessment for six month (3rd point) ahead are already covered in the interconnection study itself. The timeline for 1st and 2nd point are reproduced in **Annexure B.12.3** and **Annexure B.12.4**.

Previously, all state planning agencies (STUs) and SLDCs were requested to furnish data on an email dated 6th September, 2023 for facilitation of transfer capability assessment and interconnection study as per the timelines above. A separate email to ISTS licensees and regional generation entities has also been sent on 17th November. However, so far no inputs in form of raw data of upcoming elements, modelling such data for base case, sharing of study results of their impact in transfer capability, or even case incorporating twelve month ahead LGBR and their transfer capability has been received from concerned agencies involved. In this context, it is requested to share these data at the earliest and then on a rolling basis every month as per the timelines mentioned above such that relevant provisions in IEGC are complied. The given timelines can also be found in 'Procedure for Transfer Capability Assessment Methodology', 'Procedure for carrying Out Inter-Connection Studies of New Power System Elements' published by Grid India on September 2023.

ERLDC may update. Members may discuss.

Deliberation in meeting

*ERLDC representative shared a concise presentation (**Annexure B.12**) outlining the timelines of data submission for 6 month ahead interconnection study of upcoming elements thus facilitating base case network modelling as well as TTC assessment beforehand. To facilitate avail of T-GNA for 11 months duration, SLDCs were requested to share ATC/TTC data twelve months in advance.*

OCC advised SLDCs to abide by timelines of sharing details of upcoming intra-state transmission elements to aid in inter-connection study for enhanced system reliability.

OCC also advised all SLDCs to prepare standard operating procedure (SOP) for demand projection as well as unit-wise availability on week-ahead basis and thereby adhering to timelines of operational planning (demand and generation projection) as mandated in IEGC 2023.

2.13 Review of UFR setting- ERLDC

As per the IEGC 2023 Following to be adhered with respect to UFR.

The following shall be factored in while designing and implementing the UFR and df/dt relay schemes:

- a) The under-frequency and df/dt load shedding relays are always functional.
- b) Demand disconnection shall not be set **with any time delay in addition to the operating time** of the relays and circuit breakers.
- c) There shall be **a uniform spatial spread of feeders selected for UFR and df/dt disconnection.**

d) SLDC shall ensure that telemetered data of feeders (MW power flow in real time and circuit breaker status) on which UFR and df/dt relays are installed is available at its control centre. SLDC shall monitor the combined load in MW of these feeders at all times. SLDC shall share the above data with the respective RLDC in real time and submit a monthly exception report to the respective RPC.

e) RLDC shall inform SLDCs as well as the concerned RPC on a quarterly basis, durations during the quarter when the combined load in MW of these feeders was below the level considered while designing the UFR scheme by the RPC. SLDC shall take corrective measures within a reasonable period and 72 inform the respective RLDC and RPC, failing which suitable action may be initiated by the respective RPC.

f) RPC shall undertake a monthly review of the UFR and df/dt scheme and also carry out random inspection of the under-frequency relays. RPC shall publish such a monthly review along with an exception report on its website.

g) SLDC shall report the actual operation of UFR and df/dt schemes and load relief to the concerned RLDCs and RPCs and publish the monthly report on its website.

The load shedding for each stage of UFR operation, in percentage of demand or MW shall be as finalised by the respective RPCs.

- In 10th NPC meeting it was finalized that **Total 25% relief to be planned in all 4 stages-49.4 Hz, 49.2 Hz, 49.0 Hz & 48.8 Hz.**

- Percentage of UFR installed with respect to all time highest demand met is calculated. For Bihar it seems very low for other states also it may be reviewed.

- Based on the last event of Under frequency operation and in view of the implementation of IEGC 2023 from 01st October. State wise UFR status, additional intentional time delay other than the measurement delay, Relay type has been collected and compiled with major observations as shown below in the table which needs to be rectified as mandated in IEGC-2023.

State	STG-I/49.4	STG-II/49.2	STG-III/49	STG-IV/48.8	Total	Maximum Demand Met	Percent age load relief by UFR(%)	Remarks
DVC	122	145	147	138	552	3751	14.7	100 Ms Intentional delay, All are Numerical relay
BIHAR	84.5	104.5	133	82.5	404.5	7578	5.3	Uneven stage wise distribution, No intentional delay but mostly Static relay
WEST BENGAL	381	374	390	386	1531	11868	12.9	200 Ms Intentional Delay, with mostly Numerical type.

ODHISH A	183	184	184	186	737	7192	10.2	No time delay with mostly numerical type.
JHARKH AND	70	69	45	79	263	1923	13.7	Uneven stage wise distribution, No intentional delay with all numerical relays.
CESC	65	90	125	120	400	2606	15.349 19417	
SikkiM	No UFR Installed							

- Percentage of UFR installed with respect to all time highest demand met is calculated. **For Bihar it seems very low for other states also it may be reviewed.**
- Intentional Delay as observed should be removed as mandated in IEGC.**
- Distribution of UFR quantum should also be uniform in each stage which is not for Bihar ,CESC.**
-

ERLDC may update. Members may discuss.

Deliberation in meeting

*ERLDC representative shared a concise presentation (**Annexure B.13**) highlighting the load relief quantum deficit under UFR scheme of ER constituents and necessary actions to be taken in this regard.*

It was asserted that the installed UFR relays must not have any intentional time delay and also the load relief quantum (25% of peak demand) should be uniformly distributed across 4 distinct stages (Stage-I to Stage-IV) of under-frequency load shedding scheme.

WB SLDC submitted that revision in load relief quantum for UFR can be committed only in presence of concerned DISCOMs.

DVC representative updated that intentional time delay has now been removed from installed UFR rendering it to operate instantaneously while intentional time in UFRs under WB SLDC jurisdiction shall be removed with consent from concerned DISCOM as asserted by WB SLDC representative.

OCC advised CESC to explore possibility of ADMS installation in addition to UFR despite having chunk of critical loads and urged Bihar SLDC to submit study report for UFR installation detailing out identified locations along with necessary components.

OCC also advised all concerned utilities to demarcate separate feeders for ADMS and UFR installation to ensure sufficient load relief for specific purpose.

ADDITIONAL AGENDA

2.14 1200 MW Teesta III HE projects, Sikkim-Revision of schedule for due to force majeure on 04.10.2023

Natural disaster in the upstream catchment area of the dam of our 1200MW Teesta III HE project on the intervening night of 03.10.2023 & 04.10.2023.

IEX, through PTC, has raised a debit note on us towards penalty due to revision of schedule for 04.10.2023 amounting to RS 1.19 Cr (copy enclosed). As per the details provided in the said debit note, a compensation amount @20% of the trade price has been levied on the difference between contracted capacity quantity & delivered quantity. Such compensation has been arrived based on the following of the Business Rules of IEX:

*“7.4.2.3 Revision of schedule: Exchange shall not revise the schedule once addition for schedule has been made. IN case system operator on account of reason other than force majeure or constrains in the transmission corridor revises the schedule, a penalty shall be imposed on defaulting Member or Client, as the case may be. The penalty amount will be equivalent to shortfall in contracted quantity at the rate of 20% of the trade price i.e ([contracted quantity-delivered quantity]*trade price*0.2) or as may be specified by the exchange from time to time. The penalty amount shall be passed on to the counterparty as compensation after adjustment of applicable charges, if any, & deducting administrative charges for the Exchanges which will be 5% of the penalty amount. Subject to a maxim of Rs 10,000(including taxes). the penalty amount shall be passed on to the counterparty at the end of the contract duration or at the time of cancellation of the contract. If the duration of the contract is more than a month/30 days the compensation of penalty amount & passing on that as a compensation to the counterparty shall be carried out on a month/30 days basis.”*

As inferred from the above provision, revision of schedule by the system operator for reasons attributable to force majeure doesn't attract penalty/ compensation. Govt of Sikkim, vide its Notification has already dated 04.10.2023 has already declared the above catastrophes as a Disaster in exercise of its power under the Disaster management act 2005. copy of the said notification has already been provided to the ERLDC vide our letter under reference. Therefore, the above incident that occurred in the intervening night on 03.10.2023 & 04.10.2023 which resulted in the Teesta III HE plant getting shut down & revision of schedule for 04.10.2023, is a force majeure event.

In view of the foregoing, it is requested that ERLDC may kindly convey IEX that revision of SCHEDULE FOR 04.10.2023 WAS CARRIED OUT DUE TO force majeure event.

SUL may update. Members may discuss.

Deliberation in meeting

TUL representative briefly explained the agenda.

On request from SUL, OCC consented that the Glacial Lake outburst flow event that happened on the intervening night of 03.10.2023 & 04.10.2023 was a Force Majeure event and it adversely impacted the Teesta III project including other downstream hydro projects. The

same has already been categorised as natural disaster by Govt. of Sikkim vide notification dated 04.10.2023.

2.15 Installation of Three phase transformer bank at 400 kV New Chanditala sub-station- WBSETCL

In view of the congestion faced across ICTs at 400 KV New Chanditala sub-station in last summer and the delay in procurement process for 4th ICT due to tender cancellation etc, it is felt needed to utilise single phase transformers from own stock of WBSETCL to form a trf bank of capacity 315 MVA.

- (i) This is needed to minimise / avoid PLS in associated districts and in CESC areas also in coming crucial summer before Election 2024.
- (ii) 80 MVAR reactor has to be sacrificed to make the bay free to accommodate the trf bank. As in November 2023 one 125 MVAR already installed and will be in service even after removal of the 80 MVAR reactor, so voltage rise during winter off-peak will be well taken care of by the 125 MVAR reactor.
- (iii) Understanding the gravity of the matter, it was discussed in the meeting on Subhasgram ICT update status with WBSETCL, Power Grid, CESC, ERPC, ERLDC.
- (iv) The said 80 MVAR reactor is already out of service for oil leakage arrest work, which may not be put in service after the work at New Chanditala 400 kV s/stn on receipt of approval from OCC forum.
This step is taken on emergency, and will be extremely helpful to reduce congestion / load shedding during forthcoming summer, 2024.

WBSETCL may update. Members may discuss.

Deliberation in meeting

WBSETCL representative briefly outlined the necessity of installing three nos. of single-phase transformers to form a 315MVA transformer bank to combat congestion in ICTs at 400 kV New Chanditala S/S. He submitted that since the 80 MVAR reactor is already out of service due to oil leakage, the said transformer bank can be accommodated in its place. He suggested that any voltage rise during winter off-peak will be well taken care of by the 125 MVAR reactor.

OCC granted in-principle approval to this proposed scheme at New Chanditala S/S and also advised WBSETCL to put up the agenda in CMETS-ER meeting for further deliberation.

2.16 Long outage of reactor at Jeerat S/S imposing high VAR charges on all DISCOMs of West Bengal_WBSETCL

The 50 MVAR Reactor 2 in Jeerat (WBSETCL) 400 KV sub-stn owned and maintained by PG since beginning. This reactor is out of service from 23.04.23 due to high winding temperature. In absence of this reactor, voltage of Jeerat (WBSETCL) point was crossing 417 KV during

winter off peak hours and WBSETCL is paying even up to the extent of 1.25 Lakhs per week as MVAR charges for the Jeerat point only.

As for some technical issues developed, the said reactor was taken out of bus earlier and as Power Grid has failed to rectify the same before winter (in last 8 months), WBSETCL and state DISCOMs are paying the amount as MVAR charges for Jeerat point without having any fault at their part, so the Jeerat point VAR amount may please be relieved for WBSETCL, if Power grid fails to bring back the reactor in service shortly. Power Grid may please intimate the up to date status and the probable date of return of the reactor in service.

WBSETCL may update. Members may discuss.

Deliberation in meeting

WBSETCL representative submitted that in event of long outage of 50 MVAR reactor at Jeerat S/S , huge amount of weekly VAR charges are being imposed on WBSETCL and state DISCOMs. In this regard, he requested Powergrid ER-II to restore the 50 MVAR reactor to service at the earliest . In case of failure by Powergrid ER-II to bring the reactor to service at Jeerat S/S on time, WBSETCL appealed to OCC forum for waiving off VAR charges at the Jeerat(400 kV) point.

Powergrid ER-II representative vide mail dated 23.12.2023 apprised that the subject Reactor being aged more than 35 Years old, is subjected to many snags and based upon the information received from Jeerat site, normally time to time action is taken. After 23.04.23, the subject Reactor has been kept under Voltage Regulation causing no possibility to check the activity (Repairing done) . Further when the reactor was again tried to put into operation during present winter, different problems were observed for which necessary actions are already undertaken and expected to be finished by 15.01.2024.

OCC advised Powergrid ER-II to expedite actions for restoration of 50 MVAR reactor at 400 kV Jeerat(WBSETCL) S/S and also opined that the request of WBSETCL for waiving off of VAR charges may be taken up in the next CCM meeting for necessary deliberation.

3. PART-C: ITEMS FOR UPDATE/FOLLOW-UP

3.1. ER Grid performance during November 2023.

The average and maximum consumption of Eastern Region and Max/Min Demand (MW), Energy Export for the month October-2023 were as follows:

AVERAGE CONSUMPTION (MU)	MAXIMUM CONSUMPTION(MU)/ DATE	MAXIMUM DEMAND (MW)	MINIMUM DEMAND (MW)	SCHEDULE EXPORT	ACTUAL EXPORT
		DATE/TIME	DATE/TIME	(MU)	(MU)
432 MU	482 MU 02.11.2023	23216 MW, 02.11.2023 at 17:54 Hrs.	15135 MW, 27.11.2023 at 02:56 Hrs.	5026	5647

ERLDC/ERPC may highlight the performance of the ER grid.

Deliberation in meeting

The grid performance of ER for the month of November 2023 was highlighted.

3.2. Ensuring the healthiness of ADMS

State	Criteria for ADMS operation	Number of instances for which ADMS criteria satisfied	Number of instances for which detail received	Discussion regarding previous month performance	Update in 209 th OCC meeting
West Bengal	1. System Frequency < 49.7 Hz 2. WB over-drawl > 150 MW 3. Delay = 4 min	0	-	-	-
Jharkhand	1. System Frequency < 49.9 Hz 2. Jharkahnd over-drawl > 25 MW	117	Nil	-	-

	3. Delay = 3 min				
DVC	1. System Frequency < 49.9 Hz 2. DVC over-drawl > 150 MW 3. Delay = 3 min	10	Nil	-	-
Odisha	1. System Frequency < 49.9 Hz 2. Odisha over-drawl > 150 MW 3. Delay = 3 min	11	Nil	-	-

Members may note.

Deliberation in meeting

Members noted.

3.3. Commissioning status of ADMS

Automatic demand management scheme (ADMS) has been already commissioned in West Bengal, DVC, Odisha and Jharkhand. However for Bihar it is yet to be implemented, the last status as confirmed in the earlier meeting is as follows.

SI No	State/Utility	Logic for ADMS operation	Target Date
1	Bihar	F < 49.7 AND deviation > 12 % or 150 MW	

Bihar may update the status of the implementation of ADMS scheme.

Members may note.

Deliberation in meeting

Members noted.

3.4. Primary frequency response of generating units in ER

The availability of sufficient primary frequency response is one of the fundamental requirement of power system operation not only from reliability point of view but also from regulatory compliance point of view. Based on the assessed FRC re-testing of primary frequency response can be recommended. Therefore, the accurate and high-resolution data from generator end is extremely important in absence of which assessment of FRC is done as per low resolution ERLDC SCADA data. The plant wise data submission statistic for frequency event flagged by ERLDC upto 30.11.2023 is given below:

STATIONS	20.1 2. 202 2	12.0 1. 2023	14.01.2023				17.0 1. 2023	09.02.202 3		16.0 3. 2023	28.0 3. 2023	01.0 5. 2023	15.0 5. 2023	28.0 6. 2023	20.0 7. 2023	31.0 7. 2023	06.0 8. 2023
	06:4 8	05:5 2	12:0 6	13:0 3	14:5 5	15:1 8	09:5 6	11:4 5	12:2 9	09:1 6	10:3 7	13:2 3	11:5 1	02:2 8	13:5 3	10:3 6	09:48
ADHUNIK																	
BARH																	
BRBCL																	
DARLIPALLI																	
DIKCHU																	
FARAKKA																	
GMR																	
JITPL																	
KAHALGAO N																	
MPL																	
NPGC																	
TALCHER																	
TEESTA III																	
TEESTA V																	
North Karanpura	Not Applicable																

Deliberation in meeting

Members noted.

4. PART-D: OPERATIONAL PLANNING

4.1. Anticipated power supply position during January 2024

The abstract of peak demand (MW) vis-à-vis availability and energy requirement vis-à-vis availability (MU) for the month of January 2024 were prepared by ERPC Secretariat (**Annexure D.1**) on the basis of LGBR for 2023-24 and feedback of constituents, keeping in view that the units are available for generation and expected load growth etc.

Members may update.

Deliberation in meeting

*The updated anticipated power supply position for January 2024 is provided at **Annexure D.1***

4.2. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on 12-12-2023)

a) Thermal Generating Stations outage report:

S.No	Station	Fuel	State	Agency	Unit No	Capacity	Reason(s)	Outage
1	KHSTPP	COAL	BIHAR	NTPC	4	210	Annual Overhauling	01-Dec-2023
2	NABINAGAR (BRBCL)	COAL	BIHAR	NTPC	1	250	Annual Overhauling	01-Dec-2023
3	KBUNL	COAL	BIHAR	NTPC, BSPHCL	1	195	Due Flame failure, later unit taken under annual overhauling for 45 days	30-Nov-2023
4	FSTPP	COAL	WEST BENGAL	NTPC	3	200	Initially the unit was out due to RSD. Later, from 00:00 Hrs of 01-12-2023 the unit was taken under annual overhauling.	30-Nov-2023
5	BARH	COAL	BIHAR	NTPC	1	660	Initially the unit was out due to Boiler Tube Leakage. Later, from 00:00 Hrs of 24-11-2023 the unit was taken under annual overhauling.	19-Nov-2023
6	DSTPS	COAL	DVC	DVC	1	500	FGD related work	10-Dec-2023
7	BAKRESHWAR	COAL	WEST BENGAL	WBPDC	2	210	For Overhauling Purpose	08-Dec-2023
8	HALDIA ENERGY LTD	COAL	WEST BENGAL	HEL,CESC	1	300	For Overhauling & De-NOx modification activities	02-Dec-2023
9	SAGARDIGHI	COAL	WEST BENGAL	WBPDC	3	500	For Overhauling Purpose	23-Nov-2023
10	ADHUNIK	COAL	JHARKHAND	APNRL	1	270	Ash evacuation issue	11-Dec-2023
11	Sterlite	COAL	ODISHA	SEL	1	600	Due to Ash Handling Plant problem	11-Dec-2023
12	TENUGHAT	COAL	JHARKHAND	TVNL	2	210	BOILER TUBE LEAKAGE	10-Dec-2023

13	OPGC3	COAL	ODISHA	OPGC	3	660	Leakage in Ash slurry discharge pipe	10-Dec-2023
14	OPGC	COAL	ODISHA	OPGC	4	660	Leakage in Ash slurry discharge pipe	10-Dec-2023
15	DPL	COAL	WEST BENGAL	WBPCL	8	250	problem in the turbine gland sealing.	25-Nov-2023
16	BARAUNI TPS	COAL	BIHAR	NTPC	8	250	Boiler tube leakage later generator hydrogen leakage detected	25-Nov-2023
17	BARAUNI TPS	COAL	BIHAR	NTPC	6	110	Low vacuum	22-Jul-2023
18	BARAUNI TPS	COAL	BIHAR	NTPC	7	110	Poor condenser vacuum	19-Jul-2023

All Generating stations are requested to update expected restoration time and reason outage to ERLDC/ERPC on weekly basis in case of any change at their end.

b) Major Generating stations Out on Reserve Shutdown due to low system demand:

S.No	Station	Fuel	State	Agency	Unit No	Capacity	Reason(s)	Outage
1	BARH	COAL	BIHAR	NTPC	5	660	Reserve shutdown	01-Dec-2023
2	MEJIA TPS	COAL	DVC	DVC	1	210	Reserve shutdown	02-Dec-2023
3	SOUTHERN	COAL	WEST BENGAL	CESC	1	67.5	Reserve Shutdown	29-Nov-2023

c) Hydro Unit Outage Report:

S.No	Station	Fuel	State	Agency	Unit No	Capacity	Reason(s)	Outage
1	INDRAVATI	HYDRO	ODISHA	OHPC	2	150	Capital Maintenance	23-Nov-2023
2	RENGALI HPS	HYDRO	ODISHA	OHPC	1	50	Annual maintenance work	15-Nov-2023
3	BALIMELA HPS	HYDRO	ODISHA	OHPC	4	60	The unit taken out under R&M since 08/07/2022 for 18 months.	08-Jul-2022
4	BALIMELA HPS	HYDRO	ODISHA	OHPC	3	60	The unit taken out under R&M since 08/07/2022 for 18 months.	08-Jul-2022
5	TEESTA HPS	HYDRO	SIKKIM	NHPC	1	170	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta river and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
6	TEESTA HPS	HYDRO	SIKKIM	NHPC	2	170		
7	TEESTA HPS	HYDRO	SIKKIM	NHPC	3	170		
8	DIKCHU Hep	HYDRO	SIKKIM	SKPPL	1	48		
9	DIKCHU Hep	HYDRO	SIKKIM	SKPPL	2	48		
10	TEESTA STG III Hep	HYDRO	SIKKIM	TUL	1	200		

11	TEESTA STG III Hep	HYDRO	SIKKIM	TUL	2	200		
12	TEESTA STG III Hep	HYDRO	SIKKIM	TUL	3	200		
13	TEESTA STG III Hep	HYDRO	SIKKIM	TUL	4	200		
14	TEESTA STG III Hep	HYDRO	SIKKIM	TUL	5	200		
15	TEESTA STG III Hep	HYDRO	SIKKIM	TUL	6	200		
16	TLDP	HYDRO	WEST BENGAL	WBPDC	1	33		
17	TLDP	HYDRO	WEST BENGAL	WBPDC	2	33		
18	TLDP	HYDRO	WEST BENGAL	WBPDC	3	33		
19	TLDP	HYDRO	WEST BENGAL	WBPDC	4	33		
20	U.KOLAB	HYDRO	ODISHA	OHPC	2	80	Rotar earth Fault	25-Aug- 2023

d) Long outage report of transmission lines (As on 05.12.2023):

Transmission Element / ICT	Outage From	Reasons for Outage
400 KV IBEUL JHARSUGUDA D/C	29.04.2018	As information gathered, around 40-50 nos of towers were collapsed and conductor theft more than 400Ckm and restoration work is in progress. Interim arrangement/Power evacuation of IBEUL(2 x350MW) by LILOing between OPGC and ISTS Jharsuguda (one ckt) approved. Deadline for construction of 400KV IBEUL -Jh'da set as 30-03-2024.
220/132KV 100 MVA ICT II AT LALMATIA	22.01.2019	Commissioning work of 220/132KV, 100MVA Transformer and its associated control Panel under progress.
220 KV PANDIABILI - SAMANGARA D/C	03.05.2019	Tower Collapsed during Cyclone FANI (Restoration project is entrusted upon PGCIL & 220kV Samangara-Pandiabili ckt-I&II are anti-theft charged from Pandiabili end from loc no.01 to loc no.74)
220/132KV 100 MVA ICT 3 AT CHANDIL	30.04.2020	Due to Fire hazard ICT damaged and burnt.
400/220KV 315 MVA ICT 4 AT JEERAT	09.04.2021	Due to Fire hazard ICT damaged and burnt. It was decided that 315MVA, 400/220KV spare ICT from Regional Pool at Malda to be replaced in place of

		defective 315 MVA ICT (4th) at Jeerat 400 KV S/S(WB).Work under progress.
220KV-FSTPP-LALMATIA-I	21.04.2021	Conductor stringing 12.965 km has been completed and Stringing between Tower Loc. no. 152 to 159 is under progress. Transmission line is idle charged between Lalmatia GSS end to Tower Loc.no.169
220KV-MUZAFFARPUR(PG)-GORAUL(BH)-1	11.06.2022	Main Bay is under breakdown due to flashing in GIS module
220KV-WARIA-BIDHANNAGAR-1 & 2	08.06.2022	To control overloading of 220 kV Waria-DSTPS (Andal) D/C line
400/220KV 315 MVA ICT 2 AT PATRATU	27.09.2022	ICT tripped on few occasions due to Buchholz later DGA violation found, internal fault in transformer to be rectified. (DGA violation)
132KV-BARHI-RAJGIR-1	25.03.2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earthwire between tension tower no. 218-237 in same line.
132KV-NALANDA-BARHI(DVC)-1	25.03.2023	
220KV-TSTPP-MEERAMUNDALI-2	10.06.2023	Tower collapse at loc no 41, 42 (from Meramundali end). Ckt1 charged through ERS.
400KV-KHSTPP-BARH-1	04.08.2023	Upgradation of Bay equipments at KHSTPP
400KV/220KV 315 MVA ICT 3 AT BIDHANNAGAR	31.08.2023	For jumpering of 220 kV dropper from strung bus at 315MVA ICT-3. 220KV side Bay of said ICT is under construction at Bidhannagar.
400KV-RANGPO-TEESTA-V-1 & 2	04.10.2023	Tower near gantry of Teesta V powerhouse collapsed due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-TEESTA-III-RANGPO-1	04.10.2023	Hand tripped from Teesta-III end due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-TEESTA-III-DIKCHU-1	04.10.2023	

400KV-RANGPO-DIKCHU-1	04.10.2023	Hand tripped from Rangpo end due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV JHARSUGUDA-ROURKELA-1 &3	26.10.2023	Reconductoring work
400KV-BINAGURI-BONGAIGAON-2	27.10.2023	Reconductoring work
400KV/220KV 315 MVA ICT 2 AT MTPS-B	20.11.2023	BPI, LA, ISOLATOR Foundation civil work activities near ICT#02
400KV-KISHANGANJ-NEW PURNEA-1 & 2	17.11.2023	To facilitate the Re-routing Pile project at new pile loc no 340 & 366
400KV PATNA-BALIA-1 & 2	22.11.2023	Diversion work under consultancy services to RVNL
400KV NEW PPSP-ARAMBAGH-1	22.11.2023	Initially on voltage regulation and then rectification of Tie bay (Under Breakdown since 30.12.2022) at New PSSP
220KV DEHRI-GAYA-1	25.11.2023	Conductor re-sagging work.
220KV/132KV 100 MVA ICT 2 AT RANGPO	28.11.2023	SF6 gas leakage rectification
220KV/132KV 160 MVA ICT 2 AT MALDA	30.11.2023	For shifting of the ICT- 2 from 132KV AIS to GIS
400KV-ALIPURDUAR (PG)-JIGMELLING-1	02-12-2023	SCADA rectification work at Jigmelling

Transmission licensees/ Utilities are requested to update expected restoration date & work progress regarding restoration regularly to ERLDC/ERPC on monthly basis by 5th of each month so that status of restoration can be reviewed in OCC. Utilities are also requested to update outage of any elements within their substation premises like isolator/breaker to ERLDC/ERPC regularly. (Reported as per Clause 5.2(e) of IEGC)

Deliberation in meeting

Members noted. OCC advised all GENCOs to strictly abide by guidelines mandated in IEGC 2023 for availing unit shutdown.

4.3. Commissioning of new units and transmission elements in Eastern Grid in the month of November-2023.

The details of new units/transmission elements commissioned in the month of November - 2023 based on the inputs received from beneficiaries:

NEW ELEMENTS COMMISSIONED DURING NOVEMBER, 2023							
GENERATING UNITS							
SL. NO.	Location	OWNER/ UNIT NAME	Unit No/ Source	Capacity added (MW)	Total/ Installed Capacity (MW)	DATE	Remarks
1	Jharkhand	NORTH KARANPURA- UNIT 2	2	660	1320	11.09.2023	Unit#2 Synchronised on 09/11/2023 at 07:02 Hrs along with Generating transformer 2 and Bay No-413
ICTs/ GTs / STs							
SL. NO.	Agency/ Owner	SUB- STATION	ICT NO	Voltage Level (kV)	CAPACITY (MVA)	DATE	Remarks
1	NTPC	NORTH KARANPURA	GT-02	242.48/21	265	11.09.2023	
2	PGCIL	MALDA(PG)	ICT-1	220/132	160	20.11.2023	Code issued for 220/132 kV ICT - 1 at Malda (out since 04/01/2023 for 132KV GIS Commissioning work (GIB erection of ICT-I)) and 132 kV main bus 1 at Malda
TRANSMISSION LINES							
SL. NO.	Agency/ Owner	LINE NAME	Length (KM)	Conductor Type	DATE	Remarks	
1	BSPTCL	220KV MUZAFFARPUR(PG)- AMNOUR-1	65.4	ACSR Zebra	11.07.2023		
LILO/RE-ARRANGEMENT OF TRANSMISSION LINES							
SL. NO.	Agency/ Owner	Line Name/LILO at	Length (KM)	Conductor Type	DATE	Remarks	
NIL							
BUS/LINE REACTORS							

SL. NO.	Agency/ Owner	Element Name	SUB-STATION	Voltage Level (kV)	DATE	Remarks
1	WBSETCL	125MVAR 400KV B/R-1 AT NEW CHANDITALA	NEW CHANDITALA	400	11.09.2023	
HVDC /AC Filter bank / FACTS DEVICE associated System						
SL. NO.	Agency/ Owner	Element Name	SUB-STATION	Voltage Level (kV)	DATE	Remarks
NIL						
BAYS						
SL. NO.	Agency/ Owner	Element Name	SUB-STATION	Voltage Level (kV)	DATE	Remarks
1	PGCIL	132KV MAIN BUS - 1 AT MALDA(PG)	MALDA(PG)	132	20.11.2023	132 KV Main Bus-I through ICT-I Bay (101). After charging 132 KV Bus voltage shall be available through Bus VT (Bus-I)
2	PGCIL	132KV MAIN BUS - 2 AT MALDA(PG)	MALDA(PG)	132	20.11.2023	
3	JUSNL	220KV MAIN BAY OF 315 MVA ICT -1(Bay No-205) AT LATEHAR(JUSNL)	LATEHAR (JUSNL)	220	30.11.2023	
4	NTPC NORTH KARANPURA	MAIN BAY OF GT-2(Bay No-413) NORTH KARANPURA UNIT -2	NTPC NORTH KARANPURA	400	11.09.2023	
5	PGCIL	ICT-1 Bay (Bay No-101) at MALDA(PG)	MALDA(PG)	132	20.11.2023	
6	PGCIL	ICT-4 Bay (Bay No-106) at MALDA(PG)	MALDA(PG)	132	26.11.2023	

Members may note.

Deliberation in meeting

Members noted.

4.4. UFR operation during the month of November 2023.

Frequency profile for the month as follows:

MONTH	MAX	MIN	% LESS IEGC BAND	% WITHIN IEGC BAND	% MORE IEGC BAND
	(DATE/TIME)	(DATE/TIME)			
Nov, 2023	50.39 Hz on 27-11-2023 at 00:02 hrs	49.55 Hz on 25-11-2023 at 14:17 hrs	6.8	74.4	18.8

Hence, no report of operation of UFR has been received from any of the constituents.

Members may note.

Deliberation in meeting

Members noted.

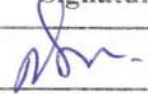
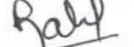
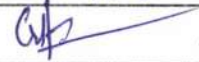
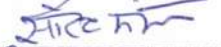
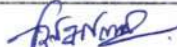
Annexure A

Participants in 210th OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 15.12.2023 (Friday)

Sl. No.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
1	N.S.Mondal	Member Secretary	ERPC	9958389967	mserpc-power@gov.in	
2	R. Sutradhar	Executive Director	ERLDC	9436302714	rajibsutradhar@posoco.in	
3	S. Konar	GM, ERLDC		9436335370	konar_s@posoco.in	
4	M. DAS	DGM, ERLDC	GRID-INDIA ERLDC	9007070925	manasdas@grid-india.in	
5	D.P. PUTANDI	Sr GM, SLDC	DVC	9434745905	deliprasad.puitandi@dvc.gov.in	
6	Rahul Arand	DGM, NTPC	NTPC	9425823430	rahulanand@ntpc.co.in	
7	T.P.Sharma	CE, THP, DGPC	DGPC	+97517663422	t.sharma777@dnsgreen.lt	
8	P.V. RAUT	Head OPN	MPL	9223501513	rautpv@tatapower.com	
9	Sauresh Jain	Asst. Manager	JSW	9988976377	Sauresh.jain@jsw.in	
10	Sunil Dave	Asst. Genl. Mgr	JSW	8875029103	sunil.dave@JSW.in	
11	Agniva Chatterjee	AEE	ERPC	8100307502	agniva.cca@gov.in	
12	Sourav Mandal	Manager (MO)	ERLDC	9402102354	souravmandal@grid-india.in	
13	CHANDAN MALLICK	manager (SO)	ERLDC	9007059660	chandam.mallik@grid-india.in	
14	Rabisanjan Jeta	Sr. Manager	MPL	9204855211	JETIR@TATAPOWER.COM	
15	Soumya Mukherjee	Asst. Manager	HEL	9051486663	Soumya.mukherjee@rpsc.in	
16	D.K. Khuntia	Asst. Manager	ERPC	7683889161	delip.khuntia.cca@gov.in	
17	Oma Nath Kwikel	Head Operations	CHP, Bhubaneswar	97517688335	O.Kwikel623@dnsgreen.lt	
18	Yeshi Dorji	Asst. Engineer	MHP, Bhubaneswar	+97517400050	y.dorji2687@dnsgreen.lt	
19	Debi Kanyo	Asst. Engineer	DCET, DGPL	+97517634920	d.kanyo144@d.green.lt	
20	PREETOSH GHOSH	Dy. Manager	SLDC, DVC	9674299618	preetosh.ghosh@dvc.gov.in	

Participants in 210th OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 15.12.2023 (Friday)

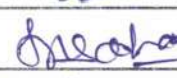
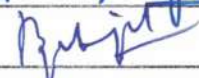
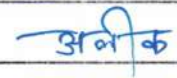
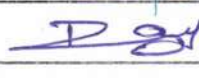
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21	Suryaprakash Mohapatra	Managen	GMR Odisha Kamalanga Energy Ltd	8249217715	Suryaprakash.mohapatra@gmrgroup.in	Surya
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40	D Mukherjee	Sr. Mgr. (OS)	WBPDCL	9830052830	d.mukherjee@wbpdcl.co.in	D Mukherjee

Participants in 210th OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 15.12.2023 (Friday)

Sl. No.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
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Annexure B.1



GUIDELINES FOR RESOURCE ADEQUACY PLANNING FRAMEWORK FOR INDIA

CONTENTS

- 1. What is Resource Adequacy (RA)?**
- 2. Why RA is required?**
- 3. Important Terminologies**
 - RELIABILITY
 - LOLP(LOSS OF LOAD PROBABILITY)
 - NENS(NORMALIZED ENERGY NOT SERVED)
 - PRM(PLANNING RESERVE MARGIN)
 - CAPACITY CREDIT
 - MARGINAL COST OF REDUCING LOAD SHED
 - RESOURCE ADEQUECY REQUIREMENTS
- 4. Institutional mechanism for Resource Adequacy and Compliance Monitoring**
- 5. Resource Adequacy Implementation Timeline**

WHAT IS RESOURCE ADEQUACY(RA)?

Resource Adequacy is generally defined as a mechanism to ensure that there is an adequate supply of generation to serve expected peak demand reliably.

Resource Adequacy=Availability+ Accessibility + Affordability

In exercise of the powers conferred under the Rule 16 of Electricity (Amendment) Rules, 2022, the Ministry of Power, Government of India, in consultation with Central Electricity Authority (CEA) hereby issues the guidelines for Resource Adequacy for the Indian electricity sector.

WHY RA IS NEEDED ?



IMPORTANT TERMILOGIES

- **Reliability:**

The resource adequacy planning should be such that outages (loss of load events) are minimized.

IMPORTANT TERMILOGIES

- **Loss of Load Probability (LOLP):**

Measure of the probability that a system's load may exceed the generation & firm power contracts available to meet that load in a year.

IMPORTANT TERMILOGIES

- **Expected Energy Not Served (EENS):**
 - It is a Energy centric metric that considers both the duration & magnitude of energy not served in a particular period.
 - It is a summation of the expected number of MWh of demand that may not be served for the year.
 - The metric can be normalized (i.e. divided by system load)to create a **Normalized energy not served (NENS)**.
- The LOLP and NENS values adopted by CEA for the purposes of the National Electricity Plan (NEP) are **0.2%** and **0.05%**, respectively.

IMPORTANT TERMILOGIES

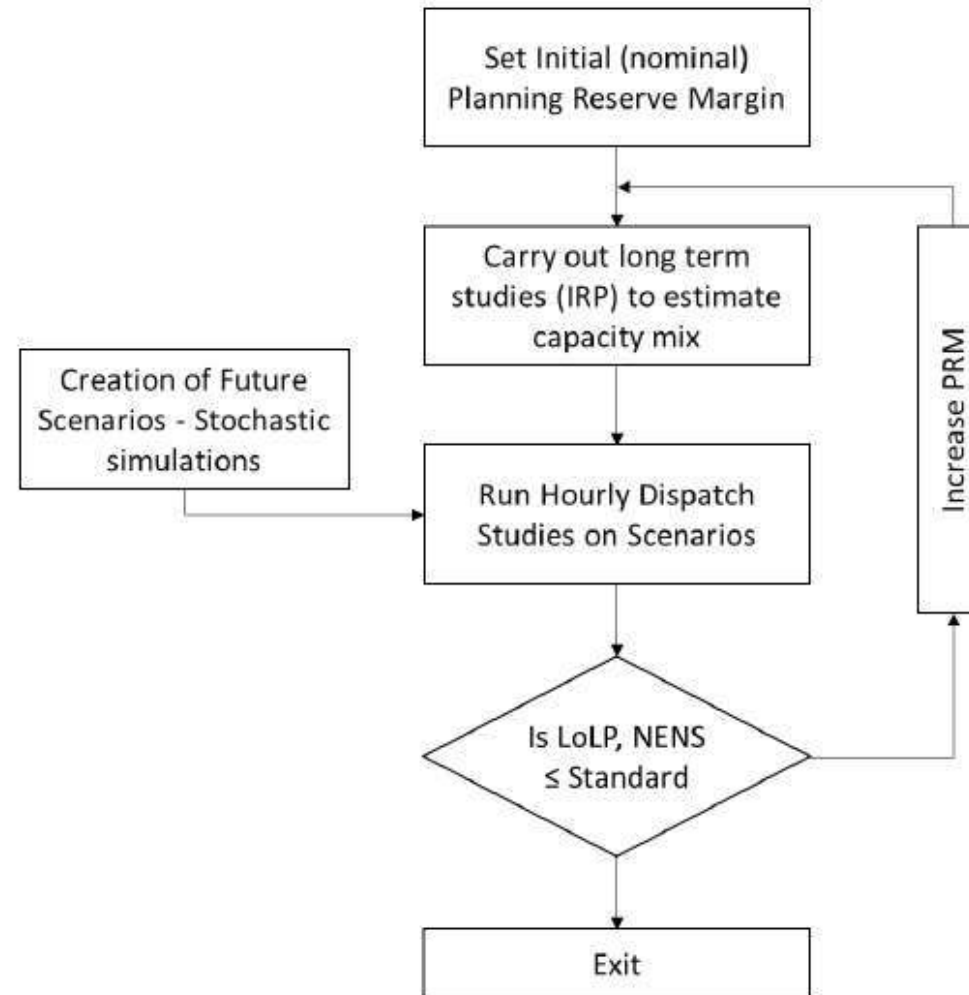
- **Planning Reserve Margin (PRM):**

- It is an indicator of adequacy of generation resources in the system to meet the load demand.
- The most important design parameter for long term planning model.
- PRM in a power system is expressed as a certain % of peak load forecast of the system

$$PRM = \frac{\text{Firm Capacity} - \text{Peak Demand}}{\text{Peak Demand}}$$

DETERMINATION OF LOLP/NENS

Figure 1: Flowchart of the Optimal Reserve Margin Study



IMPORTANT TERMILOGIES

- **Marginal Cost of Reducing Load Shed:**

- The marginal cost of reducing load shed is the effective increase in cost for every unit of load shed reduced.

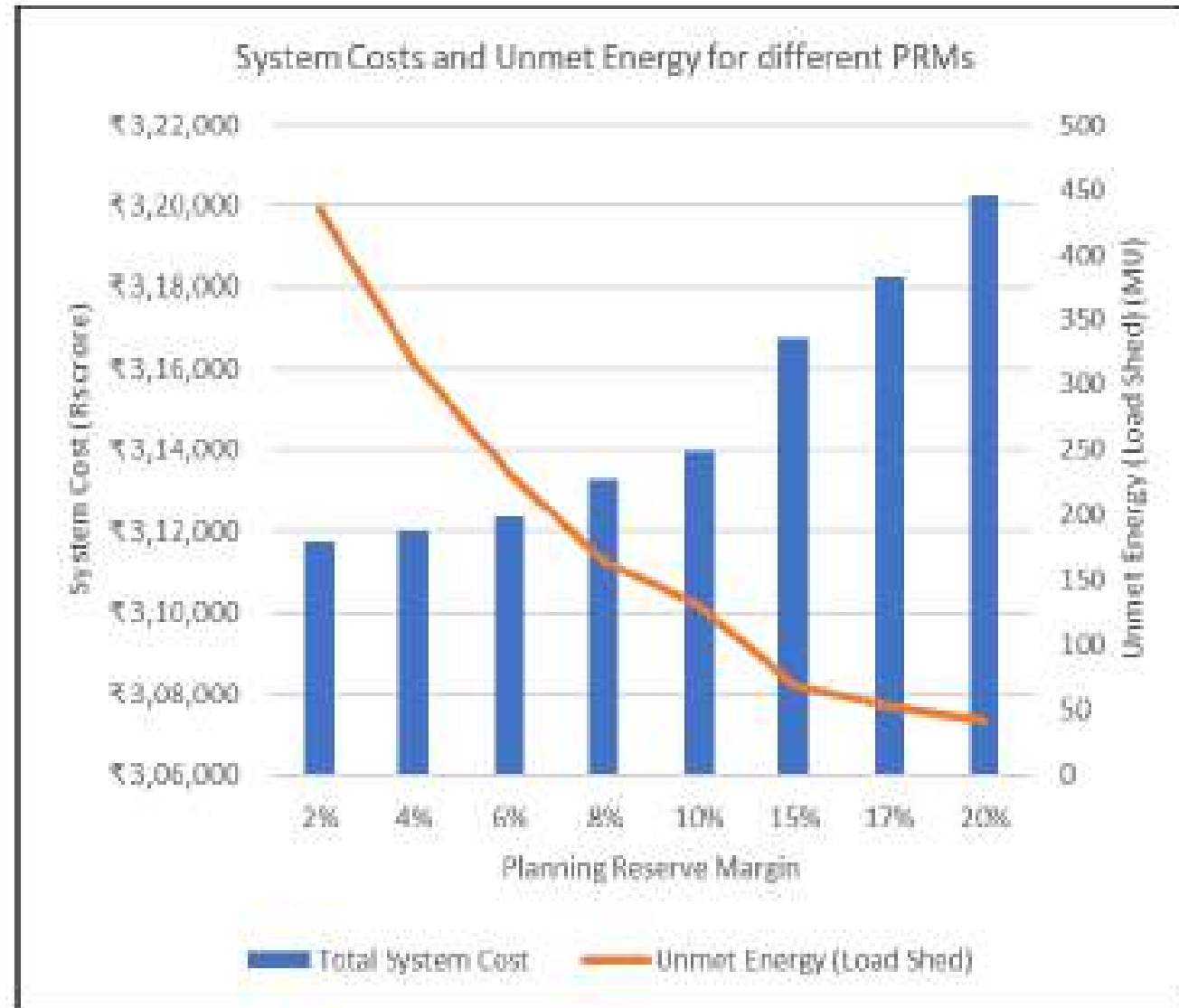
$$\text{Marginal Cost} = \frac{\text{System Cost}_{PRM_{i+1}} - \text{System Cost}_{PRM_i}}{ENS_{PRM_i} - ENS_{PRM_{i+1}}}$$

- The economic optimal planning reserve margin is the planning reserve margin at which:

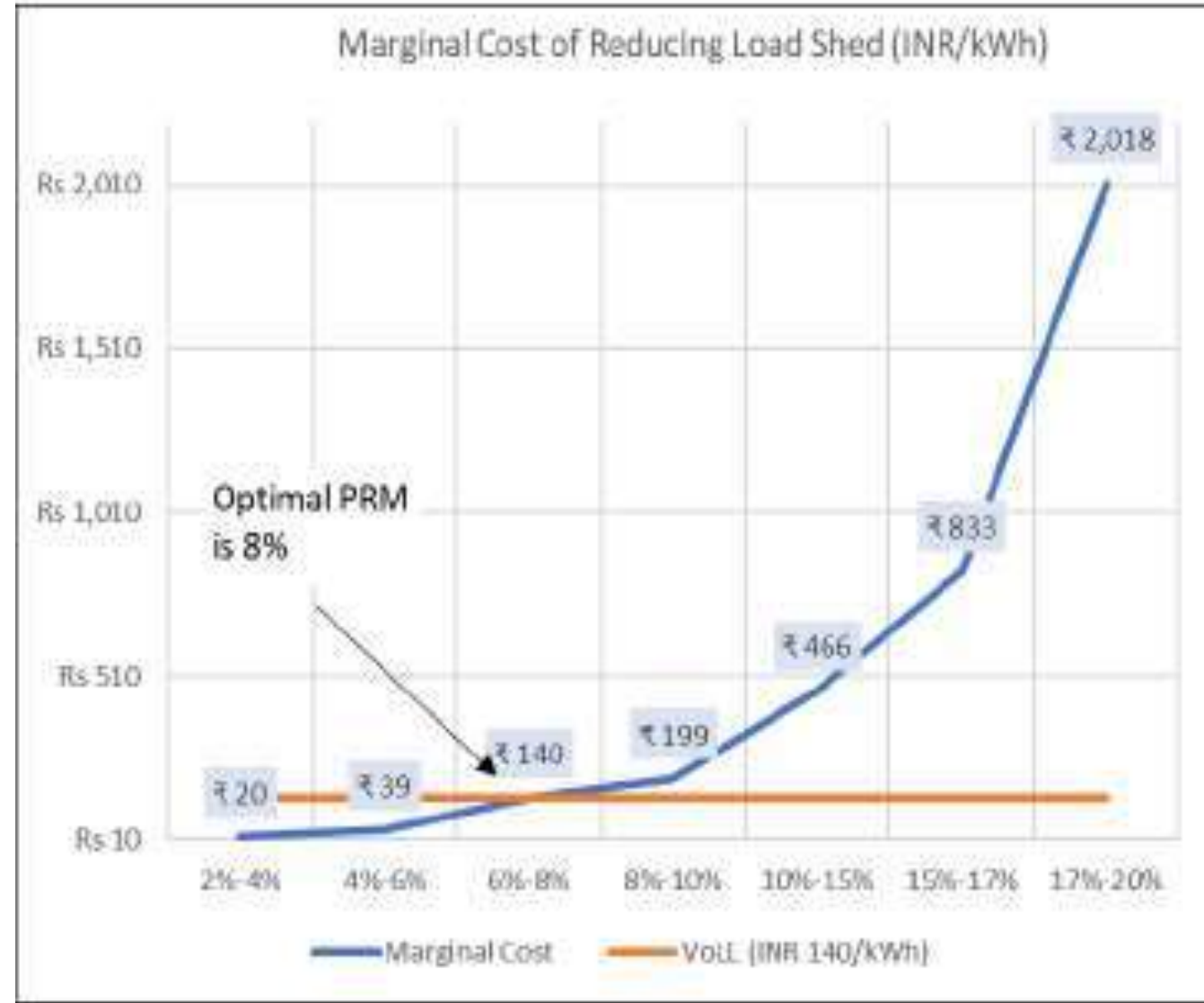
Marginal cost of reducing load shed = The value of lost load

- **Value of Lost Load (VoLL)** represents the loss that is incurred by customers in case of electricity service interruption.
- **Value of Lost Load (VoLL)** is the estimated amount that customers receiving electricity with firm contracts would be willing to pay to avoid a disruption in their electricity service.

IMPORTANT TERMILOGIES



IMPORTANT TERMILOGIES



IMPORTANT TERMILOGIES

An illustration of the calculation of marginal costs of reducing load shed is shown in Figure 2.

- The capacity expansion planning model is run for different PRMs between 2% and 20%. Based on the capacities obtained, the system costs are calculated.
- Demand-supply matching using these capacities on future scenarios created using stochastic simulations are used to obtain the total load shed (unmet energy).
- Based on the system costs and unmet energy (graph on the left), the marginal cost of reducing load shed (graph on the right) is calculated using the formula mentioned above.

Assuming a Value of Lost Load (VoLL) of INR 140/kWh, the optimal PRM would be around 8%.

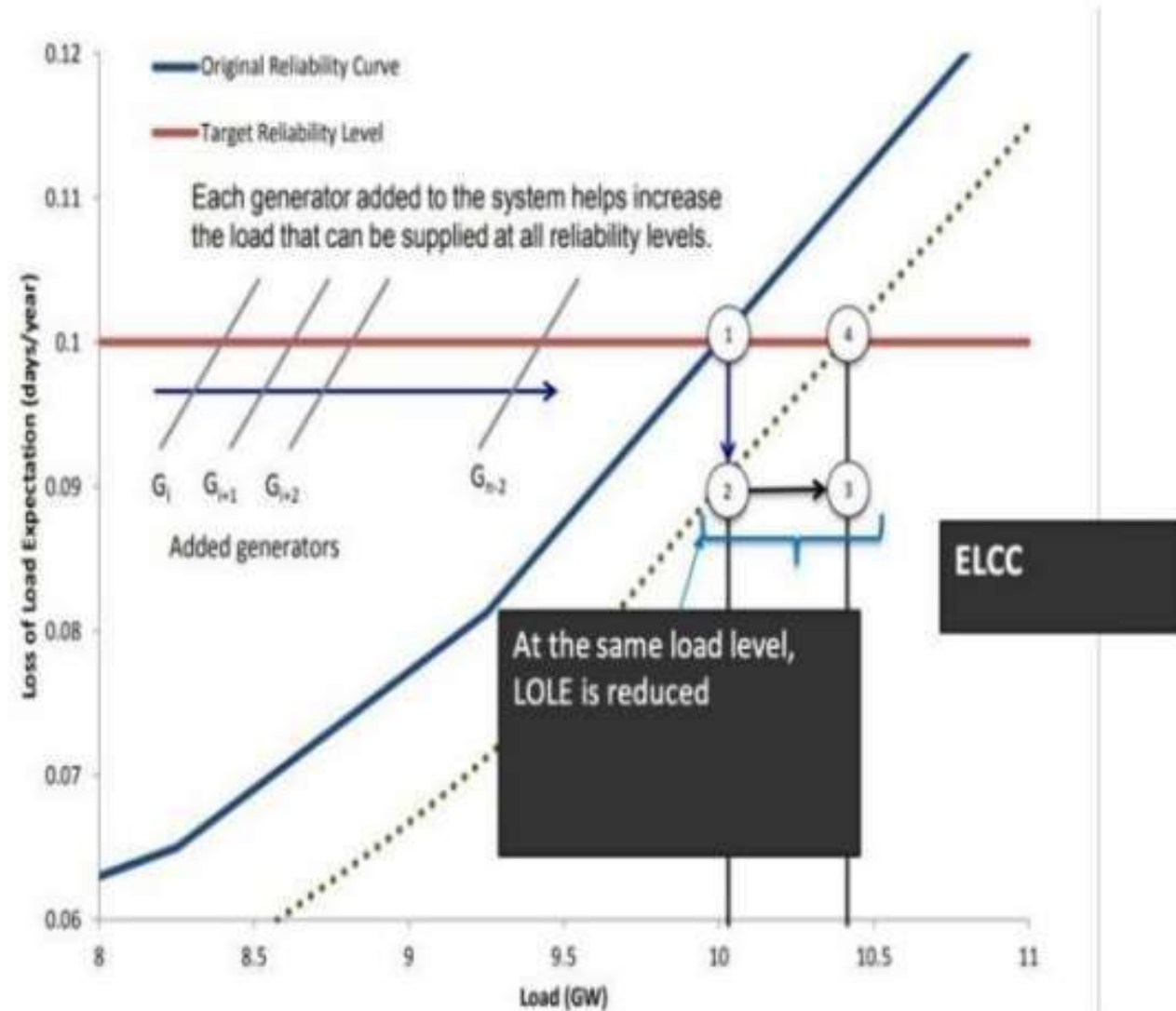
IMPORTANT TERMILOGIES

- **Capacity credit (CC)**

- Capacity credit (CC), sometimes referred to as capacity value, is a metric used to indicate an electric generator's ability to meet peak demand in a Power system.
- It is also the measure of amount of conventional power that can be replaced by RE without causing any disturbance to system reliability w.r.t. to RE sources.
- With more accurate CC calculations, better lowest cost options to maintain reliable service can be found.
- The capacity credits for generating resources and demand response resources to meet the national peak shall be estimated by CEA.

Capacity credit (CC) calculation

IMPORTANT TERMILOGIES



IMPORTANT TERMILOGIES

- **Resource Adequacy Requirement (RAR):**
 - The resource adequacy requirement for each distribution licensee is computed as:

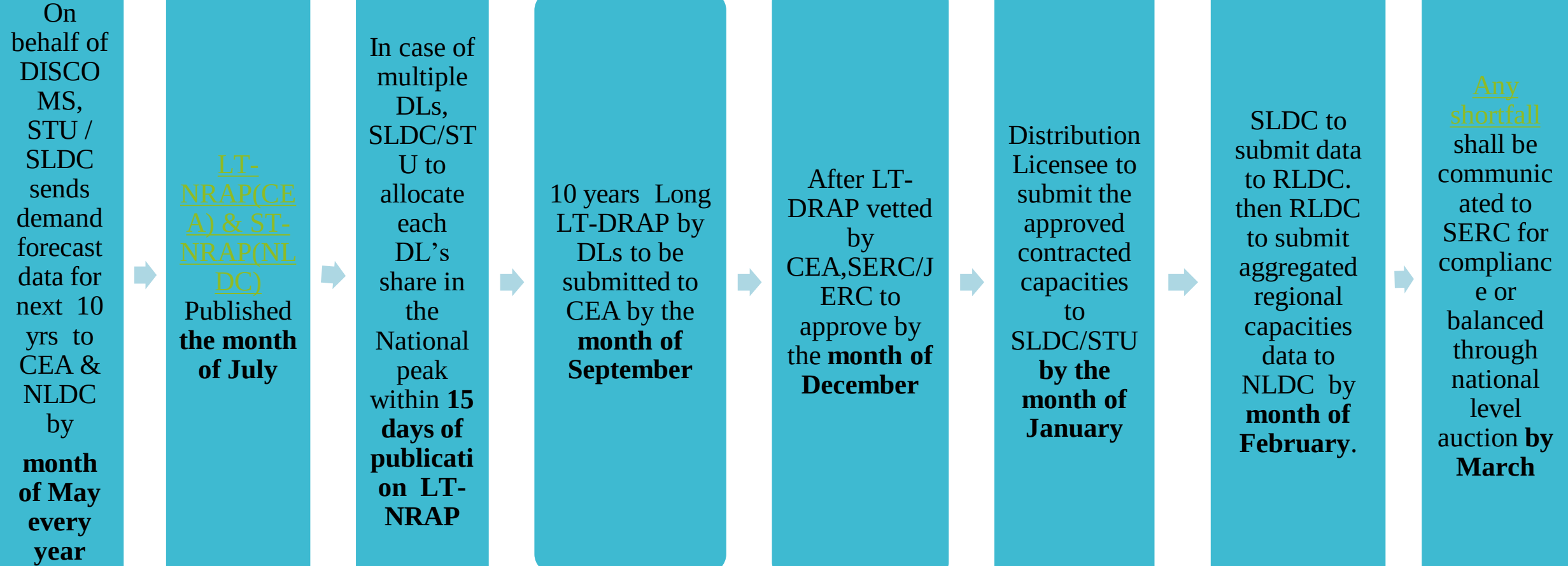
$$\text{RAR} = \text{Contribution to forecasted national peak in GW} * (1 + \text{PRM})$$

- From the supply side, the RAR is the sum of the “firm capacity” or “capacity credits” of contracted / planned capacities (including renewables, storage, other resources such as demand response) along with derated interconnection limits (imports).

IMPORTANT TERMILOGIES

$$\begin{aligned} RAR = & \sum_{i=1}^{num_solar} Solar_Capacity * Solar_Capacity_Credit \\ & + \sum_{i=1}^{num_wind} Wind_Capacity * Wind_Capacity_Credit \\ & + \sum_{i=1}^{num_hydro} Hydro_Capacity * Hydro_Capacity_Credit \\ & + \sum_{i=1}^{num_thermal} Thermal_Capacity * Thermal_Capacity_Credit \\ & + \sum_{i=1}^{num_nuclear} Nuclear_Capacity * Nuclear_Capacity_Credit \\ & + \sum_{i=1}^{num_storage} Storage_Capacity * Storage_Capacity_Credit \\ & + \sum_{i=1}^{num_other} OtherResource_Capacity * OtherResource_Capacity_Credit \\ & + \sum_{i=1}^{num_other} Import_limit * capacity_credit \end{aligned}$$

4. Institutional mechanism for Resource Adequacy and Compliance Monitoring



Institutional mechanism for Resource Adequacy and Compliance Monitoring

- The STUs/SLDC shall prepare one-year look ahead ST-DRAP (Short term Distribution Resource Adequacy Plan), on an annual basis for operational planning, at the state level based on the LT-DRAP study results.
- The SLDC shall review the ST-DRAP on a daily, monthly and quarterly basis based on actual availability of generation resources.

Resource Adequacy Implementation Timeline

Entity	Description	May'xx	Jun'xx	Jul'xx	Aug'xx	Sep 'xx	Oct'xx	Nov'xx	Dec'xx	Jan'(xx+1)	Feb'(xx+1)	Mar'(xx+1)
STU/SLDC	STU/SLDC, on behalf of distribution licensees shall provide to CEA and NLDC the details regarding demand forecasts for the next 5 years, assessment of existing generation resources and other details required for LT-NRAP and ST-NRAP											
CEA	To publish LT-NRAP containing National PRM, Reliability Metrics, Coincident peak, capacity credits and Optimal Generation mix for 10 years horizon.											
NLDC	To publish ST-NRAP.											
Discoms	LT-DRAP exercise for long term horizon(10 years) which is RA compliant as per coincident peak to be submitted to CEA											
CEA	Vetting of discom's contracting plan for coincident peak contribution and to meet their own energy and peak											
SERC	SERC to approve of discom's contracting plan for coincident peak contribution and to meet their own energy and peak											
Discoms	To contract capacities as per approved plans.											
	Submit contract capacities to STU/SLDC											
STU/SLDC	STU/SLDC to submit state-level aggregated capacities to RLDC											
RLDC	RLDC submit regional-level aggregated capacities to national level											
POSOCO/NLDC	POSOCO/NLDC to check RA compliance at national level											
	Any Shortfall shall be communicated to the SERC for compliance or is balanced through a national level auction mechanism											

Delivery Period(Apr'(xx+1) - Mar'(xx+2))

LT-NRAP	ST-NRAP
Prepared by CEA	Prepared by NLDC
State/UT's contribution towards national peak. Capacity credits for different resource types on a regional basis. National-level PRM as a guidance for all the States/UTs	Shall include parameters such as demand forecasts, resource availability based on under-construction status of new projects, planned maintenance schedules of existing stations, station-wise historic forced outage rates and decommissioning plans.

**OPTIMAL
CONTRACT
CAPACITY**

Contract	% RAR
LONG TERM	75-80
MEDIUM TERM	10-20
SHORT TERM	5-10

Resource	Long Term	Medium Term
Coal/Lignite Based Capacity	7	2
Hydro	9	2
Solar	2	1
Wind	3	1
PSP	5	3
Other Storage	2	1
Nuclear	9	3

BALANCE CAPACITY

capacity shortfall =

$(1 + \text{National PRM}) \times \text{Licensee Demand during National Peak} - \text{sum of contracted capacities by the licensee}$

balance capacity = $(1 + \text{National PRM}) \times \text{National Peak} - \text{sum of contracted capacities}$

- NLDC shall come out with a methodology to carry out national level auction for the procurement of the balance capacity.

THANK YOU

Annexure B.2

ANNEXURE

Details of Under Construction Central Sector Thermal Capacity Addition Target (Financial Year Wise)						
S.No	Project Name/ Impl. Agency	Sector	State	Unit No.	Capacity (MW)	Anticipated Trial Run Date
F Y. 2023-24						
1	Telangana STPP, St-I (NTPC)	CENTRAL	Telangana	U-2	800	Nov-23
2	North Karanpura STPP (NTPC)	CENTRAL	Jharkhand	U-2	660	
3	Ghatampur TPP (NUPPL)	CENTRAL	Uttar Pradesh	U-1	660	
4	Ghatampur TPP (NUPPL)	CENTRAL	Uttar Pradesh	U-2	660	Feb-24
5	Khurja SCTPP (THDC)	CENTRAL	Uttar Pradesh	U-1	660	
✓6	Buxar TPP (SJVN)	CENTRAL	Bihar	U-1	660	Mar-24
Sub-Total (FY 2023-24)					4100	
F Y. 2024-25						
1	Ghatampur TPP (NUPPL)	CENTRAL	Uttar Pradesh	U-3	660	May-24
✓2	North Karanpura STPP(NTPC)	CENTRAL	Jharkhand	U-3	660	Jun-24
✓3	Buxar TPP (SJVN)	CENTRAL	Bihar	U-2	660	Jul-24
✓4	Patratu STPP (PVUNL)	CENTRAL	Jharkhand	U-1	800	
✓5	Barh STPP St-I (NTPC)	CENTRAL	Bihar	U-3	660	Aug-24
6	Khurja SCTPP (THDC)	CENTRAL	Uttar Pradesh	U-2	660	
✓7	Patratu STPP (PVUNL)	CENTRAL	Jharkhand	U-2	800	Dec-24
Sub-Total (FY 2024-25)					4900	
F Y. 2025-26						
✓1	Patratu STPP (PVUNL)	CENTRAL	Jharkhand	U-3	800	May-25
Sub-Total (FY 2025-26)					800	
F Y. 2026-27						
✓1	Talcher TPP St-III (NTPC)	CENTRAL	Odisha	U-1	660	Nov-26
Sub-Total (FY 2026-27)					660	
F Y. 2027-28						
✓1	Talcher TPP St-III (NTPC)	CENTRAL	Odisha	U-2	660	May-27
2	Lara STPP St-II (NTPC)	CENTRAL	Chhattisgarh	U-1	800	Dec-27
Sub-Total (FY 2027-28)					1460	
F Y. 2028-29						
1	Lara STPP St-II (NTPC)	CENTRAL	Chhattisgarh	U-2	800	Jun-28
Sub-Total (FY 2028-29)					800	
Grand Total						12720

Details of Central Sector Candidate Thermal Power Plants expected to be commissioned by 2030

Sl.	Name Of Project	Sector	Developer	Capacity in MW	State
1	Singrauli STPP-III	Central	NTPC	2x800=1600	Uttar Pradesh
✓2	Darlipalli-II			1x800=800	Odisha
3	Sipat-III			1x800=800	Chhattisgarh
4	TPS-II 2nd Expansion	Central	NLCIL	2x660=1320	Tamil Nadu
5	NLC Talabira STPS			3x800=2400	Odisha
✓6	Raghunathpur TPS, PH-II	Central	DVC	2x660=1320	West Bengal
✓7	Durgapaur TPS			1x800=800	West Bengal
✓8	Koderma TPS			2x800=1600	Jharkhand
✓9	Chandrapura Extn TPS			1x800=800	Jharkhand
10	Meja-II	Central	U.P. - NTPC-JV	3x800=2400	Uttar Pradesh
11	Obra Extn			2x800=1600	Uttar Pradesh
✓12	Buxar TPP-II	Central	SJVN	1x660=660	Bihar
TOTAL				31010	

Details of Central Sector Additional Identified Candidate Thermal Power Plants

Sl.	Name Of Project	Sector	Developer	Capacity in MW	State
1	Telangana Stage II	Central	NTPC	2400	Telangana
✓2	New Nabi Nagar	Central	NTPC	2400	Bihar
✓3	Patratu Stage II	Central	NTPC	1600	Jharkhand
4	Gadarwara Stage II	Central	NTPC	1600	Madhya Pradesh
5	Anpara E	Central	NTPC-UP-JV	1600	Uttar Pradesh
6	NLC Talabira STPS	Central	NLCIL	800	Tamil Nadu
✓7	Mahanadi Basin Power	Central	MCL	1600	Odisha
TOTAL				12000	

Total Central Sector Thermal Capacity Addition Target

55730 MW

Details of Central Sector Under Construction Hydro Capacity Addition Target (Financial Year Wise)						
S.No	Project Name/ Impl. Agency	Sector	State	Unit No.	Capacity (MW)	Anticipated Trial Run Date
F Y. 2023-24						
1	SUBANSIRI LOWER /NHPC	Central	Arunachal Pradesh	U-1 to 2	500	June'2024
2	Naitwar Mori /SJVN	Central	Uttarakhand	U-1 to 2	60	Oct'2023
Sub-Total (FY 2023-24)					560	
F Y. 2024-25						
✓ 1	Rangit-IV/NHPC	Central	Sikkim	U-1 to 3	120	Aug'2024
2	SUBANSIRI LOWER /NHPC	Central	Arunachal Pradesh	U-3 to 6	1000	Mar'2025
3	Parbati-II /NHPC	Central	Himachal Pradesh	U-1 to 4	800	Jun'2024
4	Tehri PSS/THDC	Central	Uttarakhand	U-1 to 4	1000	Sept'2024
Sub-Total (FY 2024-25)					2920	
F Y. 2025-26						
1	Kiru /CVPPL	Central	Jammu & Kashmir	U-1 to 4	624	Mar'2026
2	SUBANSIRI LOWER /NHPC	Central	Arunachal Pradesh	U-7 to 8	500	May'2025
3	Luhri-I /SJVN	Central	Himachal Pradesh	U-1 to 4	210	Jan'2026
4	Dhauasidh /SJVN	Central	Himachal Pradesh	U-1 to 2	66	Nov'2025
5	Tapovan Vishnugad /NTPC	Central	Uttarakhand	U-1 to 4	520	Dec'2025
6	Polavaram / Polavaram Project Authority	State	Andhra Pradesh	U-5 to 12	640	Mar'2026
7	Shahpurkandi /PSPCL	State	Punjab	U-1 to 7	206	Oct'2025
8	Lower Sileru Extension /APGENCO)	State	Andhra Pradesh	U-1 to 2	230	Mar'2026
9	Kutehr/JSW	Private	Himachal Pradesh	U-1 to 3	240	Nov'2025
Sub-Total (FY 2025-26)					3236	
F Y. 2026-27						
1	Kwar/CVPPL	Central	Jammu & Kashmir	U-1 to U-4	540	Nov'2026
2	Ratle / NHPC	Central	Jammu & Kashmir	U-1 to U-5	850	May'2026
✓ 3	Rammam - III / NTPC Ltd.	Central	West Bengal	U-1 to U-3	120	Sept'2026
✓ 4	Teesta- VI /NHPC	Central	Sikkim	U-1 to U-4	500	Aug'2026
5	Pakal Dul /CVPPL	Central	Jammu & Kashmir	U-1 to U-4	1000	Sept'2026
6	Vishnugad Pipalkoti /THDC	Central	Uttarakhand	U-1 to U-4	444	Jun'2026
Sub-Total (FY 2026-27)					3454	
F Y. 2027-28						
1	Sunni Dam /SJVN	Central	Himachal Pradesh	U-1 to U-6	382	Mar'2028
Sub-Total (FY 2027-28)					382	
Grand Total					10552	

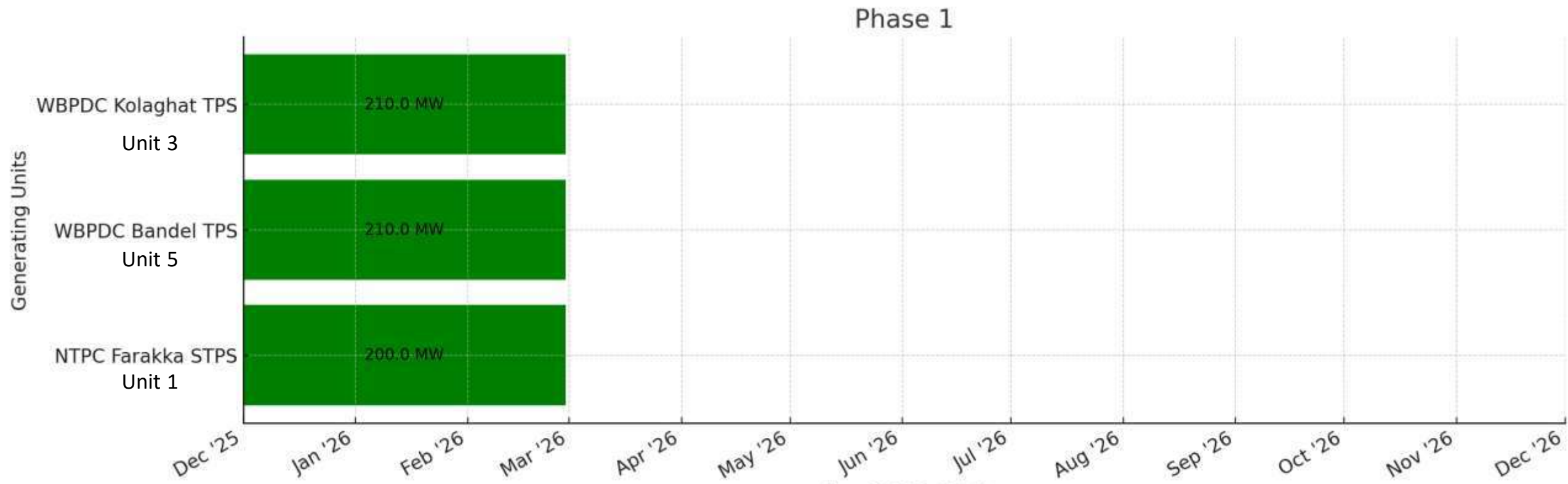
Details of Central Sector Under Planning Hydro Power Plants expected to be commissioned by 2030

Sl.	Name Of Project	Sector	Developer	Capacity in MW	State
1	Uri-I Stage-II	Central	NHPC	240	J & K
2	Nafra	Central	NEEPCO	120	Arunachal Pradesh
3	Tato-I	Central	NEEPCO	186	Arunachal Pradesh
4	Heo	Central	NEEPCO	240	Arunachal Pradesh
5	Wah Umiam Stage-III (Mawphu stage-II)	Central	NEEPCO	85	Meghalaya
6	Kirthai-II	Central	CVPPL	930	J & K
7	Dugar	Central	NHPC	500	Himachal Pradesh
8	Dulhasti Stage-II	Central	NHPC	260	J & K
Total				2561	

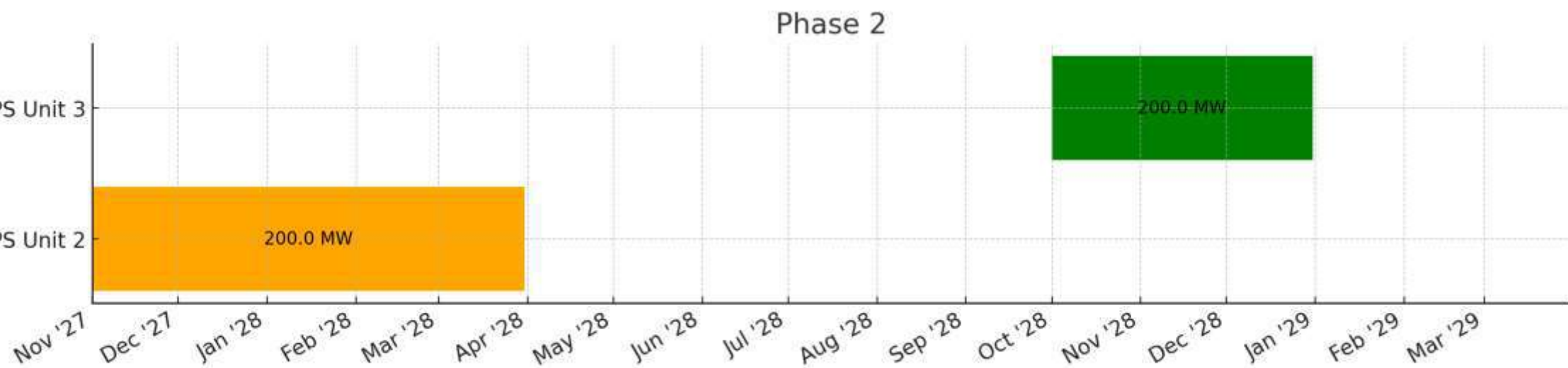
Total Central Sector Hydro Capacity Addition Target	13113 MW
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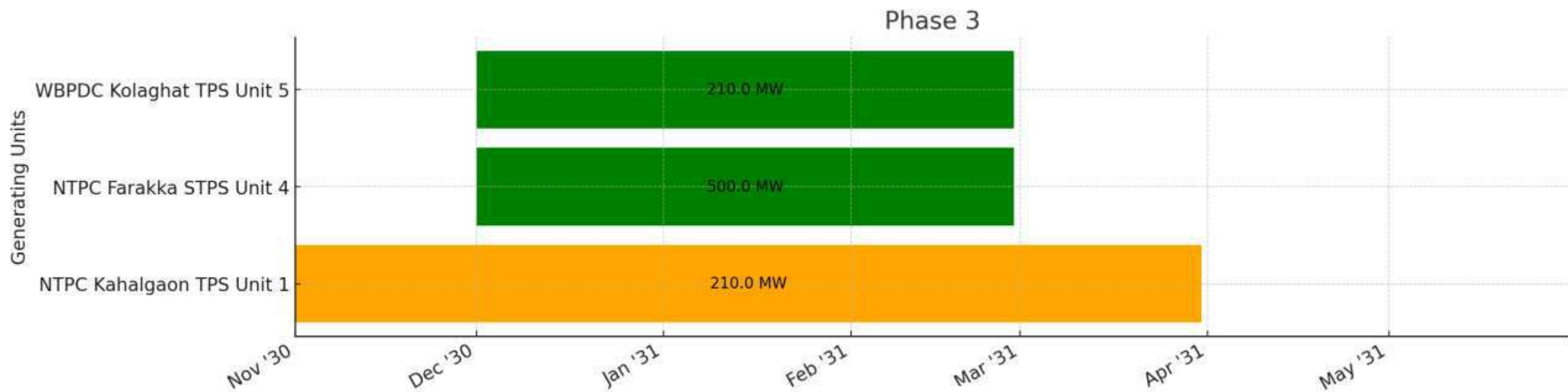
Total Capacity (thermal + Hydro) likely to be added by 2029-2030 under Central Sector	68843 MW
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Annexure B.3.1



Generating Units





Phase 4

Generating Units



Annexure B.3.2

R&M/ LE Status of the Units in reference with the phasing plans detailed below:

SRL NO.	STATION/UNIT NAME	CAPACITY	COD	Remarks sent to CEA via mail dated 03.07.23	Proposed S/D Plan from CEA (vide Letter Dated 03.11.2023)	Remarks from WBPDC (Dated 19.12.2023)
1	BANDEL TPS Unit#2	60	18.10.1965			Will be de-commissioned in Dec'25
2	BANDEL TPS Unit#5	215	08.10.1982	BTPS R&M done in 2015, Capacity enhanced from 210 to 215 MW.	S/D scheduled from Dec'25 to Feb'26 (phase-I)	EERM already done in 2015.
3	KOLAGHAT TPS Unit#3	210	12.10.1984	Total ESP replacement done in 2019.	S/D scheduled from Dec'25 to Feb'26 (phase-I)	Total ESP replacement done in 2019.
4	KOLAGHAT TPS Unit#5	210	14.05.1991	ESP R&M – Consultant engaged for DPR.	S/D scheduled from Dec'30 to Feb'31 (phase-III)	ESP & EHG to be done in FY 2024-25. Other small R&M being done under CAPEX.
5	KOLAGHAT TPS Unit#6	210	01.01.1994	ESP R&M – Consultant engaged for DPR	S/D scheduled from Sept'32 to Apr'33 – 08 Months (phase-IV)	ESP & EHG to be done in FY 2024-25. Other small R&M being done under CAPEX.
6	KOLAGHAT TPS Unit#4	210	01.04.1995	ESP R&M – Consultant engaged for DPR.	S/D scheduled from Nov'32 to Mar'33 – 05 Months (phase-IV)	ESP R&M to be done within Mar'26.

	Phase IA : Units with Age >=35 years as on 31.12.2022
	Phase IB : Units with 30<=Age <35 years as on 31.12.2022

Annexure B.5.1

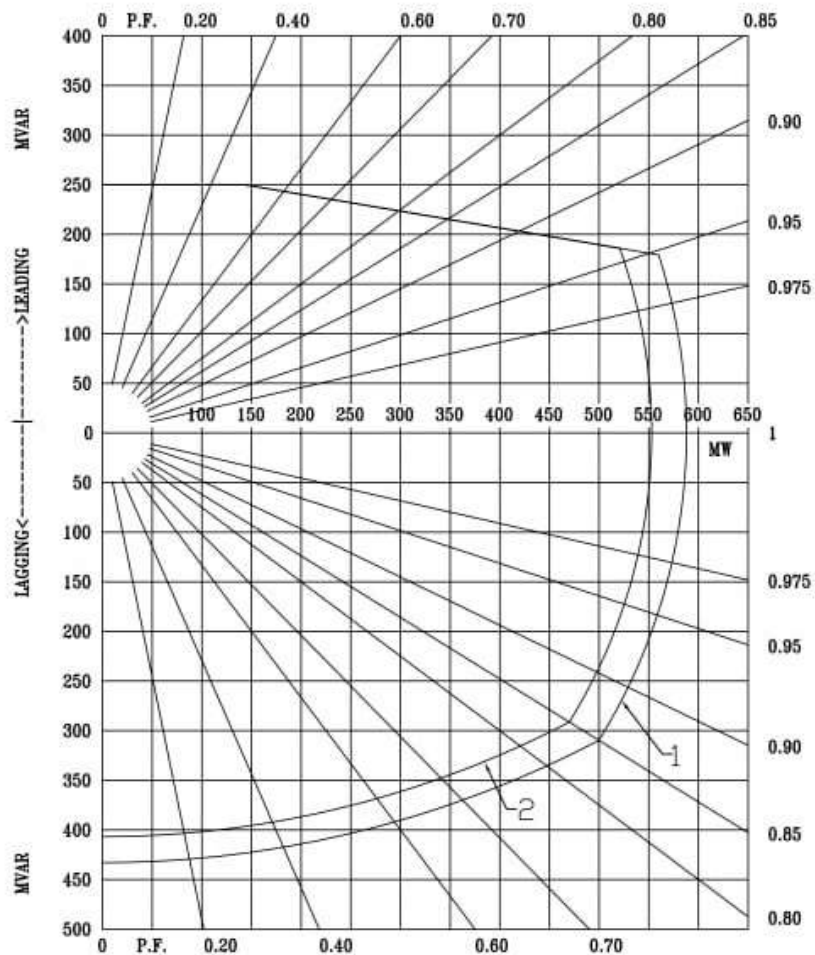


CAPABILITY DIAGRAM

588 MVA, 0.85 P.F, 21.0 KV, 3.5 BAR(g)

GENERATOR-TYPE: THDF115/59

GENERATOR OUTPUT	S	=	588.000	MVA	1 - AT 3.5 BAR(G) H2 PRESS.
ARMATURE VOLTAGE	U	=	21.000	kV	2 - AT 2.5 BAR(G) H2 PRESS.
ARMATURE CURRENT	I	=	16.166	kA	
FREQUENCY	F	=	50.0	Hz	
POWER FACTOR	P.F.	=	0.850		
RATED H2-PRESSURE	P _r	=	3.50	bar	



ELECTRICAL MACHINES ENGINEERING
BHEL , HARDWAR

AKG	-SD-	02.06.2004
NAME	SIGN.	DATE
TG - THDF115/59	-	085

	DURGAPUR.400 KV.B415 GT- 1.Metering BUS- 1_RY	DURGAPUR.400 KV.B415 GT- 1.Metering -1_P	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_Q	DURGAPUR.400 KV.B415 GT- 2.Metering GT- 2_P	DURGAPUR.400 KV.B410 GT- 2.Metering GT- 2_Q	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_P	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_Q	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_P	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_Q	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_P	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_Q	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_P	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_Q	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_P	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_Q	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_P	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_Q	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHED PUR-1_P	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHED PUR-1_Q	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_P	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_Q
Timestamp	BUS-1_RY	GT-1_P	GT-1_Q	GT-2_P	GT-2_Q	ICT-1_P	ICT-1_Q	ICT-2_P	ICT-2_Q	ST-1_P	ST-1_Q	ST-2_P	ST-2_Q	RAGHUNATHPU R-1_P	RAGHUNATHPU R-1_Q	RAGHUNATHPU R-2_P	RAGHUNATHPU R-2_Q	JAMSHEDPUR- 1_P	JAMSHEDPUR- 1_Q	JAMSHEDPUR- 2_P	JAMSHEDPUR- 2_Q
MMddyyyy HH:mm:ss	KV	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar
04-25-2023 23:30:00	400.75	-480.45	156.71	-483.03	120.87	165.25	-53.28	164.49	-51.83	4.42	3.2	6.91	5.43	69.69	-33.38	67.58	-37.43	238.03	-56.86	229.97	-58.59
04-23-2023 0:00:00	410.4	-480.29	163.87	-483.75	123.33	157.48	-54.64	156.43	-54.09	4.38	3.2	7.06	5.41	77.47	-34.18	72.37	-34.18	59.73	-59.73	222.61	-61.69
04-23-2023 0:30:00	410.4	-480.18	163.2	-487.27	126.71	166.14	-49.9	164.77	-48.37	4.83	3.7	7.27	5.55	73.27	-33.83	69.67	-33.83	237.39	-64.04	229.09	-65.87
04-23-2023 1:00:00	410.03	-489.17	166.33	-477.3	128.47	155.83	-51.55	155.01	-51.63	4.6	3.65	7.33	5.62	74.41	-35.24	70.17	-38.38	242.53	-63.09	233.5	-65.01
04-23-2023 1:30:00	410.99	-463.18	166.13	-451.85	126.08	174.19	-50.08	172.85	-48.71	4.92	3.75	7.1	5.6	46.7	-37.26	45.57	-38.29	228.45	-64.35	220.11	-65.58
04-23-2023 2:00:00	411.44	-427.91	169.42	-434.52	132.43	162.81	-56.36	160.93	-57	4.24	3.56	6.96	5.8	35.67	-31.59	34.61	-33.13	226.62	-64.87	218.01	-67.33
04-23-2023 2:30:00	410.93	-426.22	164.33	-408.9	117.15	164.93	-53.53	165.17	-52.01	4.75	3.61	7.17	5.41	10.6	-26.27	12.92	-26.13	235.39	-64.86	226.8	-67.67
04-23-2023 3:00:00	411.31	-420.81	165.2	-403.63	119.65	156.38	-53.85	155.27	-53.17	4.4	3.13	6.89	5.35	8.85	-25.74	11.08	-26.92	238.54	-65.28	229.64	-66.97
04-23-2023 3:30:00	411.32	-436.79	170.44	-408.45	122.1	167.87	-53.22	166.89	-52.45	4.12	3.1	6.86	5.36	10.34	-26.69	11.94	-26.92	236.76	-69.45	230.77	-71.72
04-23-2023 4:00:00	411.32	-432.43	167.11	-408.09	122.51	181.24	-48.65	179.47	-48.21	4.28	3.05	6.72	5.38	0	-28.55	0	-28.33	233.87	-70.47	225.46	-72.25
04-23-2023 4:30:00	411.33	-445.84	167.12	-441.1	143.18	169.56	-57.08	168.84	-57.74	4.7	3.19	7.28	5.4	17.05	-30.29	18.17	-30.1	247.74	-69.81	239.53	-72.77
04-23-2023 5:00:00	410.69	-457.31	166.7	-458.72	173.29	173.29	-52.92	-52.92	4.57	3.18	7.04	5.35	21.64	-33.29	22.72	-33.88	253.12	-71.47	244.76	-73.75	
04-23-2023 5:30:00	410.6	-467.02	169.46	-459.67	142.41	164.33	-52.83	163.16	-51.68	4.3	3.06	6.69	5.3	37.52	-33.24	36.89	-33.63	254.21	-73.01	246.07	-74.03
04-23-2023 6:00:00	410.95	-460.84	170.84	-468.09	134.49	155.88	-56.94	155.03	-56.16	4.42	3.09	6.93	5.39	48.22	-27.74	46.48	-30.22	251.78	-70.69	242.62	-72.57
04-23-2023 6:30:00	410.55	-472	169.1	-466	130.32	149.73	-57.5	148.71	-56.12	4.63	3.19	6.96	5.75	48.54	-27.51	46.3	-29.99	257.85	-68.44	248.26	-71.47
04-24-2023 5:30:00	417.54	-365.01	98.25	-358.87	78.86	107.8	-72.32	105.75	-68.29	4.37	3.36	4.52	3.62	61.3	-15.14	49.71	-17.83	195.41	-5.31	188.82	-6.0
04-24-2023 6:00:00	418.33	-360.96	101.44	-350.91	81.6	95.93	-75.5	96.31	-73.31	4.47	3.48	4.84	3.95	54.41	-14.77	50.96	-16.85	197.74	-6.02	190.97	-6.54
04-24-2023 6:30:00	418.1	-362.24	99.91	-355.42	81.76	95.88	-72.87	95.51	-71.07	4.43	3.4	4.97	3.89	58.86	-14.56	56.21	-17.91	196.22	-7.5	189.41	-10.54
04-24-2023 7:00:00	418.35	-355.83	100.14	-354.85	80.92	91.91	-74.82	91.48	-71.44	4.59	3.54	5.04	3.86	55.75	-15.94	53.15	-18.82	198.97	-4.28	192.53	-7.93
04-24-2023 7:30:00	418.35	-355.8	98.7	-355.75	82.97	90.42	-74.65	89.34	-73.11	4.07	3.15	4.61	3.84	62.36	-16.09	59.13	-19.08	196.65	-4.16	190.24	-7.35
04-24-2023 8:00:00	418.94	-354.34	106	-352.66	84.89	102.3	-73.11	101.61	-71.68	4.31	3.3	4.52	3.67	65.05	-19.18	61.88	-20.47	178.68	-8.34	172.65	-10.07
04-24-2023 8:30:00	418.63	-366.72	104.01	-365.17	86.68	113.46	-73.62	113.33	-72.4	3.79	2.8	4.63	3.36	72.06	-18.11	65.51	-20.8	172.61	-6.14	166.37	-10.87
04-24-2023 9:00:00	419.12	-377.1	109.79	-355.61	87.88	125.17	-74.73	123.35	-71.82	3.9	2.77	4.79	3.63	68.15	-17.62	65.21	-19.83	165.06	-10.5	159.84	-13.23
04-24-2023 9:30:00	418.99	-353.01	105.44	-361.98	89.31	126.9	-69.88	126.38	-68.05	3.76	0	4.58	3.26	51.83	-16.23	50.73	-17.89	169.39	-13.45	163.63	-16.19
04-24-2023 10:00:00	418.98	-347.15	105.95	-350.91	87.39	121.12	-74.02	121.53	-71.35	3.82	0	4.54	3.22	46.33	-14.75	43.43	-16.72	174.66	-11.83	168.68	-14.62
04-24-2023 10:30:00	417.99	-345.9	100.34	-356.44	84.33	125.83	-73.32	126.35	-72.52	3.71	0	4.23	2.99	40.84	-15.3	39.17	-16.29	177.05	-4.73	174.89	-7.38
04-24-2023 11:00:00	418.03	-356.82	102.56	-356.78	83.27	128.02	-71.31	127.27	-69.22	3.73	0	3.96	2.83	42.3	-18.59	39.44	-19.24	175.92	-6.87	164.7	-9.98
04-24-2023 12:00:00	418	-359.03	97.31	-386.04	87.36	137.33	-69.89	135.75	-67.72	3.65	0	4.1	2.91	41.59	-19.18	38.8	-20.16	187.56	-5.54	181.54	-6.27
04-24-2023 12:30:00	417.32	-357.82	95.29	-414.09	90.27	138.95	-71.52	138.02	-69.15	3.56	0	3.97	0	47.69	-18.1	44.68	-19.15	194.04	-5.91	187.69	-9.42
04-24-2023 13:00:00	417.75	-358.18	94.81	-414.04	93.63	133.12	-72.66	132.1	-70.3	3.62	0	4.06	2.79	53.18	-18.29	49.28	-19.55	192.94	-7.42	186.41	-11.14
04-24-2023 13:30:00	417.72	-370.83	96.8	-406.76	87.65	130.98	-73.2	130.78	-69.87	3.6	0	4.33	3.22	49.7	-17.52	47.01	-20.05	200.34	-5.28	193.63	-8.54
04-24-2023 14:00:00	417.54	-387.14	92.53	-380.32	81.92	136.14	-70.58	134.05	-67.58	3.83	0	4.27	3.15	43.29	-20.79	41.06	-20.4	196.2	-2.19	190.26	-5.85
04-24-2023 14:30:00	416.93	-360.7	93.31	-336.92	70.9	139.34	-68.7	138.59	-68.4	3.55	0	4.26	3.61	32.62	-14.44	22.1	-15.48	180.86	0	176.03	-2.53
04-24-2023 15:00:00	416.82	-347.04	90.12	-336.07	72.21	143.73	-73.07	144.17	-71.19	3.6	0	4.27	3.61	16.33	-13.66	14.93	-14.98	175.03	3.88	170.03	-1.65
04-24-2023 15:30:00	417.32	-373.02	95.53	-372.01	78.69	137.2	-72.06	137.59	-70.37	3.81	0	4.89	3.74	32.61	-19.18	29.61	-19.99	196.03	0	189.68	-3.25
04-24-2023 16:00:00	417.54	-366.12	102.3	-389.17	85.15	136.88	-74.05	136.02	-73.73	3.99	0	4.79	3.58	50.38	-16.87	47.8	-20.39	197.48	-5.28	190.96	-8.61
04-24-2023 16:30:00	417.14	-417.88	101.81	-441.13	95.54	142.33	-73.89	141.36	-72.43	3.92	0	4.69	3.57	73.31	-22.1	69.44	-24.22	205.69	-4.91	198.86	-7.59
04-24-2023 17:00:00	418.53	-419.6	111.58	-449.24	105.56	132.86	-73.97	130.78	-72.1	4.21	2.83	4.66	3.52	89.75	-25.96	85.03	-30.36	203.53	-10.65	196.69	-13.82
04-24-2023 17:30:00	418.58	-475.66	112.2	-468.17	109.38	135.66	-72.3	135.42	-69.73	4.23	2.9	4.73	3.58	121.1	-31.02	114.93	-36.02	204.96	-15.3	198.07	-17.83
04-24-2023 18:00:00	417.95	-461.96	121.63	-480.35	109.27	142.38	-72.16	141.65	-68.48	4.55	3.7	4.88	3.7	104.21	-29.94	116.92	-35.68	203.81	-15.34	198.37	-18.62
04-24-2023 18:30:00	417.53	-474.5	126.78	-481.18	105.45	147.02	-70.83	147.12	-67.38	4.06	0	5.11	3.94	118.65	-26.48	112.25	-32.01	200.64	-15.44	192.94	-18.51
04-24-2023 19:00:00	416.3	-477.79	101.36	-478.85	93.19	156.46	-72.93	154.77	-69.55	4.06	0	5.07	3.87	103.9	-24.96	98.43	-29.6	208.94	-4.8	202.25	-8.01
04-24-2023 19:30:00	415.8	-454.63	92.5	-473.25	87.48	158.76	-70.49	157.69	-67.28	3.99	0	4.61	3.76	102.7	-20.96	97.94	-26.51	192.33	0	185.97	-4.25
04-24-2023 20:00:00	415.23	-455.67	90.14	-456.72	78.96	159.68	-71.24	157.73	-68.6	4.45	3.81	5.03	3.77	91.31	-17.17	86.62	-21.6	197.14	0	190.73	-3.04
04-24-2023 20:30:00	414.75	-469.75	86.04	-469.73	75	161.48	-69.02	159.72	-65.78	4.44	3.79	5.09	3.86	90.26	-18.47	85.13	-22.64	210.19	5.44	203.63	0
04-24-2023 21:00:00	414.89	-475.88	86.65	-480.61	81.81	164.48	-72.76	161.2	-68.39	5.24	4.19	5.19	3.85	96.78	-19.13	93.52	-23.78	206.78	4.17	200.17	0
04-24-2023 21:30:00	414.46	-481.35	84.66	-480.99	74.2	165.87	-69.74	165.61	-67.61	4.9	3.98	4.74	3.58	93.47	-18.56	87.91	-22.73	214.15	0	205.41	0
04-24-2023 22:00:00	414.44	-480.26	93.31	-481.84	73.59	160.52	-70.77														

	Timestamp MMdd/yyyy HH:mm:ss	DURGAPUR.400 KV.B415 GT- 1.Metering BUS-1_RV	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_P	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_Q	DURGAPUR.400 KV.B415 GT- 2.Metering GT- 2_P	DURGAPUR.400 KV.B415 GT- 2.Metering GT- 2_Q	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_P	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_Q	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_P	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_Q	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_P	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_Q	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_P	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_Q	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_P	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_Q	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_P	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_Q	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHE DPR-1_P	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHE DPR-1_Q	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_P	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_Q
		BUS-1_RV KV	GT-1_P MW	GT-1_Q MVar	GT-2_P MW	GT-2_Q MVar	ICT-1_P MW	ICT-1_Q MVar	ICT-2_P MW	ICT-2_Q MVar	ST-1_P MW	ST-1_Q MVar	ST-2_P MW	ST-2_Q MVar	RAGHUNATHPU R-1_P MW	RAGHUNATHPU R-1_Q MVar	RAGHUNATHPU R-2_P MW	RAGHUNATHPU R-2_Q MVar	JAMSHEDPUR- 1_P MW	JAMSHEDPUR- 1_Q MVar	JAMSHEDPUR- 2_P MW	JAMSHEDPUR- 2_Q MVar
04-25-2023 18:30:00		417.53	-474.5	116.7	-478.18	105.45	147.02	-70.83	147.12	-67.38	4.06	0	5.11	3.94	118.65	-26.48	112.25	-33.01	200.64	-15.44	192.94	-18.51
04-25-2023 19:00:00		415.3	-477.79	101.38	-478.85	101.73	156.46	-72.83	154.77	-69.55	4.06	0	3.87	3.87	104.77	-24.95	98.43	-24.95	208.47	-8.8	203.25	-9.01
04-25-2023 19:30:00		415.8	-454.63	92.5	-473.25	87.48	158.76	-70.49	157.69	-67.28	3.99	0	4.61	3.78	102.7	-20.96	97.94	-26.51	192.33	0	185.97	-4.25
04-25-2023 20:00:00		415.23	-455.67	90.14	-456.72	78.96	159.68	-71.24	157.73	-68.6	4.45	3.81	5.03	3.77	91.31	-17.17	86.62	-21.6	197.14	0	190.73	-3.04
04-25-2023 20:30:00		414.75	-469.75	86.04	-469.73	75	161.48	-69.02	159.72	-65.78	4.44	3.79	5.09	3.86	90.26	-18.47	85.13	-22.64	210.19	5.44	203.63	0
04-25-2023 21:00:00		414.89	-475.88	86.65	-480.61	81.81	164.48	-72.76	161.2	-68.39	5.24	4.19	5.19	3.85	98.78	-19.13	93.52	-23.78	206.78	4.17	200.17	0
04-25-2023 21:30:00		414.46	-481.35	84.66	-480.89	74.2	165.67	-69.74	165.61	-67.61	4.9	3.99	4.74	3.58	93.47	-18.56	87.91	-22.73	214.15	5.7	205.41	0
04-25-2023 22:00:00		414.44	-480.26	83.01	-481.84	73.59	160.52	-70.77	158.65	-69.76	4.87	3.99	4.77	3.73	80.96	-16.46	76.38	-20.67	230.65	6.64	222.98	0
04-25-2023 22:30:00		414.33	-478.03	78.2	-480.91	57.07	161.89	-68.89	160.72	-68.72	5.03	4.16	4.83	3.77	79.54	-14.29	74.55	-16.67	230.78	12.36	223.82	5.45
04-25-2023 23:00:00		413.79	-484.07	79.88	-478.56	48.17	174.89	-63.86	172.75	-61.65	5.18	4.3	4.65	3.59	98.18	-16.92	84.22	-19.73	231.26	13.86	223.88	8.15
04-25-2023 23:30:00		414.51	-464.91	81.58	-482.74	54.45	168.14	-66.11	166.89	-63.53	4.45	3.05	4.48	3.55	72.61	-14.84	68.81	-18.9	224.79	9.86	217.78	3.98
04-26-2023 0:00:00		414.95	-481.9	86.93	-478.92	56.09	175.01	-68.2	172.2	-66.39	4.37	3.05	4.6	3.53	76.52	-16.26	72.97	-19.63	221.41	7.62	214.33	2.03
04-26-2023 0:30:00		414.29	-465.44	82.81	-469.74	52.27	173.37	-67.25	172.81	-65.52	4.18	2.9	4.7	3.58	71.11	-12.6	68.04	-16.83	215.87	8.76	208.98	3.67
04-26-2023 1:00:00		415.17	-456.93	85.27	-454.72	51.81	178.12	-64.6	176.95	-62.77	4.4	3.88	4.59	3.49	60.53	-13.67	56.27	-16.41	210.92	5.89	204.4	0
04-26-2023 1:30:00		415.2	-479.46	91.1	-432.45	50.42	176.35	-67.53	175.01	-63.75	4.99	4.14	4.89	3.67	59.77	-13.58	56.82	-15.34	207.33	5.82	199.84	0
04-26-2023 2:00:00		415.17	-475.03	93.27	-461.33	61.33	175.21	-70.26	171.77	-65.73	5.21	4.35	4.91	3.67	77.17	-15.73	73.01	-19.99	217.37	5.84	210.7	0
04-26-2023 2:30:00		415.69	-459	93.52	-453.64	62.29	169.95	-69.8	168.15	-67.94	5.07	4.19	5.04	3.81	68.78	-13.76	63.66	-16.41	211.42	0	204.4	-3.04
04-26-2023 3:00:00		415.68	-458.98	96.45	-412.14	54.97	164.02	-68.8	162.94	-67.74	4.93	4.12	4.75	3.53	55.11	-13.4	51.47	-15.75	209.21	0	202.33	-2.79
04-26-2023 3:30:00		416.61	-469.19	100.14	-384.24	49.07	161.84	-70.1	160.76	-69.68	4.19	3.72	4.46	3.33	48.61	-13.6	46.56	-14.91	209.84	4.2	203.23	0
04-26-2023 4:00:00		416.68	-459.53	98.86	-435.78	63.25	160.31	-70.69	160.49	-68.14	4.31	3.83	4.78	3.52	64.59	-15.07	61	-18.4	215.11	0	208.02	-4.53
04-26-2023 4:30:00		416.18	-462.02	95.89	-447.01	61.16	168.41	-68.42	166.65	-68.14	4.42	3.91	4.72	3.5	73.34	-16.19	69.54	-19.53	204.96	0	198.43	-3.19
04-26-2023 5:00:00		416.68	-458.75	106.17	-412.25	62	151.21	-70.6	150.43	-69.49	4.67	4.04	4.78	3.66	87.06	-18.5	82.42	-21.88	189.79	0	181.68	-3.8
04-26-2023 5:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 6:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 6:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 7:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 7:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 8:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 8:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 9:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 9:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 10:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 10:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 11:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 11:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 12:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 12:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 13:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 13:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 14:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 14:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 15:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 15:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 16:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 16:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 17:00:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-2023 17:30:00		409.94	-469.99	107.83	-483.32	67.75	230.54	-39.08	228.57	-37.84	4.81	4.12	7.09	5.85	70.6	-20.88	62.63	-20.4	204.37	0	192.84	-4.81
04-26-																						

Timestamp MMddyyyy HH:mm:ss	DURGAPUR.400 KV.B415 GT- 1.Metering BUS- 1.RY	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_P	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_Q	DURGAPUR.400 KV.B410 GT- 2.Metering GT- 2_P	DURGAPUR.400 KV.B410 GT- 2.Metering GT- 2_Q	DURGAPUR.400 KV.B404 ICT1.Metering IC T1-P	DURGAPUR.400 KV.B404 ICT1.Metering IC T1-Q	DURGAPUR.400 KV.B406 ICT2.Metering IC T2-P	DURGAPUR.400 KV.B406 ICT2.Metering IC T2-Q	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_P	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_Q	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_P	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_Q	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_P	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_Q	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_P	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_Q	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHED PUR-1_P	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHED PUR-1_Q	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_P	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_Q
	BUS-1_RY	GT-1_P	GT-1_Q	GT-2_P	GT-2_Q	ICT-1_P	ICT-1_Q	ICT-2_P	ICT-2_Q	ST-1_P	ST-1_Q	ST-2_P	ST-2_Q	RAGHUNATHPU R-1_P	RAGHUNATHPU R-1_Q	RAGHUNATHPU R-2_P	RAGHUNATHPU R-2_Q	JAMSHEDPUR- 1_P	JAMSHEDPUR- 1_Q	JAMSHEDPUR- 2_P	JAMSHEDPUR- 2_Q
	KV	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar
04-30-2023 10:00:00	410.14	-487.59	142.43	-317.01	90.46	202.16	-48.13	201.45	-46.74	4.85	3.92	6.58	5.52	-52.69	-15.9	-49.87	-13.53	239.8	-56.85	232.2	-59.28
04-30-2023 10:05:00	409.54	-475.90	141.95	-308.9	87.7	196.43	-49.38	196.44	-47.97	4.49	3.78	6.62	5.58	-52.69	-14.73	-49.87	-14.73	240.7	-54.36	240.5	-54.36
04-30-2023 11:00:00	409.15	-480.15	139.89	-318.31	89.55	202.22	-49.78	201.19	-47.96	4.33	3.04	6.6	5.78	-60.28	-17.41	-55.05	-14.07	251.76	-52.14	243.81	-54.74
04-30-2023 11:30:00	408.65	-390.58	121.24	-335.86	90.3	197.47	-46.46	196.5	-45.74	4.26	2.95	6.55	5.56	-77.5	-13.59	-72.18	-9.94	241.42	-48.14	234.11	-51.05
04-30-2023 12:00:00	409.13	-388.95	118.31	-351.21	93.08	205.93	-47.1	204.88	-46.37	4.42	3.06	6.37	5.41	-73.97	-14.61	-68.99	-12.71	235.6	-48.23	226.61	-50.81
04-30-2023 12:30:00	409.27	-397.88	125.49	-341.74	93.63	197.01	-50.28	196.01	-50.62	4.31	3.02	6.63	5.48	-66.49	-14.13	-61.62	-12.67	233.74	-50.58	225.39	-52.65
04-30-2023 13:00:00	409.74	-393.05	128.67	-310.75	94.08	179.53	-53.17	177.85	-52.84	4.67	3.22	6.81	5.65	-66.17	-15.51	-61.66	-12.14	232.64	-46.65	227.27	-50.03
04-30-2023 13:30:00	410.55	-375.03	130.25	-306.38	109.92	154	-54.63	154.51	-54.59	5.35	4.2	6.67	5.62	-58.96	-17.84	-56.09	-15.98	241	-51.63	232.94	-54.23
04-30-2023 14:00:00	411.4	-341.42	134.21	-290.64	99.95	143.22	-57.48	142.38	-57.4	4.89	3.89	6.66	5.51	-65.85	-13.36	-60.28	-11.87	226.9	-55.76	218.91	-57.81
04-30-2023 15:00:00	410.86	-348.85	131.03	-289.82	96.56	155.53	-56.11	155.53	-55.39	5.31	4.25	6.79	5.62	-70.96	-11.99	-65.95	-9.11	231.23	-61.96	224	-54.89
04-30-2023 15:30:00	411.35	-345.11	133.58	-287.79	100.96	147.6	-57.47	146.51	-57.71	5.45	4.43	6.11	5.2	-62.48	-10.37	-57.95	-7.4	227.59	-54.4	220.45	-56.5
04-30-2023 16:00:00	410.43	-350.98	131.14	-318.8	103.35	151.91	-54.8	150.88	-53.91	5.29	4.4	6.19	5.26	-65.42	-12.52	-51.39	-9.94	235.6	-52.4	227.89	-54.9
04-30-2023 16:30:00	410.06	-367.6	131.02	-363.54	109.29	157.9	-50.43	157.31	-50.42	4.91	4.26	6.01	5.11	-42.9	-15.92	-39.21	-16.39	246.31	-57.48	237.55	-60.7
04-30-2023 17:00:00	410.43	-392.97	125.31	-410.06	107.6	165.92	-49.94	165.7	-49.41	5.46	4.41	6.68	5.37	-27.69	-12.93	-24.02	-13.38	254.3	-57.95	245.9	-60.2
04-30-2023 17:30:00	411.61	0	0	-431.94	133.18	123.36	-43.84	122.36	-44.14	10.92	8.27	14.53	14.23	-106.39	-14.23	-106.39	-14.23	212.23	-48.03	206.33	-49.18
04-30-2023 18:00:00	411.58	0	0	-456.55	126.25	125.5	-42.71	124.93	-41.194	11.99	8.83	6.2	4.55	-118.37	-13.9	-111.78	-18.27	214.42	-47.08	214.43	-49.06
04-30-2023 18:30:00	411.08	0	0	-480.61	134.3	148.03	-39.2	147.31	-38.91	14.22	12.32	6.77	5.55	-129.77	11.54	-123.36	15.14	216.1	-48.92	208.75	-51.79
04-30-2023 19:00:00	410.31	0	0	-470.7	123.96	145.29	-36.25	144.3	-36.08	17.78	14.76	6.7	5.59	-127.25	13.16	-120.03	18.86	209.82	-48.51	202.88	-51.24
04-30-2023 19:30:00	409.8	0	0	-481.77	122.87	144.85	-40.35	145.74	-40.82	16.79	15.37	6.14	5.36	-128.41	12.89	-120.94	19.01	216.11	-45.55	210.66	-47.71
04-30-2023 20:00:00	409.94	0	0	-480.22	121.66	147.44	-42.3	146.66	-41.79	18.27	16.35	6.43	5.38	-141.77	17.17	-136.19	21.14	228.29	-47.9	220.39	-49.5
04-30-2023 20:30:00	410.35	0	0	-485.05	125.29	154.72	-41.37	154.22	-39.08	22.13	16.84	6.66	5.31	-140.4	15	-131.95	23.96	221.62	-48.18	212.68	-51.1
04-30-2023 21:00:00	410.43	0	0	-485.05	125.29	154.72	-41.37	154.22	-39.08	22.13	16.84	6.66	5.31	-140.4	15	-131.95	23.96	221.62	-48.18	212.68	-51.1
04-30-2023 21:30:00	410.05	0	0	-439.14	134.25	143.95	-40.98	143.15	-41.25	22.08	17.86	6.68	5.43	-151.58	16.45	-145.4	20.92	219.63	-47.32	210.97	-47.3
04-30-2023 22:00:00	411.43	-96.09	3.13	-429.12	117.81	144.51	-44.36	143.82	-44.81	22.4	18.05	6.04	4.37	-123.05	14.7	-117.54	17.94	234.71	-43.63	227.1	-46.09
04-30-2023 22:30:00	409.98	-181.62	60.15	-438.75	110.59	149.95	-47.06	148.1	-46.77	11.61	8.94	6.01	4.28	-94.91	4.12	-89.94	9.43	250.37	-47.67	241.92	-50.66
04-30-2023 23:00:00	409.49	-209.67	82.05	-435.86	117.9	168.11	-44.54	166.97	-44.97	12.46	9.48	6.1	4.29	-74.37	-8.74	-70.09	-9.7	224.24	-53.62	216.1	-55.91
04-30-2023 23:30:00	409.41	-270.27	78.57	-438.34	116.22	177.59	-41.06	176.8	-41.33	12.41	9.41	6.29	5.31	-63.81	-13.33	-60.08	-10.47	231.67	-51.5	224.27	-53.8
05-01-2023 0:00:00	410.3	-270.01	86.3	-445.36	126.42	165.33	-51.34	165.71	-50.88	9.56	8.82	6.74	5.42	-51.53	-8.66	-47.93	-9.57	236.32	-53.27	227.65	-55.65
05-01-2023 0:30:00	409.21	-351.15	90.65	-440.39	119.05	182.97	-43.84	180.63	-42.79	7.16	6.17	6.68	5.36	-43.67	-11.96	-41.87	-7.88	257.17	-54.6	229.88	-57.78
05-01-2023 1:00:00	409.51	-389.51	96.87	-446.46	120.74	184.72	-46.89	183.84	-46.69	6.14	5.35	6.61	5.21	-43.7	-15.95	-41.12	-15.95	258.94	-54.53	249.89	-57.34
05-01-2023 1:30:00	409.9	-343.65	97.61	-427.99	122.5	183.69	-47.12	182.55	-46.52	6.79	5.19	6.4	5.18	-41.44	-11.43	-39.03	-7.24	240.69	-56.19	232.47	-59.85
05-01-2023 2:00:00	410.3	-364.47	107.87	-378.66	116.31	174.7	-48.45	174.7	-48.45	6.81	5.21	6.56	5.24	-45.66	-9.11	-42.19	-5.92	237.65	-58.89	229.64	-61.25
05-01-2023 2:30:00	410.23	-345.58	115.38	-333.04	118.35	169.92	-50.87	169.78	-51.3	6.88	5.22	6.43	5.16	-69.77	-12.7	-55.48	-9.55	223.63	-57.44	215.32	-60.24
05-01-2023 3:00:00	410.51	-346.77	119.01	-344.6	122.17	173.18	-48.13	172.91	-48.24	6.24	4.96	6.53	5.17	-53.38	-14.07	-49.96	-11.71	219.4	-61.53	210.92	-64.6
05-01-2023 3:30:00	410.17	-344.22	123.88	-342.49	118.63	166.09	-51.55	166.13	-50.62	6.42	4.97	6.48	5.17	-49.92	-15.26	-46.71	-12.97	222.66	-60.86	214.46	-62.44
05-01-2023 4:00:00	409.63	-365.76	123.4	-341.24	114	184.3	-40.91	183.47	-41.71	6.28	4.9	6.19	5	-51.34	-18.23	-48.05	-16.53	218.44	-64.66	209.34	-66.72
05-01-2023 4:30:00	410.11	-365.4	122.94	-347.79	117.32	185.54	-45.05	170.42	-44.95	6.26	4.96	6.42	5.21	-43.7	-15.95	-41.12	-15.95	219.4	-62.84	211.07	-64.92
05-01-2023 5:00:00	409.42	-363.37	123.15	-369.44	116.57	171.08	-43.01	171.86	-42.78	6.6	5.07	6.59	5.29	-39.05	-16.73	-36.07	-15.34	228.16	-63.51	219.27	-65.91
05-01-2023 5:30:00	408.84	-377.17	117.81	-411.62	119.59	183.39	-42.04	181.88	-41.64	6.6	5.06	6.57	5.24	-29.02	-17.12	-26.38	-15.55	233.71	-64.87	225.83	-66.97
05-01-2023 6:00:00	409.82	-371.45	122.47	-427.07	128.89	178.98	-47.98	178.5	-47.79	6.35	4.84	6.46	5.17	-22.8	-13.72	-21.35	-12.27	236.63	-68.2	228.81	-70.58
05-01-2023 6:30:00	409.53	-379.8	122.44	-392.08	120.83	183.75	-47.16	181.21	-46.14	6.59	5.02	6.28	5.17	-39.44	-14.5	-36.47	-12.86	238.45	-64.48	229.28	-66.48
05-01-2023 7:00:00	409.53	-371.97	126.42	-364.84	116.91	176.39	-48.84	175.68	-48.26	6.57	5.15	6.38	5.24	-45.28	-13.56	-42.4	-10.79	230.32	-65.29	222.42	-66.99
05-01-2023 7:30:00	409.55	-355.71	123.31	-355.03	114.02	172.68	-44.92	172.07	-44.36	6.15	4.74	6.13	5.2	-45.45	-11.89	-41.97	-10.79	232.62	-64.69	219.32	-66
05-01-2023 8:00:00	409.67	-369.93	123.76	-341.8	113.26	187.54	-42.1	186.66	-41.9	6.18	4.73	6.18	5.2	-45.45	-11.89	-41.97	-10.79	232.62	-64.69	219.32	-66
05-01-2023 8:30:00	409.62	-320.17	116.78	-319.9	109.76	184.63	-43.33	184.2	-41.76	6.13	4.72	6.57	5.49	-75.34	-13.86	-70.43	-10.56	207.71	-63.23	200.11	-64.51
05-01-2023 9:00:00	410.45	-326.77	124.14	-320.67	114.04	181.42	-45.87	179.06	-45.89	6.25	4.72	6.53	-67.46	-14.11	-63.18	-12.05	207.43	-63.62	200.7	-64.91	
05-01-2023 9:30:00	410.09	-337.45	120.16	-317.27	108.12	185.87	-39.75	184.65	-38.83	6.07	4.81	6.19	5.34	-69.39	-14.45	-64.67	-11.44	206.12	-63.77	198.9	-65.77

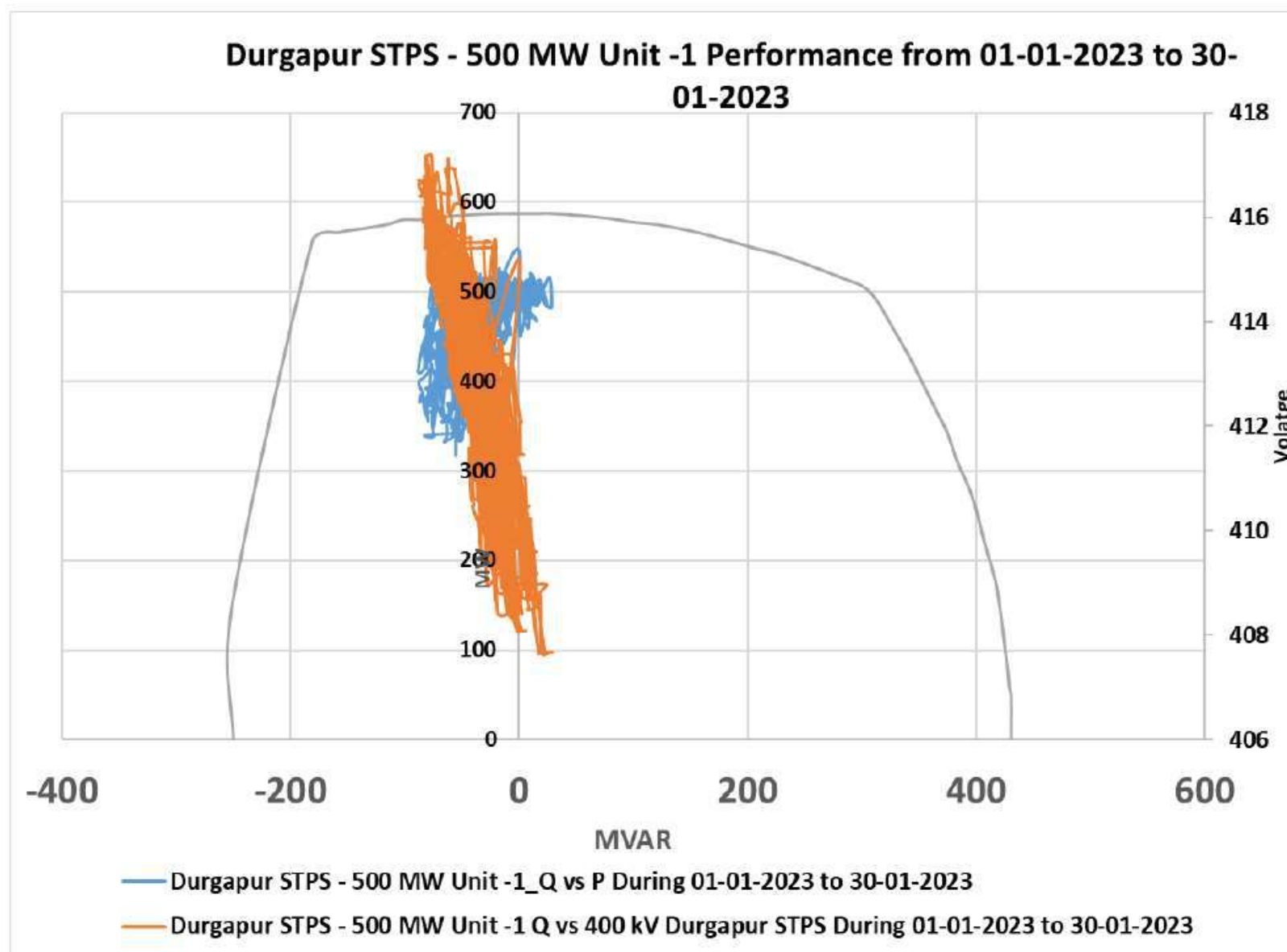
Timestamp MMdd/yyyy HH:mm:ss	DURGAPUR.400 KV.B415 GT- 1.MV	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_P	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_Q	DURGAPUR.400 KV.B410 GT- 2.Metering GT- 2_P	DURGAPUR.400 KV.B410 GT- 2.Metering GT- 2_Q	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_P	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_Q	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_P	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_Q	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_P	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_Q	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_P	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_Q	DURGAPUR.400 KV.B401 RAGNATH. L1.M FT.RAGHUNATH PUR-1_P	DURGAPUR.400 KV.B401 RAGNATH. L1.M FT.RAGHUNATH PUR-1_Q	DURGAPUR.400 KV.B403 RAGNATH. L2.M FT.RAGHUNATH PUR-2_P	DURGAPUR.400 KV.B403 RAGNATH. L2.M FT.RAGHUNATH PUR-2_Q	DURGAPUR.400 KV.B407 JSR LINEZ.MFT.JAM SDRUP-1_P	DURGAPUR.400 KV.B407 JSR LINEZ.MFT.JAM SDRUP-1_Q	DURGAPUR.400 KV.B409 JSR LINEZ.MFT.JAM SDRUP-2_P	DURGAPUR.400 KV.B409 JSR LINEZ.MFT.JAM SDRUP-2_Q
	BUS-1_RV	GT-1_P	GT-1_Q	GT-2_P	GT-2_Q	ICT-1_P	ICT-1_Q	ICT-2_P	ICT-2_Q	ST-1_P	ST-1_Q	ST-2_P	ST-2_Q	RAGHUNATHPU R-1_P	RAGHUNATHPU R-1_Q	RAGHUNATHPU R-2_P	RAGHUNATHPU R-2_Q	JAMSHEDPUR- 1_P	JAMSHEDPUR- 1_Q	JAMSHEDPUR- 2_P	JAMSHEDPUR- 2_Q
	KV	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar
05-02-2023 6:30:00	408.15	-408.46	131.65	-390.72	121.82	190.47	-53.97	189.48	-52.52	6.56	5.8	5.94	5.09	-27.49	-16.62	-25.5	-14.67	227.41	-60.88	218.98	-62.76
05-02-2023 7:00:00	410.72	-394.31	99.24	-384.3	83.29	190.81	-51.68	190.34	-51.68	6.81	5.94	5.88	5.09	-27.42	-16.62	-25.41	-14.67	220.49	-55.25	219.41	-57.83
05-02-2023 7:30:00	410.55	-418.59	98.66	-399.84	80.33	190.58	-45.26	190.49	-44.11	6.37	5.75	5.8	5.09	-14.73	3.94	-13.64	-0.21	225.98	-54.19	218.42	-56.43
05-02-2023 8:00:00	409.68	-415.83	93.67	-413.01	78.03	192.5	-40.34	190.52	-39.68	6.28	5.68	5.8	4.26	-21.26	0	-18.46	0	234.34	-50.06	227.71	-51.92
05-02-2023 8:30:00	409.7	-431.64	96.59	-428.97	83.79	200.03	-40.66	199.48	-40.79	6.14	5.75	6.01	4.48	-16.88	-1.53	-15.01	-2.04	237.09	-53.6	228.86	-55.8
05-02-2023 9:00:00	409.99	-430.59	100.06	-433.36	85.79	201.85	-39.53	201.58	-38.76	6.5	5.11	6.59	5.09	-23.12	-4.96	-21.43	-2.18	246.75	-53.57	238.63	-56.5
05-02-2023 9:30:00	410.03	-419.29	98.79	-409.59	84.88	196.26	-42.65	194.76	-42.55	6.4	5.11	6.06	4.95	-17.83	-1.35	-15.82	0	229.13	-53.27	221.06	-56.4
05-02-2023 10:00:00	411.16	-416.52	110.27	-379.34	88.2	195.24	-42.75	193.49	-42.56	7.12	6.28	6.53	5.26	-21.54	6.44	-18.74	7.09	219.49	-45.98	209.53	-47.68
05-02-2023 10:30:00	410.6	-356.4	94.42	-361.58	107.93	190.79	-44.86	189.87	-44.55	7.26	6.28	6.46	5.17	-50.83	3.79	-47.12	4.35	214.81	-44.83	205.96	-46.44
05-02-2023 11:00:00	411.07	-358.19	98.79	-362.88	114.13	185.85	-47.95	184.82	-47.74	7.95	6.51	6.24	4.97	-44.48	2.25	-41.04	4.69	213.99	-45.84	205.76	-47.77
05-02-2023 11:30:00	410.67	-347.29	94.4	-319.83	99.28	191.44	-41.86	189.67	-41.39	6.94	6.22	6.03	4.91	-60.83	3.12	-58.55	3.74	201.26	-63.09	194.64	-64.9
05-02-2023 12:00:00	410.62	-310.98	88.21	-321.09	101.08	187.81	-42.57	187.53	-42.67	6.51	5.84	5.85	4.73	-73.88	5.18	-68.77	6.35	197.4	-62.52	190.82	-64.1
05-02-2023 12:30:00	410.55	-323.26	90.79	-313.23	100.83	174.22	-44.73	172.85	-43.93	6.67	5.99	6.07	5.01	-72.42	5.71	-66.98	8.66	212.26	-61.14	206.02	-62.54
05-02-2023 13:00:00	411.2	-327.46	95.81	-304.41	102.25	181.17	-41.96	179.55	-41.43	7.3	6.3	6.48	5.12	-86.85	3.12	-80.84	4.07	219.48	-63.65	212.74	-65.28
05-02-2023 13:30:00	411.15	-331.93	98.9	-333.15	107.86	169.83	-46.62	168.32	-45.98	6.92	6.21	6.43	5.19	-70.15	2.55	-64.85	6.14	224.16	-63.84	219.01	-66.02
05-02-2023 14:00:00	411.62	-334.96	91.54	-327.51	111.72	170.85	-50.24	169.35	-49.63	6.77	6.08	6.49	5.21	-71.86	3.57	-66.36	5.19	227.04	-65.13	219.71	-67.05
05-02-2023 14:30:00	410.16	-334.13	94.13	-320.13	100.85	172.47	-43.77	172.58	-42.26	6.89	6.11	6.47	5.17	-82.74	2.87	-77.82	5.22	233.15	-61.55	224.46	-63.47
05-02-2023 15:00:00	410.98	-342.55	95.22	-319.54	102.6	170.84	-46.07	170.06	-46.39	6.58	5.89	6.67	7.73	-79.05	2.26	-73.18	8.38	232.65	-60.5	226.48	-62.07
05-02-2023 15:30:00	410.87	-332.95	97.86	-316.89	105.45	161.62	-49.66	160.91	-48.33	6.58	5.28	6.51	5.13	-68.43	4.47	-63.33	4.43	228.31	-61.6	221.19	-62.97
05-02-2023 16:00:00	410.77	-336.82	96.08	-305.5	102.02	160.7	-46.13	159.5	-44.72	6.49	5.22	6.01	4.84	-61.66	5.2	-56.77	5.84	219.59	-62.7	211.7	-64.28
05-02-2023 16:30:00	410.76	-366.83	96.12	-332.11	103.72	149.97	-48.13	150.06	-47.92	6.49	5.1	6.02	4.81	-44.81	6.19	-41.12	7.74	241.04	-61.17	232.53	-62.99
05-02-2023 17:00:00	410.69	-419.02	103.65	-374.19	110.01	168.87	-42.87	167.86	-42.51	6.49	5.21	6.44	5.18	-19.13	0	-16.98	0	239.6	-70.77	230.56	-72.6
05-02-2023 17:30:00	410.62	-403.7	101.14	-403.7	115.29	168.37	-44.83	168.37	-43.16	6.34	5.82	6.34	5.38	-8.79	1.89	-7.16	1.26	252.11	-71.87	242.26	-74.98
05-02-2023 18:00:00	410.61	-452.39	110.45	-393.64	115.97	176.8	-44.3	175.94	-43.16	5.94	4.82	6.1	5.49	-10.43	1.89	-7.16	1.26	252.11	-71.87	242.26	-74.98
05-02-2023 18:30:00	410.1	-495.06	110.76	-413.79	113.45	191.1	-45.38	189.23	-44.14	6.21	4.86	6.59	5.59	-8.8	-6.37	-6.83	-6.32	266.58	-67.88	255.31	-70.57
05-02-2023 19:00:00	408.89	-479.36	98.02	-471.33	115.75	211.69	-42.25	210.97	-41.97	6.36	5.88	6.82	5.81	-15.26	-6.67	-13.98	-5.22	233.73	-64.82	262.74	-68.11
05-02-2023 19:30:00	407.7	-474.86	84.42	-469.76	105.63	219.71	-42.19	217.58	-40.87	6.81	5.81	6.73	5.21	-27.67	-10.62	-25.52	-7.51	273.64	-50.16	263.55	-53.71
05-02-2023 20:00:00	408.67	-447.31	90.42	-430.98	102.62	216.28	-42.82	215.52	-42.03	6.58	5.78	6.25	5.14	-41.33	-12.24	-37.88	-7.51	257.55	-49.6	248.76	-52.62
05-02-2023 20:30:00	408.67	-453.72	93.9	-405.24	99.87	212.67	-49.33	211.32	-40.77	6.57	5.74	6.51	5.04	-45.86	-11.45	-42.1	-11.88	257.22	-49.13	248.71	-51.95
05-02-2023 21:00:00	408.31	-458.31	94.23	-413.35	104.26	213.15	-49.23	212.56	-40.52	6.48	5.98	6.32	5.38	-37.42	-12.66	-34.51	-7.78	256.41	-52.04	247.64	-54.7
05-02-2023 21:30:00	409.2	-430.85	91.2	-402.35	100.62	208.1	-44.24	206.79	-43.96	6.58	6.15	6.57	5.07	-33.34	-10.86	-30.38	-7.46	239.5	-52.96	231.52	-55.87
05-02-2023 22:00:00	409.9	-414.25	92.39	-383.47	104.22	190.05	-51.05	189.66	-50.19	6.16	5.72	6.73	6.07	-41.81	-8.65	-38.71	-4.83	245.28	-48.78	237	-51.62
05-02-2023 22:30:00	409.39	-387.15	87.85	-383.33	86.09	194.32	-43.46	194.42	-43.74	6.38	5.88	6.76	6.08	-56.18	-4.49	-52.63	0	244.02	-46.01	234.96	-49.05
05-02-2023 23:00:00	409.62	-383.08	85.66	-376.8	86.02	189.79	-46.05	187.82	-45.79	6.11	5.09	6.48	5.95	-48.92	-1.76	-45.03	0	234.13	-43.44	227.15	-46.26
05-02-2023 23:30:00	409.09	-370.66	84.54	-379.32	84.51	194.55	-45.59	192.72	-44.81	6.06	4.86	6.5	5.74	-53.64	-1.87	-49.12	1.95	228.9	-42.4	221.63	-45.05
05-03-2023 0:00:00	409.2	-389.61	88.7	-393.37	86.83	203.04	-40.86	201.63	-38.79	5.63	4.45	6.91	5.94	-47.16	-5.06	-43.91	-4.51	230.2	-44.2	221.34	-50.99
05-03-2023 0:30:00	409.63	-385.06	91.46	-404.25	83.22	193.92	-46.86	191.7	-45.99	5.53	4.41	6.55	4.98	-46.81	-7.45	-40.28	-5.44	250.22	-44.52	242.43	-48.13
05-03-2023 1:00:00	409.99	-415.97	98.67	-413.98	99.87	201.79	-40.39	-39.24	5.57	4.07	6.31	5.02	4.97	-34.01	-9.02	-31.78	-5.75	243.95	-52.13	235.17	-54.76
05-03-2023 1:30:00	409.93	-410.01	98.94	-416.4	102.33	192.94	-46.77	197.53	-45.67	5.83	4.7	6.31	4.97	-34.01	-9.02	-31.78	-5.75	243.95	-52.13	235.17	-54.76
05-03-2023 2:00:00	410.5	-427.12	102.92	-414.83	105.1	192.31	-47.13	190.48	-46.01	5.76	4.63	6.28	4.99	-26.4	-10.21	-23.72	-8.48	247.57	-52.69	238.97	-55.23
05-03-2023 2:30:00	410.52	-417.43	105.87	-417.07	102.85	195.64	-48.65	193.52	-46.79	5.79	4.51	6.19	4.67	-28.86	-8.44	-26.71	-4.8	245.78	-54.48	236.56	-55.84
05-03-2023 3:00:00	410.89	-391.27	102.45	-388.57	106.74	185.79	-49.48	184.34	-49.27	5.67	4.67	6.15	5.04	-40.05	-6.6	-36.79	-3.59	240.77	-53.78	231.96	-56.27
05-03-2023 3:30:00	410.97	-383.67	103.98	-400.68	108	189.37	-48.44	187.77	-49.73	5.52	4.53	6.36	5.81	-35.84	-4.79	-31.83	-3.24	234.76	-57.5	226.82	-59.93
05-03-2023 4:00:00	410.97	-386.79	104.34	-396.21	108.82	188.18	-50.28	186.02	-48.41	5.99	4.81	6.74	5.98	-38.43	-7.33	-35.75	-4.44	236.91	-56.69	228.59	-59.01
05-03-2023 4:30:00	410.98	-410.36	109.33	-409.78	108.76	194.32	-46.5	194.82	-45.92	6.02	4.83	6.67	6.02	-44.42	-9.81	-41.04	-6.03	241.19	-59.8	238.52	-63.17
05-03-2023 5:00:00	410.06	-446.54	105.77	-447.58	109.46	200.57	-46.18	199.15	-45.5	5.7	4.51	7	5.92	-16.15	-10.37	-14.14	-9.76	257.28	-58.58	247.98	-61.18
05-03-2023 5:30:00	409.6	-460.24	103.12	-453.1	107.05	202.23	-51.04	201.83	-49.93	5.4	4.61	6.89	5.9	-8.37	-6.34	-6.96	-5.5	254.91	-56.36	245.38	-59.31
05-03-2023 6:00:00																					

Timestamp MMddyyyy HH:mm:ss	DURGAPUR.400 KV.B415 GT- 1.Metering BUS- 1_RV	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_P	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_Q	DURGAPUR.400 KV.B415 GT- 2.Metering GT- 2_P	DURGAPUR.400 KV.B410 GT- 2.Metering GT- 2_Q	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_P	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_Q	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_P	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_Q	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_P	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_Q	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_P	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_Q	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_P	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_Q	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_P	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_Q	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHED PUR-1_P	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHED PUR-1_Q	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_P	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_Q
	BUS-1_RV	GT-1_P	GT-1_Q	GT-2_P	GT-2_Q	ICT-1_P	ICT-1_Q	ICT-2_P	ICT-2_Q	ST-1_P	ST-1_Q	ST-2_P	ST-2_Q	RAGHUNATHPUR- R-1_P	RAGHUNATHPUR- R-1_Q	RAGHUNATHPUR- R-2_P	RAGHUNATHPUR- R-2_Q	JAMSHEDPUR- 1_P	JAMSHEDPUR- 1_Q	JAMSHEDPUR- 2_P	JAMSHEDPUR- 2_Q
	KV	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar
05-04-2023 3:00:00	410.6	-410	110.25	-439.62	111.75	181.41	-49.44	182.09	-47.52	6.07	5.36	7.19	6.42	-29.38	-10.79	-27.24	-9.54	266.88	-58.59	256.68	-61.11
05-04-2023 3:30:00	410.65	-390	110.44	-417.84	112.44	178.23	-49.96	179.69	-48.96	5.61	5.12	7.69	6.54	-29.21	-9.53	-33.09	-6.05	255.73	-60.65	255.73	-60.65
05-04-2023 4:00:00	410.96	-427.96	116.84	-421.85	107.88	183.52	-51.7	183.28	-51.02	5.94	5.28	7.38	6.43	-33.15	-8.34	-30.41	-7.25	-59.69	-257.42	-62.84	-62.84
05-04-2023 4:30:00	410.47	-418.95	110.74	-422.26	108.68	185.26	-49.27	184.22	-47.63	5.78	5.31	6.93	6.33	-36.9	-8.51	-33.98	-5.67	268.11	-57.51	258.52	-60.35
05-04-2023 5:00:00	410.13	-439.48	108.33	-455.72	111.29	192.84	-43.03	192.94	-41.85	6.38	5.6	7.19	6.58	-26.75	-10.37	-23.72	-9.55	273.96	-61.42	268.71	-64.31
05-04-2023 5:30:00	409.88	-437.96	107.99	-451.16	106.31	180.92	-51.32	180.09	-50.17	6.43	5.64	7.8	6.57	-16.22	-5.44	-13.92	-1.8	271.96	-58.59	261.79	-61.28
05-04-2023 6:00:00	409.82	-436.31	105.27	-454.3	106.46	187.75	-48.77	186.08	-49.29	6.38	5.7	7.76	6.52	-20.46	-4.83	-18.23	-1.95	271.05	-59.03	260.22	-62.38
05-04-2023 6:30:00	409.68	-457.05	108.44	-457.18	105.2	180.07	-51.64	179.27	-51.19	6.3	5.64	7.65	6.44	-13.88	-5.67	-12.3	-3.77	283.56	-56.11	272.94	-59.35
05-04-2023 7:00:00	409.01	-455.77	105.23	-453.4	102.9	195.01	-48.53	175.38	-48.45	6.35	5.67	7.21	6.37	-9.23	-4.03	-7.61	-6.73	279.09	-55.85	269.08	-58.79
05-04-2023 7:30:00	409.19	-449.29	101.94	-460.87	104.63	178.59	-50.03	178.36	-49.07	6.02	5.38	6.88	6.31	0	-7.17	0	-7.04	271.78	-54.52	260.96	-57.95
05-04-2023 8:00:00	409.44	-428.55	102.88	-452.12	102.97	183.48	-45.15	183.01	-44.95	6.1	6.12	7.28	6.14	-18.32	-9.69	-16.38	-9.09	267.23	-56.33	258.5	-59.25
05-04-2023 8:30:00	409.43	-451.14	105.73	-458.94	105.21	188.19	-46.48	186.36	-45.53	5.89	5.18	7.83	6.4	0	-10.09	0	-9.68	266.41	-55.67	256.01	-58.49
05-04-2023 9:00:00	409.82	-442.22	109.17	-427.81	101.65	187.21	-47.06	187.3	-46.86	6.29	5.43	7.46	6.37	-14.32	-6.69	-13.41	-9.14	253.11	-55.82	245.08	-58.65
05-04-2023 9:30:00	409.84	-393.95	98.67	-396.52	93.89	188.51	-39.83	187.97	-39.27	6.35	5.47	7.77	6.38	-36.53	-7.73	-33.41	-4.77	238.75	-53.71	228.8	-56.79
05-04-2023 10:00:00	409.94	-380.48	97.7	-395.38	92.11	177.24	-44.16	177.51	-44.4	6.18	5.48	7.42	6.24	-36.61	-7.04	-33.96	-4.11	239.93	-50.91	231.39	-54.13
05-04-2023 10:30:00	409.91	-363.33	96.39	-352.33	86.23	177.86	-38.75	177.13	-38.74	6.04	5.63	6.78	6.02	-44.48	-7.78	-40.7	-4.57	244.64	-47.9	241.54	-50.4
05-04-2023 11:00:00	409.82	-369.19	96.57	-347.87	83.48	172.32	-40.97	171.72	-41.05	6.22	5.47	6.33	6.25	-43.59	-7.28	-40.3	-4.17	222.61	-49.49	214.6	-52.11
05-04-2023 11:30:00	409.74	-368.87	95.26	-360.56	83.96	169.01	-41.62	168.19	-41.05	6.67	5.81	7.01	5.86	-37.2	-9.43	-33.15	-6.86	226.73	-45.51	217.74	-48.04
05-04-2023 12:00:00	409.28	-367.88	94.89	-356.35	80.98	171.14	-41.82	170.31	-40.71	6.25	5.55	6.49	5.67	-40.93	-8.44	-37.27	-6.31	227.16	-43.89	218.98	-46.73
05-04-2023 12:30:00	409.38	-371.03	92.31	-367.84	85.23	173.12	-43.44	173.41	-41.38	6.3	5.6	6.82	5.79	-37.89	-8.97	-35.13	-6.24	231.11	-44.09	222.1	-46.62
05-04-2023 13:00:00	410.29	-346.74	97.54	-345.91	87.61	172.58	-46.18	171.65	-45.36	6.27	5.81	7.18	6.12	-50.4	-7.49	-46.81	-5.49	218.38	-45.71	210.77	-48.15
05-04-2023 13:30:00	410.17	-314.87	93.07	-320.88	83.91	181.14	-45.03	179.55	-43.4	5.94	5.5	7.1	6.05	-67.35	-3.64	-63.09	0	201.23	-45.27	193.5	-47.84
05-04-2023 14:00:00	409.44	-364.36	94.74	-360.83	86.23	174.53	-46.83	174.63	-45.96	6.07	5.63	6.98	6.07	-44.68	-7.96	-40.7	-4.57	220.24	-47.89	214.69	-48.04
05-04-2023 14:30:00	409.4	-350.82	91.68	-330.9	91.13	165.88	-44.79	164.67	-44.82	6.14	5.7	7.06	5.95	-39.33	-9.94	-36.02	-7.88	239.75	-42.88	231.5	-45.49
05-04-2023 15:00:00	408.89	-370.97	89.16	-400.08	89.39	177.41	-35.5	176.84	-34.46	6.3	5.6	7.38	6.03	-47.12	-13.74	-43.49	-12.1	249.95	-45.37	242.32	-48.2
05-04-2023 15:30:00	409.2	-375.55	89.71	-390.48	88.32	155.61	-44.68	156.38	-43.79	6.25	5.62	7.23	5.58	-36.17	-11.78	-33.02	-11.85	257.56	-40.08	248	-43.88
05-04-2023 16:00:00	409.79	-396.68	103.76	-393.61	94.45	151.68	-52.72	149.37	-51.66	6.35	5.53	7.21	5.63	-24.62	-8.14	-22.25	-8.95	262.63	-43.3	253.85	-47.22
05-04-2023 16:30:00	409.37	-378.32	96.63	-383.26	94.59	149.52	-49.28	147.85	-49.23	6.35	5.78	6.88	5.72	-27.65	-7.57	-24.92	-4.41	254.47	-43.6	245.52	-46.14
05-04-2023 17:00:00	409.69	-367.53	92.54	-385.96	89.4	150.34	-46.82	149.13	-46.47	6.39	5.93	6.83	5.87	-37.09	-7.52	-34.19	-5.7	255.78	-45.09	247.07	-48.01
05-04-2023 17:30:00	409.22	-385.2	96.88	-402.78	92.12	155.2	-48.45	155.21	-46.82	6.39	5.85	6.94	5.83	-32	-4.56	-29.16	-5.32	264.65	-47.14	255.48	-50.13
05-04-2023 18:00:00	409.71	-403.66	103.69	-403.66	99.37	159.37	-52.09	158.38	-52.14	6.43	5.93	7.01	6.21	-11.43	-7.84	-9.4	-40.2	267.78	-49.2	278.54	-52.07
05-04-2023 18:30:00	408.88	-437.75	98.37	-427.78	92.86	157.1	-55.76	155.1	-55.08	6	5.44	7.15	6.03	-25.1	-8.82	-22.25	-3.45	294.26	-40.95	284.32	-44.5
05-04-2023 19:00:00	408.37	-462.07	96.42	-476.74	98.08	179.28	-51.02	178.91	-50.9	6.42	5.59	7.57	6.03	-13.79	-10.75	-12.28	-10.06	296.36	-41.03	285.85	-44.94
05-04-2023 19:30:00	408.24	-477.85	96.99	-450.37	88.55	190.32	-49.06	189.3	-48.28	5.97	5.46	6.83	5.72	-17.68	-7.35	-15.3	-8.6	285.71	-43.42	275.63	-46.77
05-04-2023 20:00:00	407.85	-476.2	98.86	-477.39	96.34	201.67	-49.31	199.72	-49.38	5.95	5.47	6.71	5.6	-26.67	-8.55	-23.92	-5.89	296.33	-46.47	285.13	-50.23
05-04-2023 20:30:00	407.74	-477.19	94.59	-490.07	95.29	210.9	-47.36	208.69	-46.54	6.4	5.66	7.54	6.05	-34.62	-7.64	-31.75	-3.75	302.21	-47.34	292.78	-50.64
05-04-2023 21:00:00	407.91	-481.9	96.78	-482.45	96.38	206.23	-47.77	207.11	-46.7	6.13	5.52	7.38	6.34	-20.97	-6.28	-18.49	-9.16	291.22	-48.2	280.66	-50.76
05-04-2023 21:30:00	408.34	-473.2	99.96	-471.9	98.37	205.67	-46.44	205.68	-46.44	6.37	5.74	6.44	6.21	-24.13	-7.94	-21.4	-9.09	288.53	-48.71	278.54	-52.07
05-04-2023 22:00:00	409.33	-496.18	112.61	-487.76	109.12	196.42	-54.51	196.88	-54.38	6.13	5.5	7.04	6.06	-8.78	-10.94	-6.41	-10.01	295.88	-51.67	286.64	-55.67
05-04-2023 22:30:00	409.36	-463.41	107.18	-451.97	99.24	181.3	-50.28	181.94	-48.58	6.33	5.73	7.4	6.35	-16.47	-10.32	-14.83	-9.33	275.23	-48.75	275.83	-52.07
05-04-2023 23:00:00	409.83	-462.23	107.29	-450.64	100.79	189.12	-51.54	187.84	-51.87	6.27	5.63	6.87	6.14	-19.01	-13.14	-18.07	-11.74	281.39	-46.8	271.9	-49.99
05-04-2023 23:30:00	408.7	-464.02	102.26	-460.6	101.24	200.27	-44.88	199.76	-44.59	6.08	5.61	6.99	6.12	-17.86	-16.3	-16.42	-13.8	272.39	-49.18	263.75	-52.49
05-05-2023 0:00:00	409.64	-461.02	109	-457.14	102.62	193.33	-50.66	191.4	-49.92	6.14	5.6	6.98	6.24	-18.85	-13.79	-14.58	-12.72	275.19	-49.01	265.99	-51.57
05-05-2023 0:30:00	409.13	-462.51	104.85	-462.51	99.54	195.15	-45.48	193.94	-44.61	6.09	5.64	7.09	6.13	-29.32	-12.49	-26.78	-10.72	280.32	-50.98	269.98	-54.44
05-05-2023 1:00:00	409.04	-461.54	104.74	-461.54	104.74	194.58	-46.81	194.58	-46.81	6.43	5.93	7.04	6.21	-11.81	-9.06	-12.42	-9.06	281.61	-49.2	278.54	-52.07
05-05-2023 1:30:00	409.81	-431.63	106.42	-426.11	102.02	202.07	-49.4	200.18	-47.73	6.24	5.59	7.06	6.15	-44.22	-10.08	-40.55	-6.8	285.55	-52.52	258.89	-55.22
05-05-2023 2:00:00	410.04	-420.48	109.63	-389.38	50.4	190.06	-49.95	6.4	5.82	7.02	6.09	-51.52	6.09	-51.52	-47.27	-5.77	259.28	-63.56	250.92	-55.99	
05-05-2023 2:30:00	41																				

Timestamp MMddyyyy HH:mm:ss	DURGAPUR.400 KV.B415 GT- 1.MV	DURGAPUR.400 KV.B415 GT- 1.Metering GT- 1_P	DURGAPUR.400 KV.B415 GT- 1.Q	DURGAPUR.400 KV.B410 GT- 2.Metering GT- 2_P	DURGAPUR.400 KV.B410 GT- 2.Metering GT- 2_Q	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_P	DURGAPUR.400 KV.B404 ICT1.Metering IC T-1_Q	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_P	DURGAPUR.400 KV.B406 ICT2.Metering IC T-2_Q	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_P	DURGAPUR.400 KV.B413 ST- 1.Metering ST- 1_Q	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_P	DURGAPUR.400 KV.B412 ST- 2.Metering ST- 2_Q	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_P	DURGAPUR.400 KV.B401 RAGNATH L1.M FT.RAGHUNATH PUR-1_Q	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_P	DURGAPUR.400 KV.B403 RAGNATH L2.M FT.RAGHUNATH PUR-2_Q	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHE DPR-1_P	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHE DPR-1_Q	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPR-2_P	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPR-2_Q
	BUS-1_RV	GT-1_P	GT-1_Q	GT-2_P	GT-2_Q	ICT-1_P	ICT-1_Q	ICT-2_P	ICT-2_Q	ST-1_P	ST-1_Q	ST-2_P	ST-2_Q	RAGHUNATHPUR- R-1_P	RAGHUNATHPUR- R-1_Q	RAGHUNATHPUR- R-2_P	RAGHUNATHPUR- R-2_Q	JAMSHEDPUR- 1_P	JAMSHEDPUR- 1_Q	JAMSHEDPUR- 2_P	JAMSHEDPUR- 2_Q
	KV	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar
05-05-2023 23:30:00	407.97	-488.48	93.41	-475.05	86.95	201.88	-51.39	201.24	-50.77	6.35	5.89	7.3	6.42	-18.1	-7.25	-16.82	-6.06	290.5	-4.2	280.34	-45.26
05-05-2023 0:00:00	408.4	-487.57	93.41	-475.05	87.9	211.26	-44.61	210.61	-43.96	6.1	5.73	7.15	6.28	-19.15	-7.24	-17.34	-6.21	285.47	-4.18	284.39	-44.39
05-05-2023 0:30:00	408.4	-495.88	94.25	-478.51	88.24	216.19	-47.51	215.32	-39.79	6.19	5.7	7.2	6.3	-19.99	-10.33	-17.34	-6.26	285.09	-3.49	275.3	-50.48
05-05-2023 1:00:00	408.93	-492.73	96.17	-467.89	87.42	217.44	-45.5	216.55	-44.81	6.47	5.87	7.05	6.21	-27.94	-8.85	-25.26	-7.72	286.33	-45.92	276.43	-49.15
05-05-2023 1:30:00	409.39	-478.41	98.05	-474.08	92.32	220.04	-46	218.87	-44.73	6.49	5.96	7.07	6.38	-24.62	-7.3	-22.53	-4.64	275.85	-49.8	267.75	-52.97
05-05-2023 2:00:00	409.33	-483.88	97.13	-474.82	94.94	216.95	-49.65	215.78	-47.97	6.16	5.79	6.92	6.24	-15.86	-8.85	-14.02	-9.18	274.45	-47.45	265.8	-50.56
05-05-2023 2:30:00	409.34	-478.15	99.31	-473.29	91.99	206.78	-47.5	205.57	-48.17	6.38	5.82	7.09	6.38	-16.37	-8.82	-13.83	-5.76	277.64	-49.35	267.36	-52.11
05-05-2023 3:00:00	409.57	-478.05	102.47	-480.56	98.48	211.04	-46.03	209.5	-45.15	6.19	5.66	7.57	6.45	-8.72	-6.62	-8.39	-8.24	271.24	-53.11	262.65	-56.69
05-05-2023 3:30:00	409.49	-471.81	102.68	-482.15	96.56	204.32	-48.89	203.35	-48.24	6.02	5.52	7.33	6.31	-13.68	-7.77	-10.89	-8.44	277.76	-49.64	268.15	-52.5
05-05-2023 4:00:00	409.17	-488.1	97.36	-480.38	95	219.38	-40.05	216.88	-38.56	5.86	5.36	7.33	6.29	-22.16	-10.3	-20.5	-10.4	285.09	-55.09	258.84	-58.2
05-05-2023 4:30:00	409.7	-471.94	98.89	-476.47	94.48	209.8	-44.9	209.98	-45.21	6.56	5.87	7.4	6.34	-20.89	-8.92	-19.15	-3.35	275.95	-63.51	267.26	-57.01
05-05-2023 5:00:00	409.63	-460.38	97.23	-475.67	93.66	213.78	-44.93	212.13	-44.32	6.5	5.83	7.45	6.4	-26.18	-6.74	-23.72	-3.34	271.49	-53.68	262.63	-56.41
05-05-2023 5:30:00	409.18	-474.26	98.14	-482.66	95.48	207.1	-45.67	206.5	-45.29	6.34	5.67	6.99	6.28	-10.88	-5.64	-9.81	-6.34	275.14	-51.92	266.2	-55.26
05-05-2023 6:00:00	409.21	-471.3	98.74	-498.88	100.55	196.19	-49.29	193.68	-48.41	6.07	5.8	6.89	6.26	0	-5	-5	-5	272.28	-53.55	272.39	-56.86
05-05-2023 6:30:00	409.16	-468.97	91.47	-504.83	99.3	196.73	-43.74	194.59	-43.07	6.44	5.79	6.4	4.88	0	-4.32	0	-2.73	285.43	-53.82	276.07	-56.65
05-05-2023 7:00:00	409.49	-486.28	97.46	-480.02	83.29	190.61	-46.83	190.26	-47.34	6.42	5.85	7.12	6.34	-6.83	-4.35	0	-2.73	285.43	-53.82	276.07	-56.65
05-05-2023 7:30:00	409.09	-486.28	97.46	-480.02	83.29	190.61	-46.83	190.26	-47.34	6.42	5.85	7.12	6.34	-6.83	-4.35	0	-2.73	285.43	-53.82	276.07	-56.65
05-05-2023 8:00:00	409.52	-494.88	92.81	-475.94	82.69	196.38	-45.19	194.72	-45.33	6.22	5.86	6.69	5.46	0	-3.37	0	-2.1	282.94	-46.3	274.56	-50.09
05-05-2023 8:30:00	409.38	-461.06	87.92	-467.33	81.31	190.49	-47.15	189.25	-47.38	6.18	5.9	6.42	5.89	-8.54	2.15	-6.78	4.06	274.65	-46.06	265.39	-49.5
05-05-2023 9:00:00	409.37	-452.34	85.63	-448.68	78.37	205.76	-43.75	204.73	-43.05	6.41	5.8	7.37	6.51	-18.54	2.68	-16.39	3.45	256.26	-44.9	247.32	-47.7
05-05-2023 9:30:00	409.39	-437.89	77.5	-449.34	73.98	201.02	-41.32	198.78	-41.3	6.76	6.33	7.05	6.44	-28.2	1.79	-24.06	3.65	262.88	-43.52	254.3	-46.31
05-05-2023 10:00:00	409.31	-429.59	82.2	-431.86	70.58	190.25	-47.04	190.38	-46.52	6.18	5.85	7.37	6.62	-32.38	3.85	-30.25	6.19	265.66	-39.02	256.79	-41.55
05-05-2023 10:30:00	409.49	-486.28	97.46	-480.02	83.29	190.61	-46.83	190.26	-47.34	6.42	5.85	7.12	6.34	-6.83	-4.35	0	-2.73	285.43	-53.82	276.07	-56.65
05-05-2023 11:00:00	409.2	-454.98	81.55	-450.75	72.81	198.73	-46.36	187.24	-44.93	5.9	5.04	7.06	6.32	-17.35	0	-14.38	2.09	274.38	-38.24	264.25	-41.9
05-05-2023 11:30:00	408.95	-441.97	77.41	-455.12	74.81	198.22	-42.39	197.16	-42.17	6.09	5.11	6.9	6.37	-30.31	0	-27.36	0	274.74	-39.99	265.67	-43.38
05-05-2023 12:00:00	408.97	-457.24	82.69	-432.37	70.1	187.53	-44.65	186.45	-44.03	5.65	4.87	6.49	5.97	-27.58	1.47	-24.34	3.1	278.46	-36.4	269.15	-40.78
05-05-2023 12:30:00	408.86	-480.45	86.97	-419.91	69.29	184.51	-44.04	184.5	-42.79	5.63	4.86	6.79	6.28	-18.18	-6.43	-15.43	-4.74	276.76	-35.02	267.58	-37.63
05-05-2023 13:00:00	409.42	-461.47	88.62	-440.65	78.28	183.95	-52.33	182.68	-52.78	5.53	4.79	6.84	6.29	-19.63	-6.33	-17.26	-5.46	281.08	-33.15	271.2	-36.43
05-05-2023 13:30:00	409.64	-462.93	87.96	-435.01	77.95	189.88	-47.25	189.04	-46.41	6.01	5.21	6.94	6.49	-12.43	-7.1	-11.05	-6.72	266.56	-37.27	256.98	-40.6
05-05-2023 14:00:00	409.24	-469.78	90.93	-440.93	80.9	186.01	-50.34	186.28	-49.28	6.23	5.34	7.31	6.35	-11.54	-4.49	-9.36	-7.24	271.8	-35.29	263.83	-38.92
05-05-2023 14:30:00	409.49	-463.15	89.55	-440.5	81.25	183.19	-49.45	179.1	-49.24	5.99	4.96	7.12	6.27	-15.24	-7.16	-14.43	-7.4	269.85	-35.14	262.85	-39.51
05-05-2023 15:00:00	409.49	-444.65	81.85	-463.26	80.28	170.86	-52.33	169.19	-50.52	5.73	4.66	6.86	6.36	-20.2	-6.73	-17.86	-4.84	297.09	-29.63	266.89	-33.93
05-05-2023 15:30:00	409.99	-482.35	85.5	-467.89	77.75	187.76	-50.26	177.07	-49.53	5.6	4.42	6.87	6.23	-11.95	-7.84	-9.57	-31.42	302.75	-31.42	292.67	-35.39
05-05-2023 16:00:00	409.99	-457.04	85.88	-469.76	81.41	167.73	-50.75	166.35	-49.94	6.03	4.86	6.79	6.24	-10.22	-6.31	-8.23	-6.79	296.83	-34.04	285.34	-38.74
05-05-2023 16:30:00	408.98	-475.6	88.99	-471.62	82.3	172.66	-50.57	172.8	-50.56	6.04	4.67	6.65	5.95	-14.23	-4.47	-11.33	-4.13	308.97	-35.27	297.07	-38.8
05-05-2023 17:00:00	409.99	-488.18	86.21	-484.83	82.43	178.08	-48.33	177.22	-50.58	6.07	4.86	6.92	6.07	-7.01	-5.96	0	-3.2	305.89	-37.3	295.56	-41.15
05-05-2023 17:30:00	409.01	-491.42	90.88	-470.43	78.86	183.49	-49.22	182.05	-48.48	6.05	4.81	7.32	6.21	-12.8	-4.25	-10.72	-4.49	302.71	-39.19	293.4	-42.32
05-05-2023 18:00:00	409.77	-498.47	94.42	-485.17	80.47	188.81	-49.27	187.01	-49.17	6.1	4.58	7.12	6.27	-10.66	-4.3	-10.66	-4.31	301.16	-42.96	290.01	-42.77
05-05-2023 18:30:00	408.35	-474.03	82.94	-477.28	76.87	200.56	-48.85	199.61	-47.54	6.02	4.58	7.26	6.51	-34.52	4.47	-31.19	5.87	303.3	-41.39	293.6	-44.88
05-05-2023 19:00:00	408.36	-476.85	79.23	-482.91	78.6	209.23	-48.76	206.92	-48.02	5.97	4.67	8.13	6.61	-33.3	3.29	-30.32	4.67	301.53	-37.91	291.42	-41.7
05-05-2023 19:30:00	407.68	-465.5	75.8	-478.46	69.27	226.06	-41.72	224.76	-40.64	5.8	4.43	7.69	6.39	-43.35	4.76	-39.75	5.85	294.28	-38.82	283.65	-42.66
05-05-2023 20:00:00	408.21	-487.94	77.53	-469.85	69.99	229.58	-42.95	227.68	-41.65	5.67	4.42	6.95	6.09	-48.7	2.18	-43.28	2.63	293.24	-38.26	283.79	-41.72
05-05-2023 20:30:00	408.16	-489	80.67	-463.97	70.29	223.78	-46.35	222.45	-46.25	5.76	4.43	7.69	6.34	-47.28	4.47	-43.77	5.42	297.13	-37.67	286.71	-41.53
05-05-2023 21:00:00	408.21	-488.73	80.13	-467.82	69.87	229.44	-42.8	228.34	-42.29	5.7	4.41	7.18	6.15	-51.69	2.53	-47.49	3.41	295.02	-36.06	285.52	-39.61
05-05-2023 21:30:00	407.68	-470.7	75.78	-474.03	72.87	227.54	-46.27	227.54	-46.27	5.78	4.43	7.69	6.34	-47.28	4.47	-43.77	5.42	297.13	-37.67	286.71	-41.53
05-05-2023 22:00:00	408.17	-500.87	82.96	-484.65	76.8	220.46	-51.45	219.22	-51.18	5.81	4.57	7.29	6.2	-41.09	0	-36.77	1.88	308.64	-34.1	297.46	-38.53
05-05-2023 22:30:00	408.23	-467.17	76.05	-472.12	73.48	208.8	-48.82	207.01	-47.95	5.82	4.62	6.94	6.1	-43.7	2.93	-39.93	3.17	299.48	-31.56	288.51	-36.13
05-05-2023 23:00:00	408.14	-487.68	79.27	-471.59	67.99	214.06	-48.15	212.66	-47.96	5.74	4.49	6									

	DURGAPUR.400 KV.B415 GT- 1.Metering.BUS- 1_RV	DURGAPUR.400 KV.B415 GT- 1.Metering.GT- 1_P	DURGAPUR.400 KV.B415 GT- 1.Metering.GT- 1_Q	DURGAPUR.400 KV.B410 GT- 2.Metering.GT- 2_P	DURGAPUR.400 KV.B410 GT- 2.Metering.GT- 2_Q	DURGAPUR.400 KV.B404 ICT1.Metering.IC T-1_P	DURGAPUR.400 KV.B404 ICT1.Metering.IC T-1_Q	DURGAPUR.400 KV.B406 ICT2.Metering.IC T-2_P	DURGAPUR.400 KV.B406 ICT2.Metering.IC T-2_Q	DURGAPUR.400 KV.B413 ST- 1.Metering.ST- 1_P	DURGAPUR.400 KV.B413 ST- 1.Metering.ST- 1_Q	DURGAPUR.400 KV.B412 ST- 2.Metering.ST- 2_P	DURGAPUR.400 KV.B412 ST- 2.Metering.ST- 2_Q	DURGAPUR.400 KV.B401 RAGNATH_L1.M FT.RAGHUNATH PUR-1_P	DURGAPUR.400 KV.B401 RAGNATH_L1.M FT.RAGHUNATH PUR-1_Q	DURGAPUR.400 KV.B403 RAGNATH_L2.M FT.RAGHUNATH PUR-2_P	DURGAPUR.400 KV.B403 RAGNATH_L2.M FT.RAGHUNATH PUR-2_Q	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHE DPUR-1_P	DURGAPUR.400 KV.B407 JSR L1.MFT.JAMSHE DPUR-1_Q	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_P	DURGAPUR.400 KV.B409 JSR LINE2.MFT.JAM SHEDPUR-2_Q
Timestamp	BUS-1_RV	GT-1_P	GT-1_Q	GT-2_P	GT-2_Q	ICT-1_P	ICT-1_Q	ICT-2_P	ICT-2_Q	ST-1_P	ST-1_Q	ST-2_P	ST-2_Q	R-1_P	R-1_Q	R-2_P	R-2_Q	JAMSHPUR- 1_P	JAMSHPUR- 1_Q	JAMSHPUR- 2_P	JAMSHPUR- 2_Q
MMddyyyy HH:mm:ss	KV	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar	MW	MVar
05-07-2023 20:00:00	407.33	-489.87	100.42	-473.5	81.23	211.31	-50.06	210.48	-49.36	6.83	5.9	7.28	6.32	-46.21	-12.24	-42.57	-10	310.13	-38.91	298.07	-43.26
05-07-2023 20:30:00	406.96	-476.8	96.31	-476.41	77.2	208.4	-46.37	207.34	-46.7	6.81	5.84	7.2	6.41	-54.88	-12.33	-49.7	-10.54	316.27	-34.79	306.21	-38.68
05-07-2023 21:00:00	407.02	-497.09	100.85	-465.45	72.58	221.29	-44.8	219.98	-44.2	6.89	5.77	7.29	6.39	-55.54	-9.94	-52.92	-7.93	313.41	-41.02	302.43	-45.5
05-07-2023 21:30:00	407.41	-493.44	105.86	-469.81	53.52	213.8	-50.03	211.86	-48.37	5.97	5.36	7.41	6.44	-55.54	-4.69	-51.18	-3.54	319.27	-33.02	307.91	-37.32
05-07-2023 22:00:00	407.8	-496.37	96.89	-473.63	54.22	216.51	-49.06	215.87	-47.63	6.59	5.59	7.2	6.37	-51.8	-2.22	-47.64	2.45	315.8	-32.5	306.42	-36.48
05-07-2023 22:30:00	407.27	-500.35	92.58	-449.59	45.21	218.66	-47.66	218.19	-46.85	6.59	5.68	6.85	5.96	-59.3	0	-54.71	3.3	310.49	-27.28	301.2	-31.32
05-07-2023 23:00:00	406.77	-496.35	90.56	-470.96	48.52	218.62	-47.92	217.12	-47.01	6.18	5.59	6.88	5.78	-56.27	-4.33	-52.12	0	317.57	-21.47	305.42	-26.49
05-07-2023 23:30:00	407.05	-480.76	89.39	-467.52	53.13	226.29	-48.97	224.23	-48.57	6.55	5.97	6.92	5.76	-66.23	0	-61.28	2.2	310.4	-27.39	300.25	-31.61
05-08-2023 6:00:00	406.97	-488.08	92.36	-484.15	57.21	237.18	-42.54	235.71	-42.1	6.57	5.72	6.82	5.51	-65.28	-7.55	-59.86	-1.84	310.93	-31.98	299.95	-35.56

Annexure B.5.2

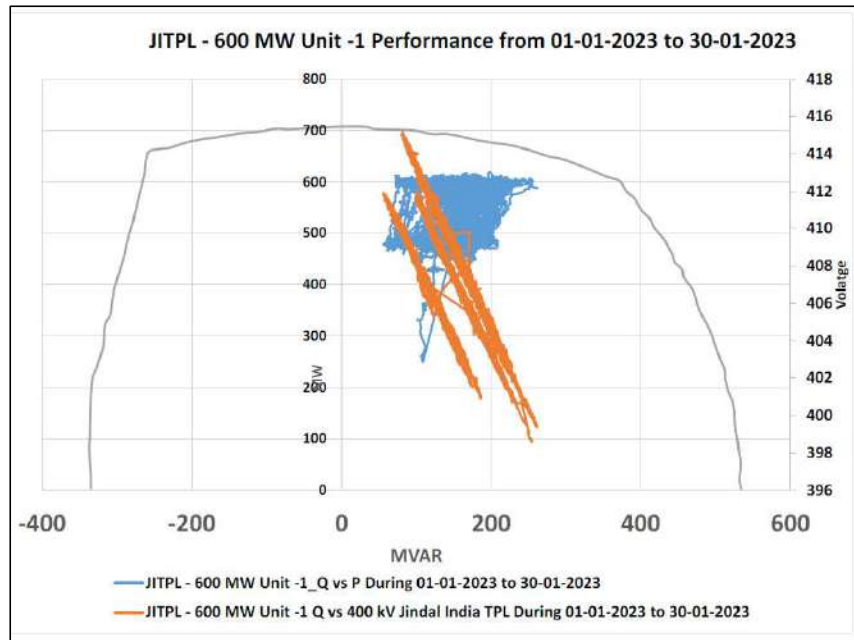


Annexure B.6

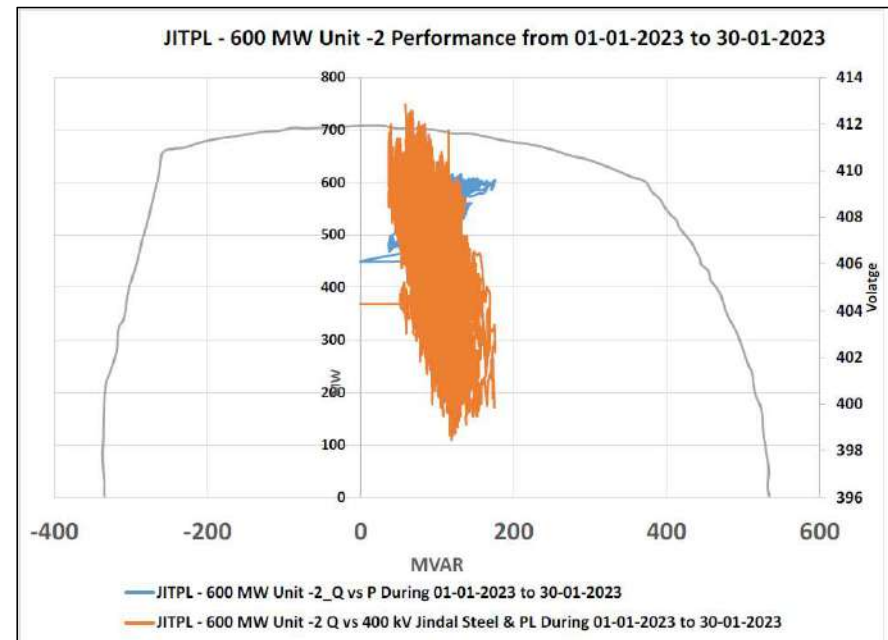
Approved Maintenance Schedule of Thermal Generating units in the month of January '2024

System	Station	Unit No.	Capacity(MW)	Period (as per LGBR 2023-24)		No. of Days	Reason	Approved period		No of days	Whether as per LGBR or not	Remarks
				From	To			From	To			
DVC	DSTPS	2	500	01.12.2023	28.12.2023	28	FGD & AOH	19.12.2023	14.01.2024	28	NO	APPROVED
DVC	RTPS	2	600	01.07.2023	20.08.2023	50	FGD & BOH	16.01.2024	12.02.2024	27	NO	APPROVED
DVC	MEJIA	5	250	25.10.2023	28.11.2023	34	FGD & COH	14.02.2024	15.03.2024	30	NO	APPROVED
DVC	MEJIA	3	210	21.01.2024	14.02.2024	25	AOH-Bir, LPT, FGD				NO	NOT AVAILED
DVC	MEJIA	7	500	01.01.2024	28.01.2024	28	AOH-Bir, LPT, Gen				NO	NOT AVAILED
NTPC	KhSTPS	7	500	10.01.2024	08.02.2024	30	Boiler Maintenance				NO	NOT AVAILED AS ALREADY AVAILED IN NOVEMBER'23
NTPC	KhSTPS	4	210	01.12.2023	04.01.2024	35	Boiler +HP+IP+Generator	01.12.2023	04.01.2024	35	YES	ALREADY APPROVED IN 209th OCC
NTPC	KhSTPS	5	500	01.12.2023	14.01.2024	45	Boiler +IP+Generator+Combustion modification				NO	NOT AVAILED
NTPC	FSTPP III	6	500				Recasting of HP turbine foundation subsequent to bearing damage	10.01.2024	25.02.2024	46		APPROVED
NTPC	Barh-I	2	660	01.12.2023	14.01.2024	45	Boiler +Generator				NO	NOT AVAILED
NTPC	Barh-II	5	660	20.01.2024	18.02.2024	30	BLR Modification Balance work				NO	NOT AVAILED
NTPC	Darlipalli STPS	2	800	15.11.2023	13.01.2024	60	COH				NO	NOT AVAILED
BRBCL	Nabinagar TPS	4	250	01.12.2023	09.01.2024	40	Boiler+ LPT O/H + Generator rotor thread out and checking +				NO	NOT AVAILED
WBPDC	Santalalih TPS	5	250	04.01.2024	23.01.2024	20	AOH/BOH	03.01.2024	22.01.2024	19	NO	APPROVED
WBPDC	Kolaghat TPS	3	210	26.01.2024	14.02.2024	20	AOH/BOH	02.02.2024	23.02.2024	21		APPROVED

Reactive power performance



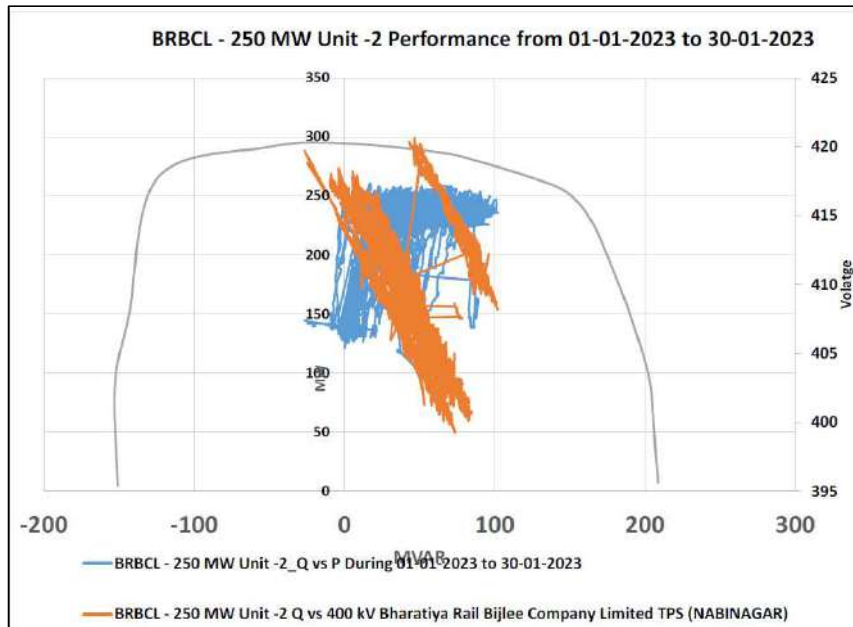
JITPL U-1



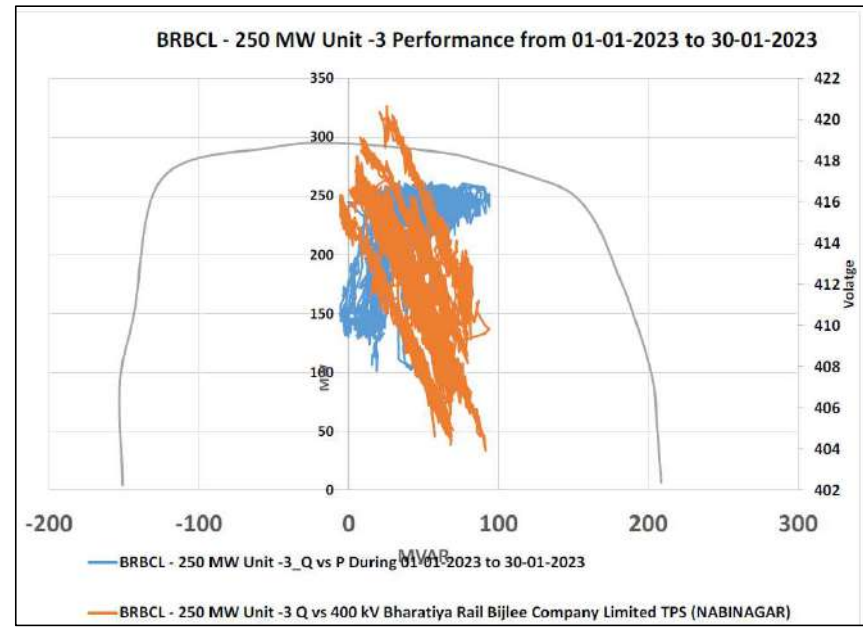
JITPL U-2

Need to perform reactive power capability test

Reactive power performance



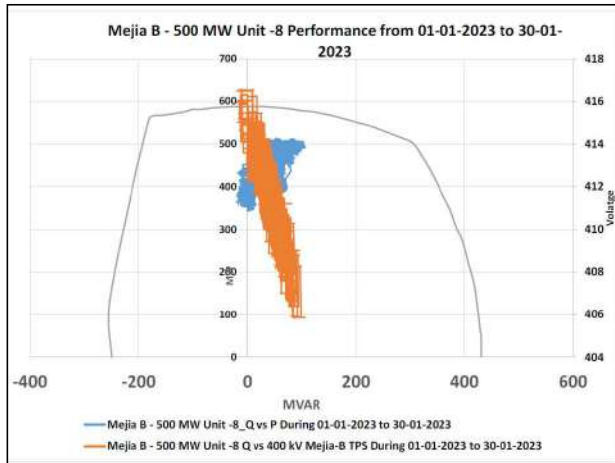
BRBCL U-2



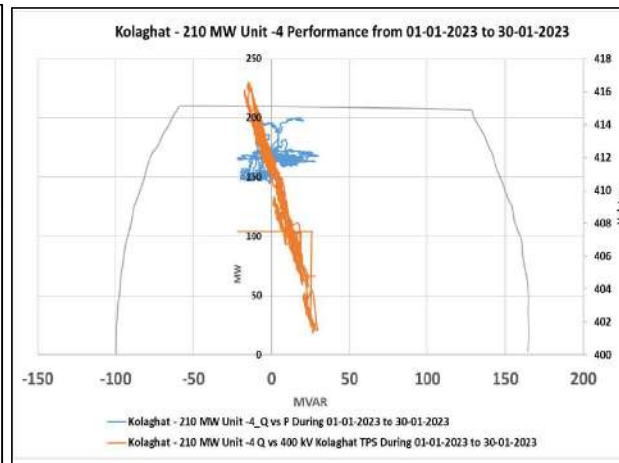
BRBCL U-3

Need to perform reactive power capability test

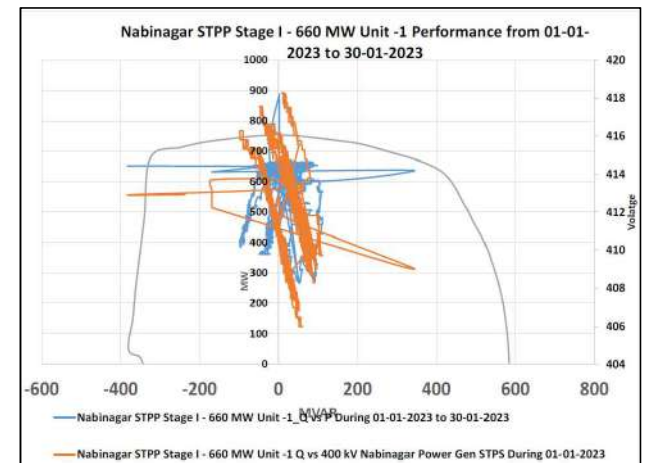
Reactive power performance



Mejia B U-8



Kolaghat U-4

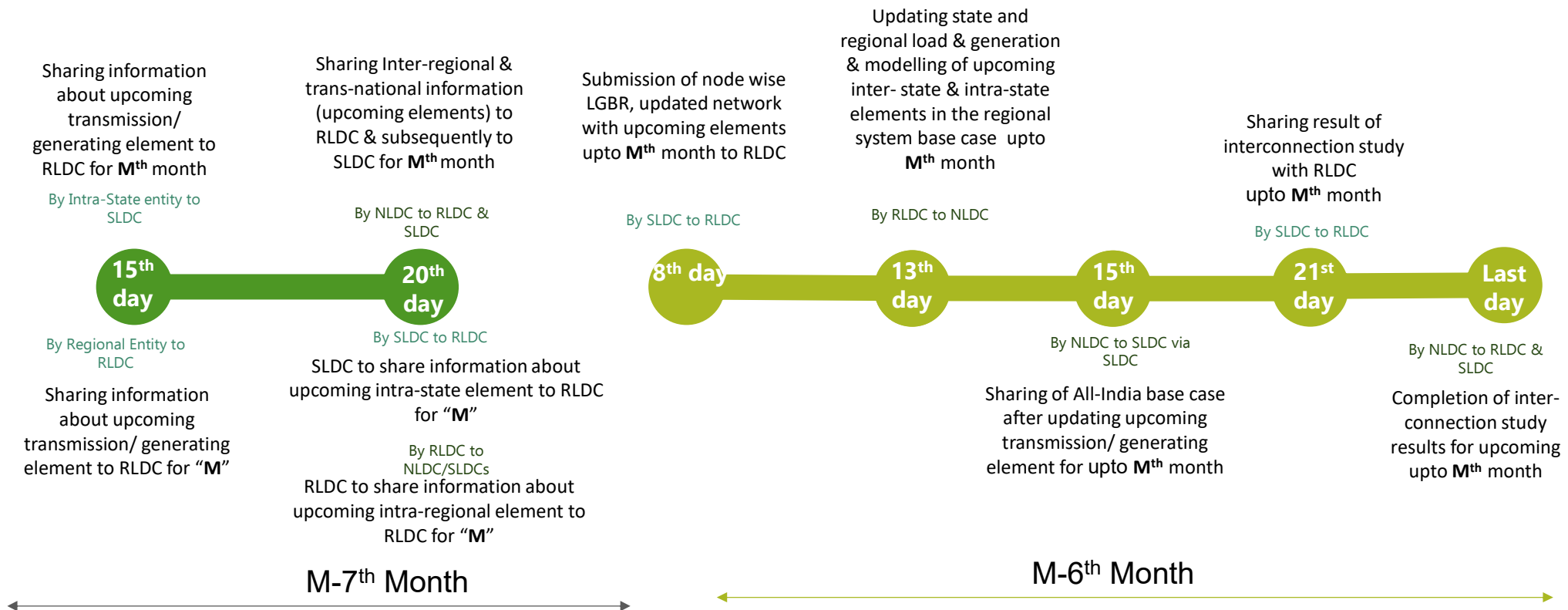


NPGC U-1

Need to perform reactive power capability test

Annexure B.12

Timeline for data submission for upcoming elements up to six months



Timeline for submission of TTC/ATC Data twelve months ahead

Submission of node wise Load and generation data along with envisaged scenarios for assessment of transfer capability, updated latest network to RLDC + Assessment of TTC/ATC revision-0 of the import/export capability of the state and intra-state system for “M+12” month

By SLDC to RLDC

10th
day

Updating state and regional load & generation & modelling of interstate& intra-state elements in the regional system base case + Assessment and declaration of TTC/ATC for the intra-regional and interstate system for “M+12” month

By RLDC to NLDC

26th
day

By SLDC to RLDC
Declaration of TTC/ATC of the intrastate system in consultation with RLDC

Update the All-India network model with inputs from RLDCs/SNA+ Assessment and declaration of inter-regional, bid area and cross-border TTC/ATC on the website for “M+12” month

By NLDC to RLDC

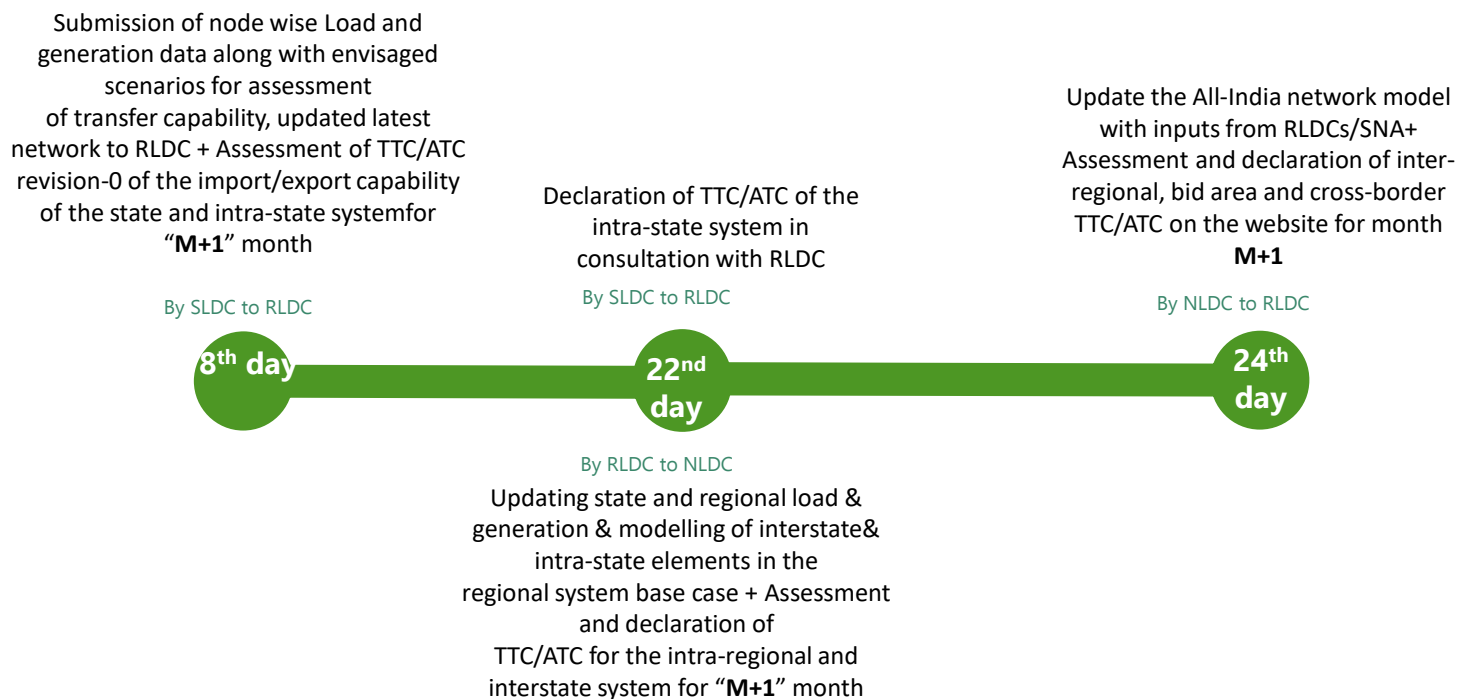
28th
day

Sharing of all india updated base case for “M+12” month

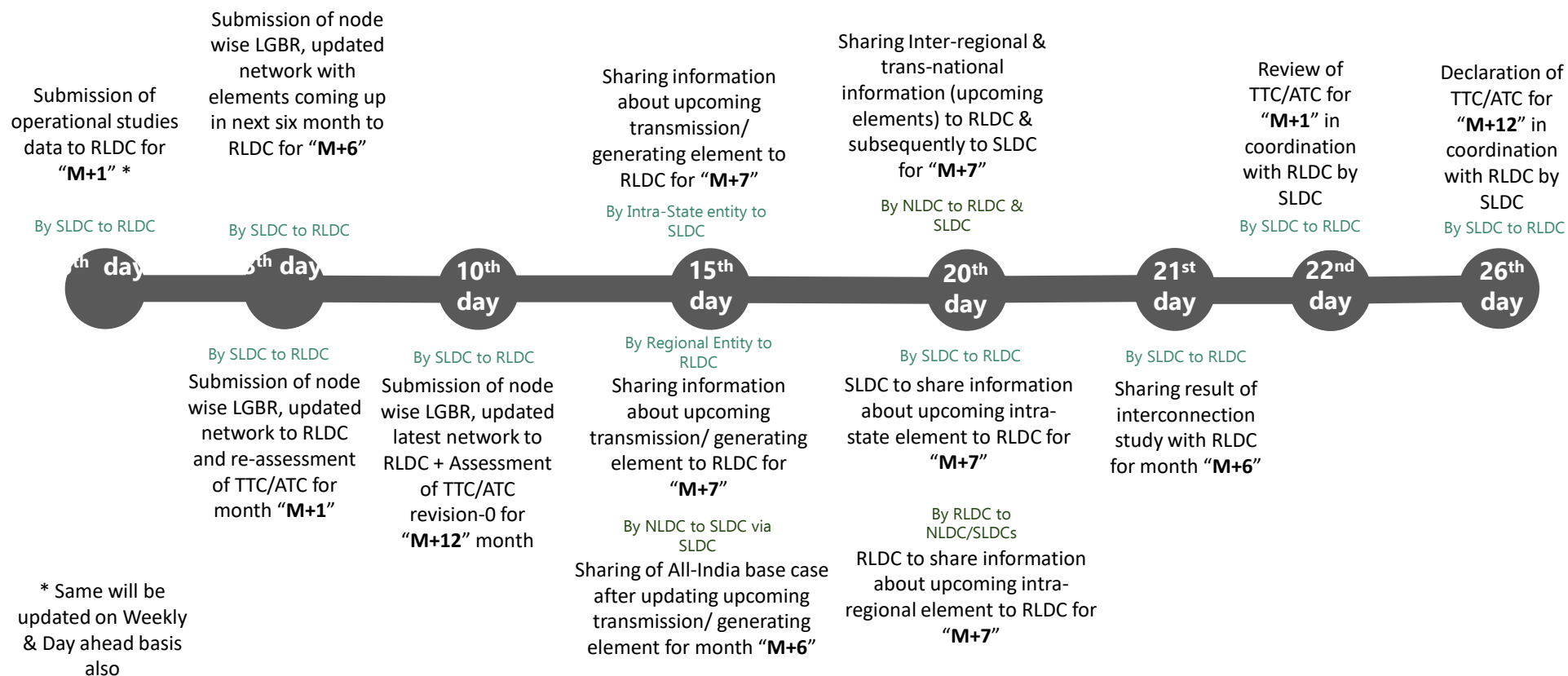
By RLDC to SLDC

By 1st day
of M+1

Timeline for submission of TTC/ATC Data One month ahead

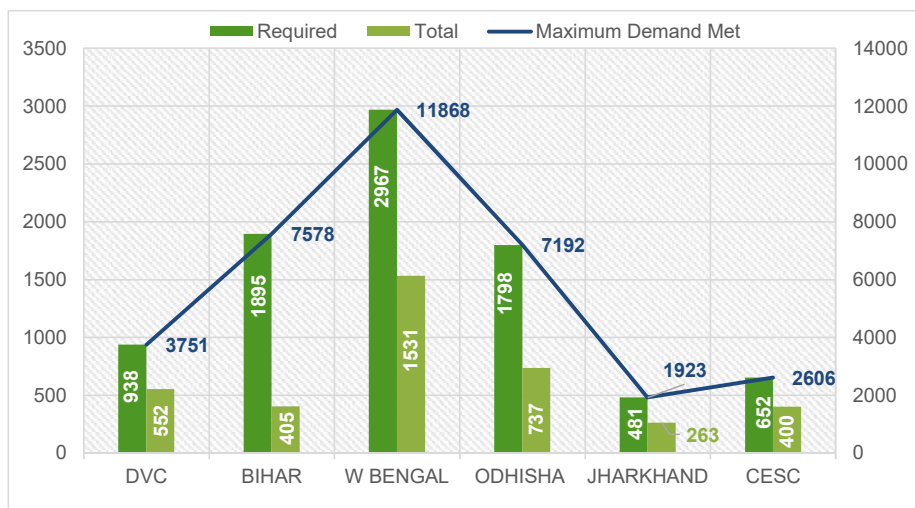


Timeline for submission data for different studies & results



Annexure B.13

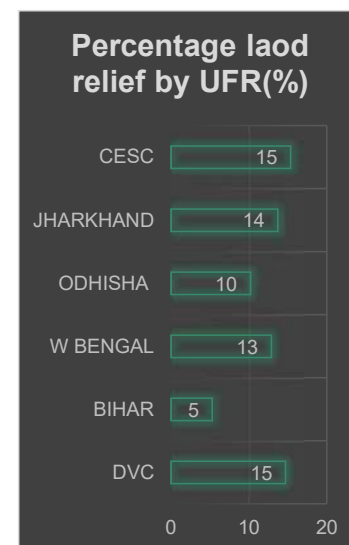
UFR Scheme



Relief under UFR Scheme

State	Delay	Relay Type
DVC	100ms	Numerical
BIHAR	No Delay	Static
W. BENGAL	200ms	Numerical
ODISHA	0	Numerical
JHARKHAND	0	Numerical

Delay & Relay type under UFR Scheme



Removal of
Intentional
Delay

Review of
configured
quantum.
Specially Bihar

Uniform
distribution
across all 4
stages

Action
Points

Updated Anticipated Peak Demand (in MW) of ER & its constituents for January 2024

1	BIHAR	Demand (MW)	Energy Requirement (MU)
	NET MAX DEMAND	5576	3212
	NET POWER AVAILABILITY- Own Sources	613	378
	Central Sector+Bi-Lateral	6323	4022
	SURPLUS(+)/DEFICIT(-)	1360	1188
2	JHARKHAND		
	NET MAXIMUM DEMAND	1845	1164
	NET POWER AVAILABILITY- Own Source	420	183
	Central Sector+Bi-Lateral+IPP	1062	576
	SURPLUS(+)/DEFICIT(-)	-363	-405
3	DVC		
	NET MAXIMUM DEMAND	3200	2022
	NET POWER AVAILABILITY- Own Source	5300	3295
	Central Sector+MPL	300	146
	Bi- lateral export by DVC	2500	1211
	SURPLUS(+)/DEFICIT(-) AFTER EXPORT	-100	207
4	ODISHA		
	NET MAXIMUM DEMAND (OWN)	4200	2827
	NET MAXIMUM DEMAND (In Case of CPP Drawal)	4900	3199
	NET POWER AVAILABILITY- Own Source	2500	1711
	Central Sector	1720	1116
	SURPLUS(+)/DEFICIT(-) (OWN)	20	0
	SURPLUS(+)/DEFICIT(-) (In Case, 600 MW CPP Drawal)	-680	-372
5	WEST BENGAL		
	WBSEDCL		
5.1	NET MAXIMUM DEMAND	6226	3661
	NET MAXIMUM DEMAND (Incl. Sikkim)	6231	3665
	NET POWER AVAILABILITY- Own Source (Incl. DPL)	5224	2659
	Central Sector+Bi-lateral+IPP&CPP+TLDP	2475	1202
	EXPORT (To SIKKIM)	5	4
	SURPLUS(+)/DEFICIT(-) AFTER EXPORT	1468	197
5.2	CESC		
	NET MAXIMUM DEMAND	1330	684
	NET POWER AVAILABILITY- Own Source	1000	419
	IMPORT FROM HEL	540	179
	TOTAL AVAILABILITY OF CESC	330	598
	DEFICIT(-) for Import	210	-86
			-86
	WEST BENGAL (WBSEDCL+CESC+IPCL)		
	(excluding DVC's supply to WBSEDCL's command area)		
	NET MAXIMUM DEMAND	7596	4345
	NET POWER AVAILABILITY- Own Source	5924	3078
	CS SHARE+BILATERAL+IPP/CPP+TLDP+HEL	2745	1381
	SURPLUS(+)/DEFICIT(-) BEFORE WBSEDCL'S EXPORT	1073	115
	SURPLUS(+)/DEFICIT(-) AFTER WBSEDCL'S EXPORT	1068	111
6	SIKKIM		
	NET MAXIMUM DEMAND	135	65
	NET POWER AVAILABILITY- Own Source	2	1
	Central Sector	56	15
	SURPLUS(+)/DEFICIT(-)	-76	-49
	EASTERN REGION		
	NET MAXIMUM DEMAND	24207	13758
	NET MAXIMUM DEMAND (In Case of CPP Drawal of Odisha)	24512	13714
	BILATERAL EXPORT BY DVC (Incl. Bangladesh)	1860	1211
	EXPORT BY WBSEDCL TO SIKKIM	5	4
	EXPORT TO B'DESH & NEPAL OTHER THAN DVC	642	478
	NET TOTAL POWER AVAILABILITY OF ER	25726	15040
	(INCLUDING CS ALLOCATION +BILATERAL+IPP/CPP+HEL)		
	SURPLUS(+)/DEFICIT(-)	1513	1279
	SURPLUS(+)/DEFICIT(-) (In Case, 600 MW CPP Drawal of Odisha)	1208	1323